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X-Fed NeuroScan

Federated Learning with Explainable AI
for Medical Image Classification

Team members

Abdelrahman Mamdouh	Abdelrahman Wahid
Goda Saber Goda	Hana Ahmed Nabhan
Mahmoud Ahmed Mahmoud	Ziad Ahmed Mohamed

Supervised by

Prof. Shaimaa Yousry
Prof. Walaa Gouda

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Abstract

Medical image analysis plays a critical role in the early detection and diagnosis of various diseases. However, manual interpretation of radiographic images is time-consuming and subject to inter-observer variability. This project presents an end-to-end deep learning–based system for the automated detection of chest diseases and bone fractures from X-ray images. The proposed solution integrates multiple transfer learning models within a unified diagnostic framework, enabling efficient and scalable analysis of heterogeneous medical images.

The system architecture consists of a deep learning pipeline that includes image preprocessing, feature extraction using pre-trained convolutional neural networks, an intelligent medical image routing model, and task-specific diagnostic models. ResNet50 is employed as a core feature extractor due to its residual learning mechanism and suitability for medical image analysis, while YOLOv8 is utilized for bone fracture detection using a one-stage object detection approach. The chest disease detection task is addressed using a dedicated classification model optimized for thoracic imaging.

Preliminary experimental results demonstrate the feasibility of the proposed approach and indicate promising performance in detecting both bone fractures and chest-related conditions. The modular design of the system allows for extensibility and supports the integration of additional diagnostic tasks. This work highlights the potential of deep learning and transfer learning techniques to assist clinical decision-making and improve the efficiency of medical image diagnosis.

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INTRODUCTION

Chapter 1 Introduction

Medical imaging plays a critical role in modern healthcare, supporting clinicians in diagnosing a wide range of conditions, from bone fractures to chest tumors and other life-threatening emergencies. As imaging technologies become more advanced and widely available, the volume of medical scans produced each day continues to grow. This rapid increase has intensified the need for efficient, reliable, and scalable diagnostic support systems that can assist healthcare professionals in interpreting complex visual data.

Recent advances in computer vision have made automated medical image analysis a promising tool for improving diagnostic accuracy and reducing clinical workload. Deep learning models, in particular, have demonstrated exceptional performance in tasks such as classification, localization, and segmentation across many imaging modalities. However, despite their capabilities, traditional centralized learning approaches often face critical limitations—especially in sensitive domains like healthcare, where data privacy, institutional policies, and regulatory constraints restrict the sharing of patient information.

To address these challenges, new paradigms in machine learning and model transparency have emerged. Federated learning enables collaborative training across multiple medical institutions without requiring direct access to patient data, offering a pathway toward more diverse and robust models while preserving privacy. Complementing this, explainable AI techniques provide insights into model decisions, helping clinicians understand why a prediction was made and increasing trust in AI-assisted diagnostics.

This project aims to integrate these three powerful components—computer vision, explainable AI, and federated learning—into a unified system designed to support medical image analysis across a variety of diseases and emergency conditions. By leveraging decentralized learning and transparent decision-making, the system seeks to improve accuracy, reliability, and interpretability while respecting the privacy and variability inherent in real-world clinical environments.

1.1 Problem Definition

Accurate diagnosis of bone fractures and chest diseases from X-ray images remains a critical challenge in modern healthcare. Many clinically important abnormalities—such as hairline fractures, early-stage pneumonia, subtle tuberculosis patterns, or small tumor-like chest masses—are visually faint, easily overlooked, or masked by overlapping anatomical structures. These challenges are intensified in high-volume clinical environments and can lead to delayed or incorrect diagnosis, negatively impacting patient outcomes.

Furthermore, deep learning models have achieved state-of-the-art performance in medical image analysis, but their clinical adoption is limited by two major obstacles: **(1) the inability to train diverse and robust models without aggregating sensitive**

patient data, and **(2) the black-box nature of deep models, which makes their predictions difficult for clinicians to trust or verify**. Hospitals typically hold isolated datasets that differ in imaging quality, patient demographics, disease prevalence, and device configurations, resulting in highly non-IID data distributions that degrade model performance when trained centrally or institutionally.

Given these challenges, the central problem addressed by this project is the following:

> **How can we build a privacy-preserving, explainable, and clinically reliable deep learning system for detecting four major chest disease classes—pneumonia, tuberculosis (TB), COVID-19, and tumor-like chest masses—using X-ray and CT images, despite highly heterogeneous data and strict privacy constraints across medical institutions?**

To contextualize this problem, the diseases and abnormalities targeted by the system are defined as follows:

- **Pneumonia:** An inflammatory lung infection characterized by alveolar consolidation, visible as opacities or infiltrates on chest X-rays.
- **Tuberculosis (TB):** A bacterial infection caused by *Mycobacterium tuberculosis*, typically presenting as lung cavities, nodular lesions, or patchy infiltrates.
- **COVID-19:** A viral infection (SARS-CoV-2) associated with ground-glass opacities, bilateral consolidations, and diffuse lung involvement on X-rays.
- **Chest masses/tumor-like abnormalities:** Suspicious opacities or lesions that may indicate malignant or benign tumors, requiring early detection for timely treatment.
- **Normal (healthy) cases:** X-rays exhibiting clear lung fields and healthy bone structures, used as the baseline class.

Addressing these conditions within a single diagnostic pipeline introduces several technical challenges:

- **Subtlety and variability of abnormalities:** Many chest pathologies present with faint or overlapping features that are difficult to detect even for experienced radiologists.
- **Data privacy and distribution constraints:** Medical data cannot be shared across hospitals due to privacy regulations, limiting model diversity and robustness.
- **Non-IID data across institutions:** Variations in imaging devices, acquisition protocols, and patient populations create distribution shifts that degrade performance.
- **Lack of interpretability:** Clinicians require transparent explanations—such as heatmaps showing fracture lines or diseased lung regions—to trust AI-assisted diagnosis.
- **Efficiency and deployability:** Models must run efficiently, with low communication costs, enabling real-world federated setups across hospitals or clinics.

Therefore, the problem tackled in this work is to design an integrated framework—*X-*

Fed NeuroScan—that combines federated learning for privacy preservation and multi-institution collaboration, with explainable AI techniques to generate clinically meaningful visual justifications for predictions, while achieving high diagnostic performance across the targeted disease categories.

1.2 Challenges in Medical Image Analysis

Medical imaging plays a fundamental role in modern healthcare, providing clinicians with valuable information for diagnosing, monitoring, and treating a wide range of diseases. Imaging modalities such as chest X-rays and Computed Tomography (CT) scans are routinely used to assess abnormalities within the thoracic region, including pneumonia, tuberculosis, pulmonary edema, and lung cancer. Despite the significant advancements in imaging technologies, the interpretation and analysis of medical images remain highly challenging tasks.

Medical image analysis traditionally relies on the expertise of radiologists and physicians who must examine large volumes of imaging data and identify subtle patterns that may indicate disease. The complexity of anatomical structures, variability among patients, and limitations of human perception can make accurate diagnosis difficult. These challenges can affect diagnostic accuracy, increase workload, and potentially delay clinical decision-making.

Complex Anatomical Structures

The human thoracic cavity contains numerous overlapping anatomical structures, including the lungs, heart, ribs, blood vessels, diaphragm, and surrounding soft tissues. In chest X-ray images, these structures are projected onto a two-dimensional plane, causing different tissues to overlap and potentially obscure important pathological findings.

The presence of overlapping structures makes it difficult for clinicians to accurately distinguish between normal anatomical variations and disease-related abnormalities. Small lesions, nodules, or early-stage infections may be hidden behind ribs or other dense structures, increasing the likelihood of missed diagnoses.

Subtle Appearance of Early-Stage Diseases

Many thoracic diseases initially manifest as subtle visual changes that are difficult to detect through routine examination. Early-stage lung cancer, for example, may appear as a small pulmonary nodule that is barely distinguishable from surrounding tissue. Similarly, mild pneumonia or early tuberculosis infections may produce only faint abnormalities within the lung fields.

Detecting these subtle findings requires significant expertise and experience. Even highly trained radiologists may encounter difficulties when abnormalities are small, poorly defined, or located in anatomically complex regions.

Variability in Disease Presentation

The same disease can appear differently across patients due to factors such as age, disease stage, genetic characteristics, and the presence of other medical conditions. Furthermore, different diseases may exhibit similar radiological appearances, making differential diagnosis challenging.

For example, pulmonary infections, inflammatory conditions, and certain types of lung cancer can produce similar opacities in chest X-rays. Likewise, benign and malignant lung nodules may share similar characteristics in CT scans. This variability increases diagnostic complexity and may lead to uncertainty during image interpretation.

Dependence on Radiologist Expertise

Accurate medical image interpretation requires extensive education, training, and clinical experience. Radiologists must possess detailed knowledge of anatomy, pathology, imaging physics, and disease progression in order to make reliable assessments.

However, diagnostic performance can vary among clinicians depending on their level of expertise and familiarity with specific conditions. Less experienced practitioners may overlook subtle findings or misinterpret complex cases. As a result, diagnostic consistency can differ between healthcare professionals and institutions.

Inter-Observer and Intra-Observer Variability

One of the well-known challenges in medical imaging is the variability in interpretation among radiologists. Different specialists may provide different assessments when evaluating the same image, a phenomenon known as inter-observer variability.

Additionally, the same radiologist may occasionally reach different conclusions when reviewing an image at different times, which is referred to as intra-observer variability. Such inconsistencies can affect diagnostic reliability and complicate treatment planning, particularly in borderline or ambiguous cases.

Large Volume of Imaging Data

Modern healthcare systems generate enormous quantities of medical images on a daily basis. Radiologists are often required to review hundreds of imaging studies during a single workday, creating significant workload pressures.

CT examinations present an additional challenge because a single scan may contain hundreds of image slices that must be carefully reviewed. Thorough examination of such large datasets is time-consuming and can contribute to fatigue, potentially increasing the risk of oversight or diagnostic errors.

Time Constraints in Clinical Practice

Healthcare professionals frequently operate under strict time constraints, particularly in busy hospitals and emergency departments. Rapid diagnosis is often essential for initiating timely treatment and improving patient outcomes.

The need to balance speed with diagnostic accuracy can be challenging, especially when dealing with complex imaging studies. Under time pressure, subtle abnormalities may be overlooked, and difficult cases may require additional consultation or follow-up examinations.

Image Quality Limitations

The diagnostic quality of medical images can be affected by several factors, including patient movement, improper positioning, low contrast, imaging noise, and equipment-related artifacts. Poor image quality may obscure important pathological findings and make interpretation more difficult.

In chest X-ray imaging, underexposure or overexposure can reduce visibility of anatomical details. In CT imaging, motion artifacts caused by breathing or patient movement can degrade image clarity. Such limitations may reduce diagnostic confidence and necessitate repeat imaging procedures.

Diagnostic Errors and Missed Findings

Despite advances in medical imaging technology, diagnostic errors remain an unavoidable challenge. Errors may occur due to perceptual limitations, cognitive biases, fatigue, distraction, or the inherently complex nature of certain cases.

Missed lung nodules, overlooked fractures, and undetected early-stage diseases represent some of the most common diagnostic challenges in thoracic imaging. Because delayed diagnosis can significantly impact patient outcomes, minimizing such errors remains a major objective within radiology.

Challenges in Lung Cancer Detection

Lung cancer detection presents particularly significant difficulties due to the diverse appearance of pulmonary nodules. Nodules may vary greatly in size, shape, texture, density, and location. Small malignant nodules can closely resemble benign lesions or normal anatomical structures.

Furthermore, certain tumors may be partially obscured by blood vessels, ribs, or surrounding tissues. Accurate identification and characterization of suspicious nodules often require careful examination across multiple CT slices and comparison with previous studies, making the diagnostic process both complex and time-intensive.

Increasing Demand for Radiological Services

The global demand for medical imaging services continues to grow due to population growth, aging demographics, and increased utilization of diagnostic imaging technologies. However, many healthcare systems face shortages of trained radiologists and imaging specialists.

This imbalance between demand and available expertise can lead to increased workloads, longer reporting times, and reduced access to timely diagnoses. The challenge is particularly pronounced in developing regions and underserved healthcare environments.

1.3 Background and Motivation

The rapid growth of medical imaging data, coupled with increased clinical workloads, has made conventional diagnostic workflows increasingly insufficient. Although automated analysis using computer vision holds considerable promise in improving diagnostic accuracy and efficiency, practical implementation is hindered by challenges such as patient privacy restrictions, limited data set diversity, and substantial computational and data transfer demands. Therefore, a thorough understanding of both the potential benefits and inherent limitations of AI in medical imaging is essential for the design of systems that are accurate, interpretable, and scalable in heterogeneous clinical environments.

Rise of Computer Vision in Automated Analysis

The rapid digitization of healthcare has led to a dramatic increase in medical imaging data, including X-ray, CT, MRI, and ultrasound scans. Radiologists now face the dual challenges of handling large patient loads and interpreting large volumes of images quickly and accurately. Computer vision has emerged as a crucial tool for addressing these demands by enhancing diagnostic efficiency and accuracy.

To understand why computer vision is now widely adopted in clinical workflows, consider the following advantages:

- **High sensitivity to subtle patterns**: AI can detect faint fractures or early-stage tumors.
- **Rapid processing**: Thousands of images can be analyzed within seconds.
- **Consistency**: Predictions remain stable regardless of fatigue or emotional state.
- **Adaptability**: Deep models can generalize to various imaging modalities.

The effectiveness of computer vision in medical imaging is reinforced by major achievements in the field. These advancements primarily arise from deep learning architectures such as CNNs and Vision Transformers. In practice, computer vision supports several core tasks:

1. Classification of abnormal vs. normal scans.

2. Localization of suspicious areas using object detection.

3. Segmentation of organs, lesions, and fracture regions.

4. Identification of patterns that deviate from healthy anatomy.

A concise summary of computer vision benefits in medical imaging is outlined in Table 1.1.

Table 1.1: Key Advantages of Computer Vision in Medical Image Automation

Computer Vision Capability	**Role in Medical Imaging**
Feature extraction	Detects subtle abnormalities beyond human perception
Automation of routine tasks	Reduces workload and radiologist fatigue
High-speed inference	Enables rapid triage in emergencies
Generalization across cases	Handles variations in anatomy and imaging conditions

Limitations of Centralized Machine Learning

Despite the promise of AI-based diagnostic systems, models trained in a centralized manner face several constraints. These limitations arise mainly from restrictions on data availability, data diversity, and practical constraints in medical settings.

Centralized learning typically suffers from:

- **Restricted data sharing**: Patient data cannot be freely exchanged due to privacy laws.
- **Lack of representation**: Single-hospital datasets fail to capture global patient diversity.
- **High storage and transfer requirements**: Medical images are large and expensive to move.
- **Data imbalance**: Rare conditions remain underrepresented, harming model performance.

These issues do not simply degrade model accuracy—they introduce significant risks into clinical AI deployment. The consequences can be better understood through the following observations:

1. Models trained in isolation often misclassify images from new machines or demographics.

2. Performance can fluctuate between institutions due to imaging protocol differences.

3. Bias can emerge when the training set fails to include certain patient groups.

Table 1.2 summarizes these core limitations in a structured manner.

Table 1.2: Challenges of Centralized Machine Learning in Healthcare

Limitation	**Impact**
Privacy restrictions	Prevent hospitals from sharing sensitive patient data
Limited dataset diversity	Reduces generalization to different populations or imaging devices
Large data transfer requirements	Increases cost and delays in model development
Imbalanced condition frequency	Degrades performance on rare diseases or subtle abnormalities

Centralized machine learning, while powerful, therefore struggles to meet the scalability, diversity, and privacy needs of real-world medical AI deployment. These challenges motivate the development of more robust learning paradigms capable of overcoming data fragmentation and institutional isolation.

1.4 Suggested Solution

The challenges identified in the medical imaging domain—including subtle abnormalities, limited dataset diversity, privacy constraints, and variability in imaging standards—necessitate a more advanced diagnostic support system. The proposed project aims to address these limitations by integrating three complementary technologies: Computer Vision, Explainable AI, and Federated Learning. Together, these components form an intelligent system capable of assisting clinicians in identifying fractures, tumors, infections, and other abnormalities more reliably and transparently.

The core idea behind the project is to design a computer vision pipeline capable of analyzing X-ray and other medical imaging modalities with high accuracy. Deep learning models will detect patterns that may be difficult for clinicians to recognize quickly, especially in emergency or high-stress situations. By augmenting image interpretation with AI-based classification and region highlighting, the system serves as a powerful second-opinion tool, reducing oversight and improving diagnostic confidence.

To ensure that the system remains trustworthy and clinically interpretable, Explainable AI techniques are incorporated to reveal how the model arrives at its predictions. This transparency allows medical professionals to understand the rationale behind the AI's outputs, compare them with clinical knowledge, and detect any mistakes or biases early.

Moreover, Federated Learning is utilized to overcome the limitations of centralized training. Instead of pooling patient data into a single server—a process that is often legally and ethically restricted—hospitals collaboratively train the model without exposing sensitive information. This approach significantly increases dataset diversity while maintaining strict compliance with privacy regulations.

In summary, the proposed solution seeks to:

- Improve diagnostic accuracy through advanced computer vision algorithms.
- Increase clinician trust via transparent and interpretable model outputs.
- Overcome data-sharing barriers by enabling collaborative, privacy-preserving training.

These three components work together to form a robust, scalable, and clinically meaningful AI-based diagnostic support framework.

Explainable AI (XAI)

Explainable Artificial Intelligence (XAI) refers to a set of methods and frameworks aimed at increasing the transparency, interpretability, and trustworthiness of machine learning models. As modern AI systems grow in complexity, their internal decision-making processes often become opaque, creating what is commonly known as the "black-box" problem. In safety-critical domains such as medicine, where clinical decisions directly affect patient outcomes, this lack of transparency is a major barrier to adoption. XAI methods attempt to address this challenge by providing human-understandable insights into how and why a model arrives at a specific prediction.

In medical imaging, explainability is particularly important due to the high stakes of diagnosis, the need for clinician trust, and the requirement for regulatory compliance. Rather than simply outputting a prediction, explainable models are capable of highlighting meaningful visual patterns, quantifying the contribution of specific features, or generating human-interpretable reasoning. These explanations serve as an additional layer of evidence that supports radiologists and clinicians during diagnostic workflows.

XAI supports clinical decision-making by:

- Identifying the precise image regions or features that contribute most strongly to the model's prediction.
- Ensuring that the model's focus aligns with clinically relevant structures rather than spurious artifacts.
- Facilitating the detection of potential model errors, such as overfitting, dataset bias, or misinterpretation of anatomical features.
- Enabling clinicians to verify, critique, or challenge AI-generated predictions using domain expertise.
- Improving trust and accountability by providing a transparent rationale that can be communicated in clinical reports.

A variety of XAI techniques are widely used in medical imaging research and practice. Visual explanation methods such as Class Activation Maps (CAM), Gradient-weighted CAM (Grad-CAM), and attention-based heatmaps highlight the spatial regions that influence a model's output. Pixel-level saliency maps capture fine-grained sensitivity to image perturbations. Beyond visual methods, some approaches provide concept-based explanations, feature attribution scores, or even natural-language descriptions that mimic clinical reasoning. Together, these techniques help bridge the gap between the sophisticated internal representations learned by deep neural networks and the need for interpretable, clinically meaningful insight within medical environments.

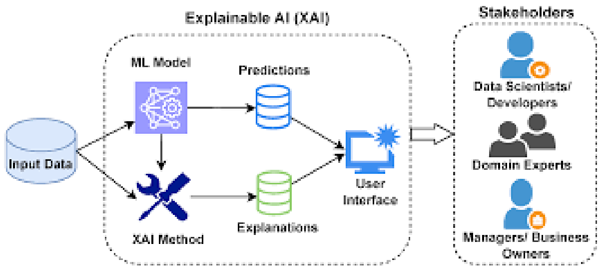


Figure 1.1: Explainble AI in action

Federated Learning

Federated Learning (FL) is a decentralized machine learning paradigm designed to enable collaborative model training without the need to share raw data between institutions. Unlike traditional centralized approaches, where data must be gathered into a single repository, federated learning allows each participating site to keep its data local while only exchanging model parameters or gradients. This architecture directly addresses major concerns in modern healthcare environments, including patient privacy, data security, legal constraints, and the challenges of aggregating large, distributed medical datasets.

In the medical field, data collected across hospitals, clinics, and imaging centers is often heterogeneous and siloed due to regulatory frameworks such as GDPR and HIPAA. Federated Learning provides a solution by ensuring that sensitive patient information never leaves its source institution. Instead, the global model is iteratively improved by aggregating updates computed locally at each site. This setup allows institutions to benefit from the collective data diversity and scale—key factors for building robust AI models—while preserving strict privacy and confidentiality guarantees.

Federated Learning benefits clinical AI development by:

- Allowing organizations to collaboratively train high-performing models without sharing **protected health information (PHI).**
- Leveraging diverse datasets from multiple geographic regions, improving the model's generalizability across patient populations.
- Reducing the risk of data breaches, since sensitive information remains stored on secure local servers.
- Supporting compliance with ethical and legal requirements governing medical data usage.
- Enabling continuous model improvement as new data becomes available in real clinical environments.

Several FL strategies have been developed to support medical imaging applications. The most widely used is *Federated Averaging (FedAvg)*, where each institution trains a local version of the model and periodically sends its parameter updates to a central aggregator. More advanced techniques—such as personalized federated learning, secure aggregation, and differential privacy—address challenges like domain shifts between institutions, communication efficiency, and protection against inference attacks. In addition, hybrid approaches integrate federated learning with techniques such as transfer learning or multimodal fusion, enabling complex models to be trained across distributed datasets while retaining strong privacy guarantees.

By enabling collaborative model development across institutions without compromising data confidentiality, Federated Learning plays a crucial role in the advancement of trustworthy and scalable AI systems in healthcare.

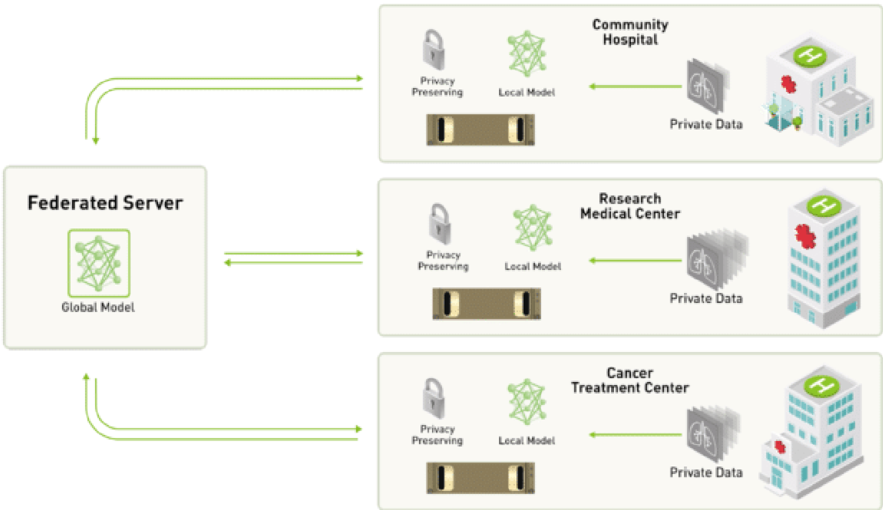


Figure 1.2: Centeralized server approach to federated learning

1.5 Project Scope

The scope of this project centers on developing a comprehensive medical image analysis system capable of diagnosing two primary categories of conditions: bone fractures and chest tumors. These conditions were selected due to their prevalence in clinical settings, their potential for severe medical consequences if missed, and the proven effectiveness of computer vision techniques in detecting abnormalities within skeletal and thoracic imaging. The system aims to support clinicians by providing fast, reliable, and interpretable diagnostic assistance.

The core of the project focuses on designing and training deep learning models that can accurately analyze X-ray images. For chest diseases and tumors, the pipeline will identify abnormal masses or opacities that may indicate malignancies or other critical conditions. The emphasis is on achieving high sensitivity and specificity while ensuring that the model remains robust across diverse imaging conditions.

To enhance usability and ensure accessibility for non-technical users, the diagnostic engine will be wrapped within a web-based application. This interface will allow clinicians, technicians, and medical staff to upload images, visualize model predictions, and interact with diagnostic outputs without requiring specialized technical knowledge. Although the interface will play an important role in facilitating real-world use, the primary focus of the project remains on the development, training, and evaluation of the underlying diagnostic models.

The system will include the following high-level functionalities:

- Automated detection of bone fractures in X-ray images.
- Identification of suspicious chest masses or tumor-like abnormalities.
- Visualization of diagnostic results and highlighted regions of interest.
- A user-friendly web interface that integrates the underlying AI engine.

The project does *not* aim to replace professional medical judgment. Instead, it seeks to provide a reliable, efficient, and interpretable tool that supports clinicians in making faster and more accurate assessments. Future extensions—such as expanding the system to additional diseases, modalities, or institutions—fall outside the current scope but remain promising directions for continued development.

1.6 Project Objectives

The objectives of this project are centered around creating a reliable, accurate, and accessible diagnostic support system for bone-fracture and chest tumor identification. The project aims to ensure that the system not only performs well in terms of machine learning metrics but also provides practical value to clinicians through usability, efficiency, and cost-effectiveness. Achieving clinical-level performance requires a well-defined set of goals that align both with technical constraints and real-world medical needs.

A primary objective is to develop a computer vision model capable of achieving high diagnostic performance. In particular, the system is designed to reach an accuracy above 90% across its targeted conditions while maintaining a recall rate that minimizes the risk of missed fractures or tumors. Since false negatives can lead to severe clinical consequences, optimizing recall is essential for ensuring the safety and reliability of the system.

In addition to predictive performance, another major objective is to deliver an intuitive and engaging web interface that enables medical personnel to interact with the system easily. The interface should allow users to upload images, view results, and interpret highlighted regions without requiring technical expertise. Ease of use is critical for facilitating adoption in real clinical workflows, where time efficiency and clarity are important.

Operational efficiency is also a key objective. The system must be designed in a manner that minimizes computational overhead and reduces ongoing costs. This includes optimizing model size, using lightweight inference strategies where possible, and ensuring that the web application remains responsive even with limited hardware resources.

To summarize, the project seeks to meet the following objectives:

- Achieve a diagnostic accuracy exceeding 90% on bone-fracture and chest tumor detection tasks.
- Maintain a high recall rate to reduce the likelihood of missed abnormalities.
- Provide a user-friendly, accessible, and engaging web interface suitable for non-technical clinical staff.
- Develop an efficient system that minimizes operational and computational costs.

Collectively, these objectives define the criteria by which the success of the project will be evaluated and guide the design decisions throughout development.

1.7 Project Challenges

Despite the remarkable progress achieved in recent years, medical image analysis remains a challenging field. The complexity of medical data, the critical nature of clinical decision-making, and the variability of imaging procedures introduce numerous obstacles that must be addressed to develop reliable and clinically applicable systems. The following subsections discuss the major challenges associated with medical image analysis, particularly in the context of chest disease detection using X-ray images and lung cancer detection using CT scans.

Limited Availability of High-Quality Medical Data

One of the most significant challenges in medical image analysis is the limited availability of large, high-quality, and well-annotated datasets. Deep learning models generally require substantial amounts of labeled data to achieve high performance and robust generalization. However, obtaining medical images with accurate annotations is often difficult and expensive.

Medical image annotation requires the expertise of trained radiologists or medical specialists who must carefully examine each image and identify pathological findings. This process is time-consuming and resource-intensive. Furthermore, patient privacy regulations and ethical considerations often restrict the sharing of medical data between healthcare institutions, limiting the availability of publicly accessible datasets.

In the context of chest X-ray analysis, some diseases may occur infrequently, leading to severe class imbalance within datasets. Similarly, for lung cancer detection from CT scans, obtaining sufficiently large collections of confirmed malignant and benign cases can be challenging. These limitations can negatively affect model training and reduce diagnostic performance.

Variability in Image Acquisition

Medical images are acquired using different imaging devices, protocols, and settings across hospitals and healthcare facilities. As a result, images may exhibit significant variations in quality, resolution, contrast, brightness, noise levels, and anatomical positioning.

Chest X-ray images obtained from different institutions may vary due to differences in machine manufacturers, patient positioning techniques, exposure settings, and image preprocessing procedures. Likewise, CT scans may differ in slice thickness, reconstruction algorithms, scanning protocols, and radiation dose levels.

Such variability can lead to distribution shifts between training and deployment environments. A model trained on data from one institution may experience performance degradation when applied to images acquired in another clinical setting. Ensuring robustness across diverse imaging conditions remains a major challenge for medical image analysis systems.

Image Noise and Artifacts

Medical images frequently contain noise and artifacts that can obscure important anatomical structures and pathological findings. These imperfections may arise from patient movement, hardware limitations, acquisition errors, or environmental factors during image capture.

In chest X-ray imaging, artifacts such as medical devices, implants, external objects, and improper positioning can interfere with disease detection. CT scans may contain motion artifacts, beam-hardening effects, metal artifacts, and reconstruction-related distortions.

The presence of noise and artifacts can significantly reduce image quality and negatively affect the performance of automated detection systems. Consequently, effective preprocessing techniques, including image enhancement, denoising, and artifact removal, are often required before analysis.

Complexity of Disease Manifestations

Many thoracic diseases exhibit highly variable visual characteristics, making automated detection particularly challenging. Diseases may present differently depending on their severity, progression stage, patient demographics, and coexisting medical conditions.

For example, pneumonia, tuberculosis, pulmonary edema, and COVID-19 may produce overlapping radiographic features in chest X-rays. Similarly, lung nodules observed in CT scans can vary considerably in size, shape, texture, density, and location. Some malignant nodules closely resemble benign abnormalities, increasing the difficulty of accurate classification.

Additionally, certain diseases may manifest as subtle changes that are difficult to detect even for experienced radiologists. Developing models capable of recognizing both obvious and subtle pathological patterns remains a critical research challenge.

Class Imbalance in Medical Datasets

Medical datasets often suffer from class imbalance, where healthy cases significantly outnumber disease-positive cases or where some diseases occur much less frequently than others. This imbalance can lead machine learning models to favor majority classes during training.

For instance, in chest X-ray datasets, normal scans may constitute a large proportion of the available data, while rare diseases may only be represented by a small number of samples. Similarly, in lung cancer screening datasets, malignant nodules may be considerably less common than benign findings.

Class imbalance can result in poor sensitivity for detecting rare but clinically important conditions. Addressing this issue often requires specialized techniques such as data augmentation, oversampling, synthetic data generation, weighted loss functions, and balanced training strategies.

Interpretability and Explainability

Healthcare professionals require transparency and trust when using artificial intelligence systems in clinical practice. Many state-of-the-art deep learning models function as black-box systems, producing predictions without providing clear explanations for their decisions.

In medical diagnosis, explainability is particularly important because clinicians must understand the reasoning behind automated recommendations before incorporating them into patient care decisions. A prediction without justification may not be sufficient in high-stakes medical environments.

To address this challenge, researchers employ explainable AI techniques such as saliency maps, Grad-CAM visualizations, attention mechanisms, and feature attribution methods. These approaches help identify image regions that contribute most strongly to model predictions, thereby improving transparency and clinical confidence.

Generalization and Robustness

A medical image analysis system must perform reliably across different patient populations, imaging devices, healthcare institutions, and geographical regions. However, models trained on specific datasets may inadvertently learn dataset-specific characteristics rather than clinically meaningful features.

This issue can lead to reduced performance when models encounter unseen data during real-world deployment. Factors such as demographic differences, variations in disease prevalence, and differences in imaging protocols can all affect model generalization.

Ensuring robustness requires diverse training datasets, rigorous validation procedures, external testing across multiple institutions, and continuous performance monitoring after deployment.

Three-Dimensional Nature of CT Data

Unlike chest X-rays, which provide two-dimensional projections, CT scans generate three-dimensional volumetric data. While this additional information improves diagnostic capabilities, it also introduces substantial computational and analytical challenges.

CT volumes may contain hundreds of slices for a single patient examination. Processing such large datasets requires considerable memory, storage capacity, and computational resources. Furthermore, accurately identifying lung nodules often requires analyzing spatial relationships across multiple slices.

Advanced techniques such as 3D convolutional neural networks, volumetric segmentation algorithms, and multi-view learning approaches are commonly employed to address these challenges. However, these methods generally demand greater computational complexity and longer training times.

Accurate Segmentation of Anatomical Structures

Segmentation is a fundamental step in many medical image analysis pipelines. It involves identifying and isolating relevant anatomical structures such as lungs, airways, lesions, tumors, and nodules.

Accurate segmentation is particularly challenging when boundaries are unclear, tissues overlap, or pathological regions exhibit irregular shapes. In chest X-ray images, overlapping anatomical structures can complicate lung segmentation. In CT scans, tumors may have poorly defined margins or be attached to surrounding tissues.

Segmentation errors can propagate through subsequent analysis stages and adversely affect disease detection and classification performance. Therefore, robust segmentation algorithms remain an important area of ongoing research.

Clinical Validation and Regulatory Requirements

Developing an accurate machine learning model is only one step toward clinical adoption. Medical image analysis systems must undergo extensive validation to demonstrate safety, reliability, and effectiveness in real-world healthcare environments.

Clinical validation typically involves testing models on large and diverse patient populations, comparing performance against expert radiologists, and assessing their impact on clinical workflows. Furthermore, healthcare technologies are subject to strict regulatory requirements and quality standards before deployment.

Meeting these regulatory and clinical validation requirements can be a lengthy and resource-intensive process, but it is essential for ensuring patient safety and maintaining trust in AI-assisted diagnostic systems.

Ethical and Privacy Considerations

Medical image analysis systems operate on highly sensitive patient information. Protecting patient privacy and ensuring compliance with healthcare regulations are critical responsibilities throughout the development lifecycle.

Issues such as data security, informed consent, data ownership, and algorithmic bias must be carefully addressed. Models trained on non-representative datasets may exhibit reduced performance for certain demographic groups, potentially leading to inequitable healthcare outcomes.

Consequently, ethical AI development requires responsible data management practices, fairness assessments, transparency, and continuous evaluation to ensure that diagnostic systems benefit all patient populations.

Computational Resource Requirements

Modern deep learning architectures used in medical image analysis often contain millions of trainable parameters and require substantial computational resources for training and inference. High-resolution chest X-ray images and volumetric CT scans further increase computational demands.

Training advanced models frequently requires powerful Graphics Processing Units (GPUs), large memory capacities, and extensive training times. These requirements may limit accessibility for smaller healthcare institutions and research groups with limited computational infrastructure.

Optimizing model efficiency while maintaining diagnostic accuracy remains an important challenge, particularly for deployment in resource-constrained clinical environments.

1.8 Summary

This chapter introduced the motivation and context behind the development of an intelligent medical image analysis system designed to support clinicians in diagnosing bone fractures and chest tumors. The rapid growth of medical imaging data, combined with the increasing complexity and subtlety of diagnostic tasks, has created a pressing need for automated tools capable of enhancing accuracy and reducing oversight. Traditional interpretation methods are challenged by factors such as faint abnormalities, variations in imaging quality, and the diverse presentation of conditions across patients and imaging devices.

To address these challenges, the chapter examined the rise of computer vision in healthcare and its effectiveness in automating image-based diagnostic tasks. While machine learning has demonstrated considerable promise in detecting complex patterns within medical images, centralized approaches face several limitations including restricted data sharing, lack of dataset diversity, and substantial computational burdens. These constraints hinder the performance and generalizability of AI models, especially in a domain where reliability is critical.

The chapter also presented the rationale for the proposed solution, which integrates advanced computer vision techniques with Explainable AI and Federated Learning. This combination aims to improve diagnostic accuracy, enhance interpretability, and overcome data privacy barriers that limit large-scale collaborative model training. By incorporating interpretability mechanisms, the system ensures that clinicians can understand and verify model decisions, while federated learning enables multi-institutional training without compromising patient confidentiality.

Furthermore, the scope and objectives of the project were clearly defined. The system aims to achieve high accuracy and recall in detecting bone fractures and chest tumors, provide a user-friendly interface suitable for non-technical users, and maintain operational efficiency to minimize computational costs. The chapter also outlined the challenges likely to arise during development, including data limitations, model robustness, explanation quality, system efficiency, and the complexities associated with implementing federated learning securely.

In summary, the introduction established the need for an intelligent, interpretable, and privacy-preserving diagnostic platform and laid the foundation for the system proposed in this project. The subsequent chapters build upon this foundation by detailing the methodologies, design choices, experiments, and evaluations that guide the development of the final system.

CHAPTER 2
LITERATURE REVIEW

Chapter 2 Literature Review

In this chapter, a literature review is provided in project-related areas. To build the framework (X-Fed NeuroScan: Federated Learning with Explainable AI for Medical Image Classification), we need to get familiar with foundations of medical image analysis, deep learning-based classification techniques, and the challenges associated with handling sensitive clinical data. We also examine the principles and methodologies of Federated Learning (FL) as a privacy-preserving approach for multi-institutional model training, as well as Explainable AI (XAI) techniques that enhance model transparency and clinical trust. Additionally, this chapter reviews existing commercial medical imaging applications, related academic research, and publicly available datasets. Finally, we evaluate the tools, platforms, and frameworks that support the development of deep learning models, federated learning workflows, and explainable AI components.

The chapter is organized as follows:

- Section 2.1: Medical image history and challenges, with subsections on bone fractures and chest diseases.
- Section 2.2: Explainable AI (XAI) in medical image detection.
- Section 2.3: Federated Learning (FL) in medical image detection.
- Section 2.4: Combined approaches integrating detection with XAI and FL.
- Section 2.5: Summary of related papers.
- Section 2.6: Similar commercial applications.
- Section 2.7: Summary of the chapter.

2.1 Medical Image Diagnosis

Medical imaging is the cornerstone of modern diagnosis, providing non-invasive visualization of the body's interior using techniques like X-rays, Computed Tomography (CT), and Ultrasound. Due to the high volume and complexity of medical scans, interpretation is often time-consuming and subject to human variability. This process is increasingly enhanced by **Artificial Intelligence (AI)**, particularly **Computer Vision (CV)** and Deep Learning (DL) methods.

While traditional Deep Learning (DL) approaches have achieved high accuracy in radiological analysis for disease detection and bone fracture localization, their clinical deployment is significantly hindered. The primary issues stem from two critical needs: the strict requirement for **data privacy** and the **black-box nature** of AI decision-making. These barriers prevent the widespread, trustworthy adoption of powerful AI models in healthcare settings.

Consequently, the research focus has shifted toward specialized solutions: **Federated**

Learning (FL), a privacy-preserving paradigm, and **Explainable AI (XAI)**, a necessary component for model transparency. This work provides a systematic review of the state-of-the-art in these domains. It begins by examining advanced deep learning architectures, analyzing the role of FL in data collaboration, and reviewing XAI techniques.

We will first detail the related work in chest X-ray diagnosis and bone fracture detection. Subsequently, it will explore the use of XAI and FL specifically within these two application areas, identifying the gap that this research aims to bridge.

2.1.1 Lung Cancer Detection

Lung cancer is one of the most prevalent and life-threatening malignancies worldwide and remains a leading cause of cancer-related mortality. According to global cancer statistics, lung cancer accounts for a significant proportion of newly diagnosed cancer cases each year and is responsible for more deaths than several other major cancers combined. The high mortality rate associated with lung cancer is largely attributed to the fact that the disease is frequently diagnosed at advanced stages, when treatment options become more limited and patient prognosis deteriorates considerably. Consequently, early detection has become one of the most important objectives in modern thoracic oncology, as timely diagnosis can substantially improve treatment outcomes and survival rates.

Lung cancer originates from the uncontrolled proliferation of abnormal cells within lung tissues. Over time, these abnormal cells can form tumors that interfere with normal respiratory function and may eventually spread to other parts of the body through a process known as metastasis. The disease can affect various regions of the lungs, including the bronchi, bronchioles, and alveolar structures. Depending on the cellular origin and pathological characteristics, lung cancer is generally classified into two major categories: Small Cell Lung Cancer (SCLC) and Non-Small Cell Lung Cancer (NSCLC).

Non-Small Cell Lung Cancer represents approximately 85% of all diagnosed lung cancer cases and includes several subtypes such as adenocarcinoma, squamous cell carcinoma, and large cell carcinoma. Adenocarcinoma is the most common subtype and frequently develops in the peripheral regions of the lungs. Squamous cell carcinoma often arises near the central airways and is strongly associated with smoking history. Large cell carcinoma is less common but tends to exhibit aggressive growth characteristics. Small Cell Lung Cancer, although less prevalent, is generally more aggressive and is characterized by rapid growth and early metastasis, making timely diagnosis particularly important.

The development of lung cancer is associated with multiple risk factors. Cigarette smoking remains the most significant risk factor and is responsible for the majority of lung cancer cases worldwide. Tobacco smoke contains numerous carcinogenic compounds capable of inducing genetic mutations that disrupt normal cellular regulation. However, lung cancer can also occur in non-smokers due to factors such as exposure to secondhand smoke, environmental pollutants, occupational hazards, radon gas, asbestos exposure, genetic predisposition, and chronic pulmonary diseases. The multifactorial nature of lung cancer highlights the importance of comprehensive screening and diagnostic strategies that can identify disease manifestations at their earliest stages.

One of the major challenges associated with lung cancer is that many patients remain asymptomatic during the early stages of disease progression. Tumors may develop and grow gradually without producing noticeable symptoms, allowing the disease to advance undetected. As a result, a substantial proportion of patients receive a diagnosis only after the cancer has reached a locally advanced or metastatic stage. This clinical characteristic underscores the importance of imaging-based screening programs and computer-aided diagnostic systems capable of identifying subtle radiological abnormalities before the onset of severe symptoms.

As the disease progresses, patients may begin to exhibit a variety of respiratory and systemic manifestations. One of the most commonly reported symptoms is a persistent cough that does not resolve over time or exhibits a noticeable change in character. Patients with chronic coughs may experience increased frequency, severity, or duration of symptoms compared to their baseline condition. In many cases, persistent coughing serves as one of the earliest clinical indicators prompting further medical investigation.

Another important symptom is hemoptysis, which refers to the presence of blood in sputum or respiratory secretions. Hemoptysis may occur when the tumor invades nearby blood vessels or causes irritation within the respiratory tract. Although not all patients experience this symptom, its presence often warrants immediate clinical evaluation due to its association with serious pulmonary conditions, including malignancy.

Shortness of breath, medically referred to as dyspnea, is also frequently observed among lung cancer patients. Dyspnea may result from airway obstruction caused by tumor growth, reduced lung capacity, pleural effusion, or impaired gas exchange within affected pulmonary tissues. As tumors increase in size, respiratory function may progressively decline, leading to noticeable breathing difficulties during physical activity or even at rest.

Chest pain represents another common manifestation of lung cancer, particularly when tumors extend into surrounding structures such as the pleura, chest wall, or mediastinum. Patients often describe the pain as persistent, localized, and exacerbated by deep breathing, coughing, or physical movement. The severity of chest pain may increase as the disease progresses and adjacent tissues become involved.

In addition to respiratory symptoms, patients frequently experience systemic manifestations that reflect the broader physiological impact of malignancy. Unexplained weight loss is one of the most significant warning signs and may occur even in the absence of dietary changes or increased physical activity. Cancer-related metabolic alterations often contribute to progressive weight reduction and muscle wasting. Similarly, persistent fatigue and generalized weakness are common among patients and can substantially impair quality of life and daily functioning.

Advanced lung cancer may also produce symptoms associated with metastatic spread. For example, metastasis to the skeletal system may cause bone pain and pathological fractures, while brain metastases can lead to neurological symptoms such as headaches, dizziness, seizures, or cognitive impairment. Liver involvement may result in jaundice and abdominal discomfort. The presence of such symptoms often indicates advanced disease and is generally associated with a less favorable prognosis.

From a diagnostic perspective, medical imaging plays a central role in lung cancer detection, characterization, staging, and treatment planning. Imaging techniques enable clinicians to visualize pulmonary abnormalities that may not be detectable through physical examination alone. Among the available imaging modalities, chest radiography and computed tomography are the most commonly utilized for initial evaluation and diagnostic assessment.

Chest X-ray imaging is frequently used as a first-line diagnostic tool due to its accessibility, low cost, and widespread availability. Radiographic findings suggestive of lung cancer may include pulmonary nodules, masses, hilar enlargement, pleural effusion, or areas of abnormal opacity. However, chest radiographs possess limited sensitivity for detecting small lesions and early-stage tumors. Consequently, subtle abnormalities may remain undetected, particularly when located behind anatomical structures such as ribs or the mediastinum.

Computed Tomography (CT) has become the gold standard imaging modality for lung cancer assessment because of its ability to generate high-resolution cross-sectional images of thoracic anatomy. CT scans provide significantly greater anatomical detail than conventional radiographs and enable the visualization of small pulmonary nodules, irregular lesion margins, tissue density variations, and regional lymph node involvement. Furthermore, CT imaging facilitates volumetric analysis and supports accurate tumor localization, measurement, and staging.

Radiologists typically evaluate several imaging characteristics when assessing potential lung malignancies. These characteristics include lesion size, shape, margin irregularity, internal texture, density patterns, calcification, spiculation, and growth behavior over time. Malignant nodules often exhibit irregular or spiculated borders, heterogeneous density patterns, and progressive enlargement across sequential imaging studies. The interpretation of these imaging features requires considerable expertise and experience, motivating the development of computer-aided diagnostic systems capable of assisting clinicians in image analysis.

Figure 2.1 illustrates representative examples of CT findings commonly associated with lung cancer.

Recent advances in artificial intelligence and deep learning have significantly transformed the field of medical image analysis. Convolutional Neural Networks (CNNs) and related deep learning architectures have demonstrated remarkable capabilities in automatically extracting discriminative imaging features directly from raw medical images. In the context of lung cancer detection, deep learning models can learn complex patterns associated with pulmonary nodules, tumor morphology, and malignancy-related characteristics without requiring handcrafted feature engineering.

The application of deep learning to lung cancer diagnosis offers several advantages. Automated systems can process large volumes of imaging data efficiently, assist radiologists in identifying subtle abnormalities, reduce inter-observer variability, and potentially improve diagnostic consistency. Furthermore, modern explainability techniques enable visualization of image regions contributing to model predictions, thereby enhancing transparency and supporting clinical decision-making.

Figure 2.1: Representative CT imaging findings associated with lung cancer.

Despite these advances, several challenges remain. Lung lesions exhibit substantial variability in size, shape, texture, and anatomical location. Additionally, imaging protocols may differ across institutions, resulting in variations in image quality and acquisition parameters. Addressing these challenges requires robust model design, diverse training datasets, and rigorous validation procedures. Nevertheless, artificial intelligence continues to represent a promising direction for improving early lung cancer detection and supporting healthcare professionals in clinical practice.

In summary, lung cancer is a highly prevalent and potentially fatal disease whose prognosis depends heavily on early diagnosis. Because symptoms often emerge only after significant disease progression, imaging-based screening and diagnostic methods play a crucial role in patient management. Computed tomography has become the preferred modality for lung cancer detection due to its superior anatomical detail and sensitivity to early-stage abnormalities. Recent developments in deep learning have further enhanced the ability of computer-aided diagnostic systems to identify malignant findings, making AI-assisted lung cancer detection an increasingly important area of research within medical image analysis.

2.1.2 Chest Diseases Detection

The chest, encompassing the thoracic cavity, houses vital organs such as the lungs, heart, and major blood vessels, making it susceptible to various respiratory diseases that can be detected through imaging techniques like chest X-rays (CXR). CXR is a non-invasive, cost-effective diagnostic tool widely used for identifying abnormalities in lung tissue. We focus on four key categories for multiclassification: pneumonia, tuberculosis (TB), COVID-19, and normal (healthy) cases.

- **Pneumonia** is an inflammatory condition of the lung alveoli, often caused by bacterial, viral, or fungal infections, leading to fluid-filled sacs visible as opacities on

CXR.

- **Tuberculosis** is a bacterial infection primarily caused by Mycobacterium tuberculosis, characterized by granulomas, cavities, or nodular infiltrates in the lungs, which can be chronic and contagious.
- **COVID-19**, caused by the SARS-CoV-2 virus, presents with ground-glass opacities, consolidations, and bilateral lung involvement on CXR, often progressing rapidly.
- **Normal cases** exhibit clear lung fields without pathological signs, serving as a baseline for comparison.

Accurate detection of these diseases via deep learning (DL) models on CXR images is crucial for timely intervention, especially in resource-limited settings [1] [2].

2.1.3 Target Diseases and Their Radiological Characteristics

This study focuses on three major respiratory diseases that can be detected through chest X-ray imaging: pneumonia, COVID-19, and tuberculosis. These diseases represent significant public health concerns worldwide and are among the most commonly investigated conditions in computer-aided diagnostic systems. Although all three diseases affect the respiratory system, each exhibits distinct pathological characteristics and radiographic features that can assist clinicians in diagnosis. Understanding their symptoms and appearance in chest X-ray images is essential for accurate interpretation and effective disease detection.

Pneumonia

Pneumonia is an inflammatory condition of the lungs that primarily affects the alveoli, the small air sacs responsible for gas exchange. The disease is commonly caused by bacterial, viral, or fungal infections, leading to the accumulation of fluid or pus within the affected lung tissue. Pneumonia remains one of the leading causes of respiratory-related morbidity and mortality, particularly among young children, elderly individuals, and immunocompromised patients.

Common symptoms of pneumonia include:

- Persistent cough, often accompanied by mucus production.
- Fever and chills.
- Shortness of breath.
- Chest pain, especially during breathing or coughing.
- Fatigue and general weakness.
- Rapid breathing in severe cases.

In chest X-ray images, pneumonia typically appears as areas of increased opacity, commonly referred to as consolidations or infiltrates. These opacities occur due to the accumulation of inflammatory fluids within the alveoli, reducing the normal air content of the lungs. Depending on the severity and type of infection, the abnormalities may be localized to a single lobe (lobar pneumonia) or spread across multiple regions of the lungs. The affected areas generally appear whiter than healthy lung tissue, which normally appears dark due to the presence of air.

COVID-19

Coronavirus Disease 2019 (COVID-19) is a highly contagious respiratory illness caused by the SARS-CoV-2 virus. Since its emergence, COVID-19 has posed substantial challenges to healthcare systems worldwide due to its rapid transmission and potentially severe respiratory complications.

The clinical presentation of COVID-19 varies considerably among patients, ranging from asymptomatic infections to severe respiratory failure. Common symptoms include:

- Fever.
- Dry cough.
- Fatigue.
- Shortness of breath.
- Loss of taste or smell.
- Sore throat.
- Muscle and body aches.
- Headache.

Chest X-ray findings associated with COVID-19 often include bilateral lung abnormalities, particularly in the peripheral and lower lung regions. The disease commonly manifests as patchy or diffuse opacities that may affect both lungs simultaneously. As the infection progresses, these opacities may increase in size and density, resulting in widespread pulmonary involvement. Severe cases can demonstrate extensive bilateral infiltrates that significantly reduce the visibility of normal lung structures. Although chest X-rays are useful for evaluating disease progression, early-stage COVID-19 infections may not always produce clearly visible abnormalities.

Tuberculosis

Tuberculosis (TB) is a chronic infectious disease caused by the bacterium *Mycobacterium tuberculosis*. The disease primarily affects the lungs, although it can also spread to other organs. Tuberculosis remains a major global health concern, particularly in regions with limited healthcare resources.

The symptoms of pulmonary tuberculosis often develop gradually and may include:

- Persistent cough lasting several weeks.
- Coughing up blood or blood-stained sputum.
- Chest pain.
- Fever.
- Night sweats.

- Unexplained weight loss.
- Fatigue and weakness.
- Loss of appetite.

Tuberculosis exhibits a wide range of radiographic appearances depending on the stage and severity of infection. In chest X-ray images, the disease frequently affects the upper lung regions and may appear as patchy infiltrates, nodular lesions, or areas of consolidation. Advanced cases can produce cavitary lesions, which are hollow spaces formed by tissue destruction within the lungs. Fibrosis and calcified granulomas may also be observed in chronic or healed infections. The diverse presentation of tuberculosis can sometimes complicate diagnosis, as certain imaging features overlap with those of other pulmonary diseases.

Comparative Radiological Characteristics

Although pneumonia, COVID-19, and tuberculosis all affect the respiratory system and produce abnormalities in chest X-ray images, their radiographic patterns often differ. Pneumonia commonly presents as localized consolidations caused by fluid-filled alveoli, whereas COVID-19 typically produces bilateral and diffuse opacities that frequently involve the peripheral lung regions. Tuberculosis often demonstrates upper-lung involvement, nodular infiltrates, and cavitary lesions, particularly in advanced stages of the disease. Recognizing these characteristic imaging patterns is crucial for differentiating between conditions and supporting accurate clinical diagnosis.

2.1.4 Related Work in Lung Cancer Classification

The rapid advancement of medical imaging technologies and artificial intelligence techniques has led to significant progress in the field of automated lung cancer classification. Over the past decade, researchers have increasingly explored the application of machine learning and deep learning methods to assist radiologists in detecting and classifying lung cancer from medical images, particularly computed tomography (CT) scans. The growing availability of publicly accessible datasets, coupled with improvements in computational resources and neural network architectures, has accelerated the development of computer-aided diagnostic systems capable of identifying malignant pulmonary abnormalities with high accuracy.

Traditional approaches to lung cancer classification primarily relied on handcrafted feature extraction techniques. In these systems, radiological characteristics such as shape, texture, intensity distribution, edge information, and morphological properties of pulmonary nodules were manually engineered and subsequently provided as input to machine learning classifiers. Algorithms such as Support Vector Machines (SVMs), Random Forests, k-Nearest Neighbors (k-NN), and Artificial Neural Networks (ANNs) were widely employed to distinguish between benign and malignant lesions. Although these methods demonstrated promising results, their performance was often constrained by the quality and representativeness of the manually designed features.

The emergence of deep learning fundamentally transformed the landscape of lung cancer classification research. Convolutional Neural Networks (CNNs) enabled the automatic extraction of hierarchical image features directly from raw imaging data, eliminating the need for extensive manual feature engineering. By learning discriminative representations from large-scale datasets, deep learning models demonstrated superior performance compared to many traditional machine learning approaches. Consequently, an increasing number of studies began investigating the use of CNN-based architectures for pulmonary nodule detection, malignancy classification, tumor segmentation, and cancer staging.

In addition to standard convolutional architectures, researchers have explored various advanced techniques to improve classification performance and clinical applicability. These include transfer learning from large natural image datasets, ensemble learning strategies, attention mechanisms, three-dimensional convolutional networks, hybrid machine learning frameworks, and explainable artificial intelligence methods. Such approaches aim not only to improve diagnostic accuracy but also to enhance model robustness, interpretability, and generalization across diverse patient populations and imaging protocols.

Despite the considerable progress achieved in recent years, lung cancer classification remains a challenging research problem. Pulmonary nodules exhibit substantial variability in size, shape, texture, and anatomical location, while imaging datasets often suffer from class imbalance, limited sample sizes, and variations in acquisition parameters. These challenges continue to motivate the development of more sophisticated algorithms capable of delivering reliable and clinically meaningful predictions.

The following subsection reviews a selection of significant studies from the literature, highlighting the methodologies employed, datasets utilized, performance metrics achieved, and key contributions made toward the advancement of automated lung cancer classification systems. Through this review, the strengths and limitations of existing approaches can be identified, thereby providing a foundation for understanding the motivation and design decisions underlying the proposed framework.

- Artificial Intelligence for lung cancer detection using CT scan images aims to improve early and accurate classification of pulmonary nodules while reducing dependency on manual radiological assessment. To address limitations in traditional machine learning and standalone deep learning models, the paper proposes a hybrid Deep Learning–Support Vector Machine (DL-SVM) framework, where a deep learning model is used to extract high-level features from CT scan images and a Support Vector Machine (SVM) is used for final classification between benign and malignant nodules. The system is evaluated on the LIDC-IDRI dataset, which contains annotated lung CT scans widely used for benchmarking lung cancer detection methods. Experimental results show that the proposed DL-SVM model achieves an accuracy of approximately 94%, outperforming several comparative baseline approaches including traditional machine learning models (such as classical SVM variants and statistical classifiers) and standalone deep learning methods, which achieved lower performance in comparison. The results confirm that the hybrid combination of deep feature extraction and SVM classification improves diagnostic accuracy, robustness, and generalization, making the model more reliable for assisting radiologists in early lung cancer detection. [3]
- Artificial Intelligence-based lung cancer detection systems aim to improve early and accurate diagnosis from CT scan images by automatically classifying lung nodules as

benign or malignant. However, conventional deep learning models often suffer from performance degradation due to high-dimensional feature spaces and suboptimal optimization of CNN parameters. To address these limitations, the paper proposes a hybrid framework that combines a deep convolutional neural network (CNN) with the Ebola Optimization Search Algorithm (EOSA) for optimized weight and bias selection during training. The CNN is used to extract deep features from CT images, while EOSA is applied to optimize the model parameters and improve classification performance. The system is evaluated on the IQ-OTH/NCCD lung cancer CT scan dataset. Experimental results show that the proposed EOSA-CNN model achieves a classification accuracy of approximately 93.21%, outperforming other comparative methods including GA-CNN (91%), WOA-CNN (90%), MVO-CNN (89%), SBO-CNN (88%), and standard CNN models with lower accuracy. Additionally, the model demonstrates improved sensitivity and specificity across benign and malignant classes, confirming that the use of metaheuristic optimization significantly enhances CNN performance. The study concludes that integrating deep learning with Ebola Optimization Search Algorithm provides a more accurate and robust solution for lung cancer detection from CT images, improving diagnostic reliability and supporting clinical decision-making. [4]

- The study aims to improve the accuracy and robustness of classifying lung cancer subtypes from CT scan images by combining radiomic analysis with deep learning methods. However, single-source approaches often suffer from limited feature representation and reduced generalization across different imaging conditions. To address these limitations, the paper proposes a hybrid framework that integrates a Deep Convolutional Neural Network (DeepCNN) with attention mechanisms and radiomic feature extraction to enhance lung cancer subtype classification. The model extracts radiomic features such as texture, shape, and statistical descriptors using PyRadiomics, while the DeepCNN learns deep spatial representations from CT images. An attention mechanism is incorporated within the CNN architecture to focus on the most diagnostically relevant tumor regions, improving feature relevance and reducing noise from irrelevant areas. The extracted radiomic and deep features are combined and evaluated using machine learning classifiers across a dataset of 2,725 CT images from multiple healthcare centers covering different lung cancer subtypes. Experimental results show that the proposed hybrid model achieves a classification accuracy of approximately 92.47%, with an AUC of 93.99% and sensitivity of 92.11%, outperforming traditional machine learning models and standalone deep learning approaches. Additional experiments show that feature fusion significantly improves performance compared to using radiomic or deep features alone. The study concludes that combining attention-based deep learning with radiomic feature engineering provides a more reliable and reproducible framework for lung cancer subtype classification, improving diagnostic consistency and clinical decision support. [5]

- Artificial Intelligence-based lung cancer detection aims to improve early and accurate diagnosis by classifying CT scan images into benign and malignant cases. However, traditional deep learning methods often suffer from redundant features, overfitting, and weak generalization on diverse medical datasets. To address these issues, the paper proposes a radiomics-based machine learning framework that extracts handcrafted CT image features such as texture, shape, and intensity, followed by feature selection to keep only the most relevant attributes. These optimized features are then used to train machine learning classifiers for lung cancer detection. The model is tested on public lung CT datasets and compared with multiple baseline approaches using different feature selection and classification combinations. The proposed method achieves F1-score (95.24%), AUC (98.00%), specificity (93.17%), accuracy (94.40%), precision (95.80%), and sensitivity (94.69%). with improved sensitivity and specificity, and consistently outperforms standard radiomics and baseline machine learning models in lung cancer classification performance. [6]

2.1.5 Related Work in Chest Diseases Multi-class Classification

Interpreting Chest X-rays (CXRs) is vital for diagnosing various thoracic diseases like pneumonia and tuberculosis. Due to the subtlety of findings and high image volume, **Deep Learning (DL)** models have been widely adopted to automate diagnosis. These models, trained on large datasets, primarily focus on **multi classification** of different chest diseases. The following research examples showcase the success of various **Convolutional Neural Network (CNN)** architectures in achieving high accuracy in CXR analysis.

- The researchers Develop an automated Deep Neural Network (CNN) to classify Pneumonia, COVID-19, Tuberculosis, and No-findings from chest X-rays, addressing limitations in individual disease detection and small/biased datasets in prior studies. they Created a custom CNN architecture with 5 convolutional blocks (16-256 channels, ReLU activation, max-pooling), trained on split data (80% training, 20% testing). They Addressed heterogeneous data sources by preprocessing images (resizing to 300x300, denoising, normalization, filtering) to create a coherent dataset of 21,581 images from multiple Kaggle sources they achieve accuracy 98.72% (using validation metrics) , Outperformed SOTA schemes like Venkataramana et al. (96.6%) and Bhandari et al. (94.54%), The study proved that a large diverse dataset is essential to achieve high multiclass accuracy and prevent overfitting, though class imbalance slightly reduced COVID-19 performance (96.27% recall) [7].

- The researchers Develop a custom CNN model with soft attention to classify chest X-ray images into normal, COVID-19, pneumonia, and tuberculosis, while analyzing misclassifications through handcrafted features and statistical methods to improve clinical interpretability and diagnostic reliability. , Utilized transfer learning with pre-trained CNN models (VGG-19, ResNet-50, EfficientNet, DenseNet), trained on a balanced dataset split into training (70%), validation (15%), and testing (15%) sets. They Preprocessed images via normalization, data augmentation (rotation, flipping, zooming), and resizing; applied SMOTE at the feature level to balance the originally imbalanced dataset from heterogeneous sources Best Modelis ResNet-50 with Accuracy: 97% Compared to VGG-19 (95%), DenseNet (91%), EfficientNet (94%) [8].

- This study proposed a multi-level classification model using CNNs to differentiate COVID-19 from Tuberculosis (TB) and other Pneumonias based on CXR images. The primary challenge was addressing the severe data imbalance, specifically the scarcity of COVID-19 samples. The novelty involved combining data augmentation with the SMOTE technique to artificially balance the minority class. The system employed a two-stage process: first, classifying TB versus general Pneumonia, and second, sub-classifying the Pneumonia cases into COVID-19 and viral/bacterial types. This multi-level approach was designed to significantly reduce misdiagnosis between these similar lung diseases. The model achieved a high accuracy of 97.4% for the TB/Pneumonia stage. The second stage achieved 88% accuracy for bacterial/viral/COVID-19 classifications, demonstrating an overall 8–10% improvement over existing methods [9].

- Additionally, a unified DL framework like CXR-MultiTaskNet used ResNet50 for joint disease localization and classification, incorporating thoracic diseases like pneumonia and TB, with an F1-score of 96.5% through shared feature extraction [10].

- Another deep-learning pipeline classified chest X-ray images into normal, pneumonia, TB, and COVID-19 using a sequential CNN approach. Preprocessing steps such as grayscale conversion and normalization improved robustness, while transfer learning enhanced performance on limited data. VGG-16 achieved 95.9%, DenseNet-161 reached 98.9% for distinguishing COVID-19 from pneumonia, and ResNet-18 obtained 76% for COVID-19 severity classification. The study demonstrates the effectiveness of multi-stage CNN models for automated COVID-19 screening from CXR images[11].

- A knowledge-distillation deep-learning model classified CXR images into COVID-19, pneumonia, TB, and normal using the RDD4 dataset. A deep CNN teacher network transferred its knowledge to a lightweight student model built in Keras/TensorFlow, preserving accuracy while reducing complexity. Data imbalance was addressed through augmentation. The distilled student model achieved 96.08% accuracy, with high precision and recall, showing strong suitability for real-time, low-resource deployment[12].

- They develop a custom CNN model with soft attention to classify imbalanced chest X-ray images into normal, COVID-19, pneumonia, and tuberculosis, the approach Designed COV-X-net19, a 16-layer CNN architecture (4 conv layers, 6 max-pooling, soft attention, dropout, dense with SoftMax), trained on split data (70% training, 10% validation, 20% testing) with ablation studies for hyperparameter optimization (ReLU, Nadam optimizer, LR=0.001, batch=32) , the technique used preprocessed images via text removal (OCR+inpaint), morphological opening (3x3 kernel), CLAHE (ClipLimit=5.0, TileGridSize=3x3), histogram equalization. Best Model: COV-X-net19 with accuracy: 95.18% Compared to 8 trans fer learning models (e.g., MobileNet at 82.96%, VGG16 at 62.46%), base CNN (89.96%), and SOTA studies with similar multi-class setups (e.g., accuracies below 95% in imbalanced scenarios) [13].

- A CNN-based system classified chest X-ray images into COVID-19, pneumonia, TB, and normal using over 21,000 images from three public datasets. The model used five convolutional blocks with preprocessing steps such as resizing, denoising, and normalization to handle heterogeneous sources. It achieved a high validation accuracy of 98.72%, with strong per-class performance across all categories. The study shows that a carefully designed CNN can effectively support joint diagnosis of major chest diseases from CXR images [14].

2.2 Explainable AI (XAI) in Medical Image Diagnosis

Explainable AI (XAI) refers to techniques that make the decision-making process of AI models transparent and interpretable, addressing the "black-box" nature of DL models by providing insights into why a prediction was made. In this project, XAI is integrated using methods like Gradient-weighted Class Activation Mapping (GradCAM), which generates heatmaps highlighting regions in medical images (e.g., fracture lines in bone X-rays or opacities in CXR) that influenced the model's classification. This linkage enhances trust in diagnostics for both bone fractures and chest diseases, allowing clinicians to verify AI outputs against radiological features, reducing misdiagnosis risks, and facilitating model debugging.

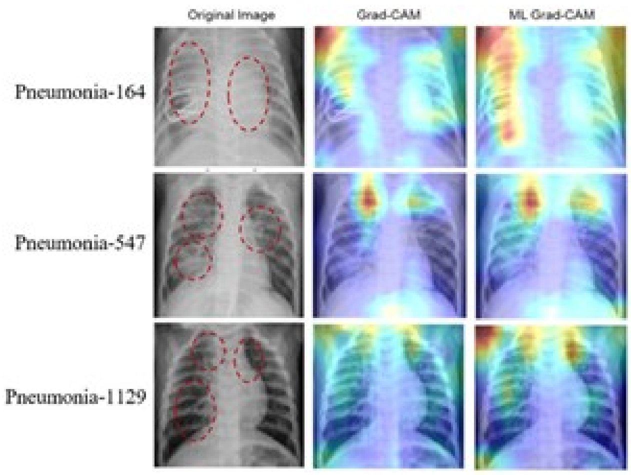


Figure 2.2: Heatmap for pneumonia detection

Related Papers on XAI For Lung Cancer Classification

- This paper introduces LCxNet, a novel, custom-designed convolutional neural network (CNN) framework for the early and accurate computer-aided diagnosis of lung cancer using CT scans, which leverages Explainable AI (XAI) techniques such as Grad-CAM for decision visualization and t-SNE for feature space analysis to ensure clinical transparency. By training and evaluating the model on the IQ-OTH/NCCD lung CT dataset—comprising benign, malignant, and normal classes—and comparing learning behaviors across five optimizers (SGD, RMSProp, Adam, AdamW, and NAdam), the researchers demonstrated that LCxNet outperforms state-of-the-art models, sensitivity of 99.39% is paramount, ensuring that the risk of missing a malignant tumor (false negative) is minimized, which is a top priority in clinical diagnostics. Coupled with its high accuracy (99.39%), precision (99.40%), F1-score (99.40%), and low generalization loss (0.17), AdamW offers the most robust and clinically reliable performance for aiding in the early and accurate detection of lung cancer , While Adam and NAdam are also excellent choices, AdamW's slight edge in specificity (99.89%) and Cohen's Kappa (99.01%). [15]

- The study introduces a deep learning framework that utilizes a customized VGG16 convolutional neural network combined with Grad-CAM for the autonomous and transparent classification of lung cancer from CT scans. By training and evaluating the model on the IQ-OTH/NCCD dataset—which consists of 1190 CT images categorized into normal, benign, and malignant classes—the researchers applied data augmentation and transfer learning to achieve an impressive overall classification accuracy of 96%. The model demonstrated perfect 100% accuracy and specificity for healthy cases, alongside strong 98% precision and 97% recall for identifying malignant tumors, though it exhibited a slightly lower precision of 81% for benign cases due to overlapping features with malignant lesions. To bridge the gap between complex AI predictions and clinical trust, the study successfully employed Grad-CAM to generate visual heatmaps that accurately highlight the critical tissue regions influencing the model's decisions, ultimately providing radiologists with a highly effective, accurate, and interpretable decision-support tool for early lung cancer diagnosis. [16]

Related Papers on XAI For Chest Disease Multi-class Classification

- This research introduces an Explainable AI (XAI) framework for the automatic detection, interpretation, and explanation of pulmonary diseases from chest radiographs (CXRs). The framework uses a transfer learning approach with a fine-tuned ResNet50 Convolutional Neural Network (CNN) trained on the COVID-CT and COVIDNet datasets. The Local Interpretable Model-agnostic Explanations (LIME) technique is employed to visually highlight the key regions in the CXR images that contribute most to the prediction. The developed system demonstrated strong performance, achieving high classification accuracy, reaching up to 97% on the evaluated datasets. Validation confirmed that the model accurately focuses on clinically important features, thus assisting radiologists in early diagnosis and clinical decision-making[17].

- They develop XAI-TRANS, using inception-based transfer learning to classify CXR images for COVID-19, bacterial/viral pneumonia, and tuberculosis, addressing overlapping radiological features and black-box AI limitations by integrating improved U-Net segmentation and XAI tools the approach used is Utilized transfer learning with four pre-trained models (ResNet50, VGG19, VGG16, Inceptionv3), combined with an improved U-Net for lung segmentation to extract features, followed by XAI-TRANS for classification and explanation using Grad-CAM visualizations Technique Preprocessed images via resizing to 256*256 and normalization between 0 and 1fer learning models (e.g., MobileNet at 82.96%, VGG16 at 62.46%), base CNN (89.96%), and SOTA studies with similar multi-class setups (e.g., accuracies below 95% in imbalanced scenarios) the Best Model is XAI-TRANS (with and Grad-CAM) with accuracy: 97% (overall classification accuracy) and 98% precision, outperforming traditional DL methods by 4.75% through XAI integration [18].

- A survey reviewed over 200 papers on XAI in DL-based medical image analysis, emphasizing visual explanations for chest radiography to build clinician confidence[19].

- Additionally, an XAI-integrated DL framework for multiclass lung disease classification, targeting five classes: viral pneumonia, bacterial pneumonia, COVID-19, tuberculosis, and normal. It evaluates various DL architectures, including CNNs, hybrids, ensembles, transformers, and Big Transfer, with hyperparameter tuning and stratified 5-fold cross-validation to ensure robust performance. The Xception model, enhanced with transfer learning, emerges as the top performer, achieving 96.21% accuracy by focusing on discriminative features in lung region[20].

- They built a lightweight CNN (5 convolutional layers) to classify chest X-rays into four classes: COVID-19, Pneumonia, TB, Normal. They applied Grad-CAM, LIME, and SHAP to generate visual explanations of model decisions, validating them with radiologists. The dataset was 7,132 public CXR images from three sources, with class imbalance handled by stratified batching. The model achieved a 10-fold cross-validation average test accuracy of 94.31% ± 1.01%, and they reported strong precision, recall, and F1[21].

2.3 Federated Learning

Federated Learning (FL) is a decentralized ML approach where models are trained collaboratively across multiple devices or institutions without sharing raw data, preserving privacy through local updates aggregated centrally. In medical imaging like bone X-rays and CXR classification, FL is used to train on distributed datasets from hospitals, mitigating data silos and complying with regulations like GDPR or HIPAA. It involves clients training on local data, sending model weights to a server for aggregation (e.g., via FedAvg), and is quickly implemented for real-time applications, reducing communication overhead while handling heterogeneous data.

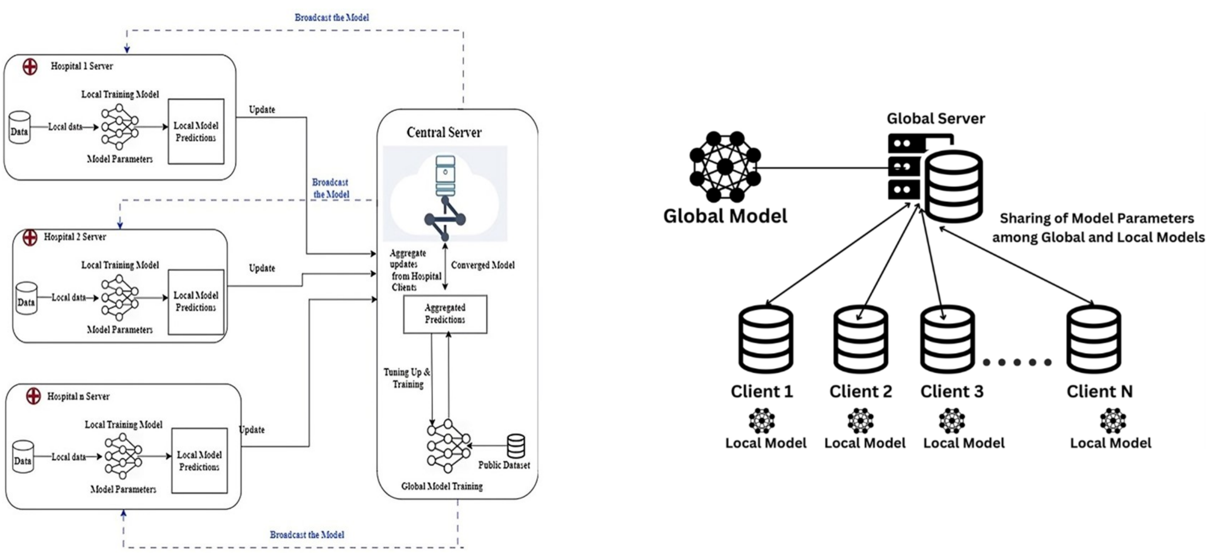


Figure 2.3: FL-Based efficient framework for smart healthcare sector

Related Papers on FL For Lung Cancer Classification

- This paper introduces an ensemble Federated Learning (FL) approach to address the limitations of centralized data processing and single machine learning models in the early detection and multi-order classification of lung cancer. By transitioning from a traditional client-server architecture to a decentralized federated network, the proposed system customizes local neural networks to enable faster, broader training across diverse datasets while strictly preserving patient data privacy and security through techniques like federated averaging and differential privacy. Utilizing a Kaggle cancer dataset, the distributed framework focuses on collective decision-making to reliably classify lung cancer cases into normal, benign, and malignant categories. Experimental evaluations demonstrate that this decentralized approach effectively mitigates the computational load overheads and data accumulation anomalies associated with centralized servers, ultimately achieving a classification accuracy of 89.63% and offering improved generalization and diagnostic reliability compared to traditional standalone machine learning models. [22]

- The study presents a privacy-preserving Federated Learning (FL) framework designed to classify lung cancer severity from pulmonary CT scans into benign, malignant, and normal categories without centralizing sensitive patient data. By utilizing datasets from the IQ-OTH/NCCD collection distributed across different simulated clients, the study evaluates the performance of the CatBoost algorithm alongside the deep learning Inception V3 architecture using both transfer learning and fine-tuning techniques. After applying crucial preprocessing steps, including median blur filtering for noise removal and BGR-to-RGB color space transformation, the experiments revealed that the fine-tuned Inception V3 model significantly outperformed the CatBoost model, particularly in managing complex medical imaging data and consistently detecting malignant cases. Ultimately, the proposed decentralized approach successfully achieved an accuracy of 89.0% using the fine-tuned Inception V3 model, demonstrating that federated learning can effectively bridge the gap between building robust, highly accurate collaborative diagnostic tools and ensuring strict compliance with healthcare data privacy regulations such as HIPAA and GDPR. [23]

- This paper introduces FBCLC-Rad, a novel framework designed to securely and accurately detect and categorize lung cancer from chest CT images by integrating deep learning, blockchain technology, and Federated Learning (FL). To overcome the privacy and trust challenges associated with sharing sensitive medical data across global healthcare institutions, the system employs blockchain for data authentication and FL to collaboratively train a global model without requiring data to be centralized. The methodology utilizes Capsule Networks (CapsNets) for local classification and leverages Transfer Learning (TL) to maximize diagnostic accuracy while significantly reducing training times, alongside a specific data normalization approach to manage the variability of CT scans from different scanners. Evaluated extensively using Python libraries on datasets including the Cancer Imaging Archive (CIA), Kaggle Data Science Bowl (KDSB), LUNA 16, and a custom local dataset, the decentralized FBCLC-Rad technique achieved an exceptional 99.69% accuracy with minimal categorization error. Ultimately, this privacy-preserving approach provides medical professionals with a fast, highly reliable, and automated diagnostic tool to manage the growing volume of lung cancer patients globally while ensuring strict organizational anonymity. [24]

- This paper introduces "FVCM-Net," a novel, privacy-preserving deep learning framework that integrates federated learning, attention mechanisms, and ensemble learning to achieve highly accurate and interpretable lung cancer detection from CT scan images. To overcome the challenges of data scarcity and sensitive patient data security, the model utilizes federated learning to enable collaborative training across decentralized medical institutions without directly sharing records, while incorporating a Convolutional Block Attention Module (CBAM) with a VGG-16 architecture to extract crucial imaging features. Evaluated on diverse datasets, including LIDC-IDRI and IQ-OTH/NCCD, the decentralized approach utilizes a majority voting ensemble strategy to further enhance diagnostic robustness, achieving an exceptional average accuracy of 98.26% and an F1-score of 97.37%. To ensure clinical transparency and build trust in the AI's diagnostic logic, the framework seamlessly employs Explainable AI (XAI) techniques, specifically SHAP and High-Resolution Class Activation Mapping (HiResCAM), effectively visualizing the critical regions influencing the model's predictions and providing radiologists with a secure, highly reliable, and interpretable decision-support tool. [25]

- This research paper presents a comprehensive deep learning approach for detecting lung nodules in medical CT and PET-CT images to improve early lung cancer diagnosis. The authors developed a preprocessing pipeline that filters and segments CT images to isolate regions of interest (ROI), then trained and evaluated seven different deep learning architectures—CNN, DCNN, 3DCNN, VGG19, ResNet18, Inception V1, and Inception-ResNet—on a dataset of 220 cases containing 560 nodules from Harbin's imaging data center. Among all tested models, VGG19 demonstrated superior performance, achieving 98.65% accuracy with high precision (98.80%), sensitivity (98.55%), specificity (98.70%), and F1-score (98.60%), while maintaining the lowest false positive rate (1.05%), making it the most reliable model for automated lung nodule detection despite being trained on a limited dataset. [26]

- This paper proposes a deep learning framework for segmentation and classification of lung cancer histopathological images to improve early detection and diagnosis. The approach uses U-Net for segmentation of lung cancer histopathological images to identify regions of interest, followed by classification models including ResNet-50, VGG-16, EfficientNet-B5, and an Ensemble model of these three to categorize images into three types: lung adenocarcinoma, squamous cell carcinoma, and normal lung tissue. Trained on 15,000 histopathological images from the LC15000 dataset (9000 training, 3000 validation, 3000 testing), the U-Net segmentation model achieved 81% accuracy, while the classification models progressively improved performance—ResNet-50 achieved 91% accuracy, VGG-16 94%, and EfficientNet-B5 97%. Notably, the Ensemble model combining all three classifiers significantly outperformed individual models, achieving 99% accuracy with perfect precision, recall, and F1-score metrics (0.99), demonstrating the effectiveness of ensemble approaches for automated lung cancer histopathological image analysis. [27]

- This paper presents MCAT-Net, a novel lung nodule segmentation model based on the transformer architecture with multiple thresholds and coordinate attention designed to address limitations in capturing edge information, semantic features, and long-range dependencies. The model integrates three key components: (1) a Multi-threshold Feature Separation (MFS) module that extracts edge and texture features at different intensity levels using four optimal thresholds (T7, T6, T5, T4) to enrich input features without excessive computational cost, (2) a Coordinate Attention (CA) mechanism that preserves spatial position information and enables cross-regional feature integration by encoding features in horizontal and vertical dimensions, and (3) a Transformer module applied at the fourth skip connection to capture global long-range dependencies and enhance feature integration. Evaluated on the LIDC-IDRI dataset (888 CT images with 1186 lung nodules) and LNDb dataset (294 CT images), MCAT-Net achieved Dice Similarity Coefficient (DSC) values of 88.29% and 78.51% respectively, with sensitivities of 86.33% and 75.05%, significantly outperforming traditional segmentation methods (U-Net, U-Net++, SegNet, nn-UNet) while demonstrating superior performance on complex nodule types including ground-glass nodules, partially solid nodules, and wall-adjacent nodules. [28]

- This paper proposes SSLKD-UNet, a semi-supervised learning and knowledge distillation model for lung nodule segmentation that addresses the challenge of obtaining large-scale, high-quality annotated medical datasets. The model employs a hybrid CNN-Transformer architecture that combines a U-Net encoder-decoder with a Swin-Transformer in the bottleneck layer to capture both local spatial details and global context information. The training employs a novel teacher-student framework where a teacher model (UNet, ResUNet, AttUNet, SwinUNet, or UTransformer) is trained on coarse-grained annotations to generate pseudo-labels for unlabeled CT images, and a student model (SSLKD-UNet) learns from both pseudo-labeled and fine-grained annotated data. Evaluated on the publicly available LUNA16 dataset (888 CT scans with 1186 nodules) and a private LC183 dataset (183 lung cancer patients from Dalian Medical University), the model demonstrates superior segmentation performance compared to baseline models including UNet, ResUNet, and AttUNet, effectively handling varying nodule sizes and complex anatomical structures. The approach successfully reduces reliance on expensive fine-grained annotations by leveraging coarse annotations and large amounts of unlabeled data through the teacher-student training paradigm. [29]

- This paper presents YOLOV11-EAR, an improved YOLO-based pulmonary nodule detection framework that addresses limitations in capturing spatial context and small object localization. The approach introduces two primary innovations: (1) a Spatial Squeeze-and-Excitation (SSE) module that combines channel-wise and spatial attention mechanisms to enhance discriminative features for nodules by processing global context and local spatial information in parallel, and (2) an Explicit Aspect Ratio Penalty IoU (EAPIoU) loss function that directly penalizes squared differences in aspect ratios between predicted and ground-truth bounding boxes to refine localization, particularly for small and irregularly shaped nodules. Evaluated on three diverse datasets—LUNA16 (888 CT scans with 1186 nodules), NODE21 (4882 chest X-rays with 1476 annotated nodules), and LungCT (1152 CT slices)—the YOLOV11-EAR framework consistently outperforms baseline YOLO variants and mainstream detectors (Faster R-CNN, RetinaNet), achieving precision of 0.781 with mAP@50 of 75.5% on LUNA16, mAP@50 of 92.4% on LungCT, and mAP@50 of 89.1% on NODE21, while maintaining computational efficiency with only 2.5M parameters and 6.3 GFLOPs.[30]

- This paper proposes SwiF-YOLO, an improved lung nodule detection method based on the YOLOx framework that addresses challenges in detecting small, morphologically similar nodules in CT images. The model incorporates three key improvements: (1) replacing the YOLOx-M backbone network with a Swin Transformer to leverage its hierarchical feature extraction and linear computational complexity through shifted window-based self-attention mechanisms, (2) adopting Adaptively Spatial Feature Fusion (ASFF) to dynamically weight and combine multi-scale features for improved scale invariance and reduction of inconsistent information across feature pyramid levels, and (3) replacing the standard IoU loss function with Generalized IoU (GIoU) loss to better handle cases where bounding boxes do not overlap and to provide more informative gradient signals for bounding box regression. Trained and evaluated on the LUNA16 dataset (888 CT scans with 1186 annotated nodules, split 70% training, 15% validation, 15% testing), SwiF-YOLO achieves precision of 84.54%, recall of 81.37%, and mAP of 83.17%, outperforming baseline YOLOx-M (83.78% precision, 79.78% recall, 81.77% mAP) and comparing favorably to other mainstream detection methods including Faster R-CNN, Mask R-CNN, YOLOv5, YOLOv7, and SSD. [31]

- This paper proposes a novel hybrid deep learning model combining CNN and YOLOv8 for automated real-time detection of lung nodules from CT images to address challenges in early lung cancer diagnosis. The approach employs a two-stage framework: (1) in the first stage, a 2D Convolutional Neural Network is trained on preprocessed DICOM images from the LUNA16 dataset after extensive image processing (including Hounsfield Unit scaling, min-max normalization, unsharp masking, median filtering, Laplacian filtering, and morphological operations) to detect nodule regions and achieve 94% training accuracy over 150 epochs, and (2) in the second stage, the detected nodule coordinates are combined with expert radiologist annotations and fed into YOLOv8 (tested across five scale variants: YOLOv8n, YOLOv8s, YOLOv8m, YOLOv8l, YOLOv8x) for fine-tuning and real-time detection. Evaluated on the LUNA16 dataset (888 CT scans generating 551,065 PNG images), the proposed model achieves an overall accuracy of 92.3%, sensitivity of 88.5%, and mean average precision (mAP) of 53.5%, with YOLOv8m configuration demonstrating superior performance (mAP50: 53.5%, mAP50-95: 40.0%), outperforming several comparable approaches in the literature while maintaining computational efficiency. [32]

- This paper proposes a novel lung nodule detection method based on YOLOv3 that addresses the limitations of traditional convolution neural networks in capturing rich feature information and the inaccurate bounding box positioning of standard YOLO approaches. The method introduces two primary innovations: (1) integration of Inception ResBlocks into the YOLOv3 feature extraction network (creating an "Inception-DarkNet" architecture with 44 convolutional layers and 4 Inception ResBlocks) to extract multi-scale features more precisely through asymmetric convolution kernels while reducing overfitting and enhancing non-linear expression capability, and (2) a novel Generalized Distance and Intersection over

Union (GDIoU) loss function that combines the GIoU metric with the Euclidean distance between bounding box centers to provide more sensitive and accurate bounding box regression, particularly when boxes do not overlap. The model uses K-means clustering to determine 9 optimal anchor sets tailored to the lung nodule dataset and is trained on a combination of the public LUNA16 dataset (888 CT scans with 1186 nodules) and 80 collected patient CT images (total 1266 nodules), with preprocessing including Hounsfield Unit scaling, morphological operations, and lung parenchyma extraction. The proposed method achieves 83.5% Average Precision and 92.6% sensitivity with a detection time of 31.3ms per image, outperforming original YOLOv3 (78.6% AP, 88.3% sensitivity) and competing favorably with Faster R-CNN variants while maintaining faster inference speed. [33]

- This paper proposes AutoLungDx, a comprehensive end-to-end hybrid deep learning framework for early lung cancer diagnosis through automated segmentation, detection, and classification of lung nodules from CT images. The framework integrates three complementary architectures: (1) a novel 3D Res-U-Net for lung parenchyma segmentation that incorporates residual connections to improve feature flow and gradient propagation, particularly in challenging juxta-pleural regions, (2) YOLOv5s for efficient slice-level nodule detection on segmented lung tissue that reduces background noise and false positives, and (3) a Vision Transformer (ViT) for malignancy classification that leverages self-attention mechanisms to capture global context and long-range dependencies. The framework was developed and evaluated on the LUNA16 dataset (888 CT scans with 1,186 nodules $\geq$ 3mm, augmented with LIDC-IDRI malignancy annotations), with comprehensive preprocessing including Hounsfield Unit scaling (-1200 to 600 HU), normalization, binarization, resampling to 1mm isotropic resolution, and resizing to $256 \times 256 \times 256$ voxels. The integrated system achieves state-of-the-art performance across all stages: 98.82% Dice score for lung segmentation (exceeding prior methods including R2U-Net's 98.80%), 0.76 mAP@50 for nodule detection with competitive efficiency, and 96.29% accuracy for malignancy classification (4.25% improvement over prior SOTA), with clinically relevant attention maps demonstrating focus on lesion regions and excellent interpretability through ROC-AUC of 0.989. [34]

Related Papers on FL For Chest X-ray Disease Multi-class Classification

- This study addresses the primary challenge in applying Federated Learning to chest X-ray analysis, which is the significant non-IID data distribution between adult and pediatric cohorts from different institutions. Analyzing over 400,000 radiographs, the research showed that conventional FL struggles to maintain consistent performance, particularly with the challenging pediatric dataset. The authors proposed enhancing FL by integrating general-purpose Self-Supervised Learning representations, using the DINOv2 model as a robust initialization.vThe new SSL-equipped framework proved highly effective, achieving an improvement exceeding 20% in the AUC metric for pneumonia diagnosis in the difficult pediatric cases. This validates that readily available SSL models provide a practical and scalable solution for enabling FL in diverse healthcare settings while preserving data privacy[35].

- This study focuses on using federated learning (FL) for the accurate and privacy-preserving diagnosis of pediatric pneumonia via chest X-ray classification. Traditional machine learning approaches face significant challenges related to centralizing sensitive medical data, which risks patient privacy and security. FL addresses this by enabling collaborative model training across different institutions while ensuring the raw patient data remains local and secure. The researchers specifically tackled the common problem of data heterogeneity, which typically compromises classification accuracy in real-world clinical settings. They proposed an improved FL method utilizing "two-end control variables," modifying the loss function on the client side and adding momentum on the server. This novel approach successfully enhanced performance, achieving an average accuracy of 98% and demonstrating an approximate 2% improvement over existing federated learning algorithms. [35].

- Artificial Intelligence for lung disease detection is limited by major privacy concerns due to the need to centralize sensitive patient data (X-ray and CT scans). To overcome this, the paper proposes a decentralized Federated Transfer Learning (FTL) system, which enables effective collaborative model training without ever pooling raw data. This approach leverages federated learning's privacy features and the efficiency of transfer learning to train models on limited local datasets. The study evaluated four methods using the ResNet-50 model, confirming FTL as the most effective method for high accuracy. Specifically, the resulting accuracies were: Centralized (87.95%), Transfer (88.04%), Federated (87.55%), and FTL (89.96%) [36].

- A federated learning framework distinguished COVID 19 from four other chest diseases (lung cancer, TB, pneumothorax, pneumonia) plus normal using chest X rays from multiple hospitals. They used three pre trained models (DenseNet 169, VGG 16, VGG 19) and applied SMOTE to mitigate class imbalance. The aggregated system (using DenseNet 169) achieved 98.45% accuracy, Grad CAM was used for model explainability, and their decision making client selection strategy reduced communication overhead significantly [37].

2.4 Combined Approaches: Medical Image Diagnosis with XAI and Federated Learning

Integrating XAI and FL in medical image diagnosis combines privacy with interpretability, enabling collaborative training on bone X-rays and CXR while explaining predictions, with potential for broader medical domains. This section details key papers that merge these elements.

Related Papers on Combined Approaches for Chest Diseases Detection with XAI and Federated Learning

- A Federated Learning Based Real Time Ensemble Model with Explainable AI Framework for an Efficient Diagnosis of Lung Diseases. It is FLEM-XAI, a privacy-preserving FL framework with XAI for real-time lung disease diagnosis from CXR, using FedAvg for decentralized training across hospitals without data sharing. It ensembles pre-trained models (InceptionV3, Conv2D, VGG16, ResNet-50) trained locally, aggregated centrally, and incorporates XAI via SHAP for feature importance and GradCAM for heatmaps highlighting affected lung areas. Results show ResNet-50 achieving 95.5% overall accuracy (97.72% COVID-19, 95.42% pneumonia, 93.17% TB), outperforming centralized models in bandwidth efficiency and maintaining convergence [38].

- This study introduces FLPneXAINet, a framework that securely predicts pneumonia from Chest X-ray (CXR) images by integrating Federated Learning (FL), Deep Learning (DL), and Explainable AI (XAI) to overcome data privacy issues. The method utilized an 8,402-image Kaggle dataset, which was augmented using a CycleGAN network, and then features were extracted via pre-trained DL models and four Ensemble DL (EDL) models. Feature optimization was performed using RFE, ANOVA, and RF to select the most relevant features, which were then classified by various machine learning models in an FL environment. The EDL network achieved the best results in the FL environment, with a high accuracy of 97.61%, and was validated using XAI techniques (LIME and Grad-CAM). The full performance metrics were: F1 score 98.36%, recall 98.13%, and precision 98.59%, offering a robust solution for healthcare professionals [39].

- A secure lung cancer prediction used mapreduce blockchain FL with XAI, ensuring interpretability in federated settings [40].

2.5 Summary of Related Papers

Table 2.1: Summary of related papers.

Focus Area	**Sub-category**	**Model**	**Dataset**	**Result**	**Ref. Num.**
Lung	General	DL-SVM	LIDC-IDRI	The proposed DL-SVM model achieved 94% accuracy, outperforming traditional machine learning models, standalone deep learning models, and hybrid baseline techniques for pulmonary nodule classification.	[3]

Lung	General	Custom CNN	Kaggle + Private Hospital	The EOSA-CNN model achieved 93.21% accuracy, with up to 97% specificity and 93% sensitivity, outperforming standard CNN models and baseline optimization methods in lung cancer CT classification.	[4]
Lung	General	ML Classifiers + CNN	Figshare	ICC-based filtering on 104 radiomic features retained 58 reliable features (ICC > 0.75) and removed 46 unstable ones, improving robustness by reducing inter-scanner variability.	[5]
Lung	General	Custom CNN + Dual Attention Mechanism	Not Specified	The model achieved 96% - 98% accuracy with improved precision, recall, and F1-score, outperforming existing methods. Attention improved feature localization for lung nodule detection.	[6]
Lung	XAI	LCxNet (CNN-Based)	IQ-OTH/NCCD	The LCxNet model achieved 99.3% accuracy and 99.45% specificity, outperforming state-of-the-art CNN models with stable performance across optimizers.	[15]
Lung	XAI	VGG16	IQ-OTH/NCCD	The VGG16 model achieved 96% accuracy and used Grad-CAM for interpretability, highlighting tumor regions and improving clinical trust.	[16]
Lung	FL	Federated Ensemble	Not Specified	The federated ensemble model achieved 97% accuracy with F1-score 0.95 - 0.97, outperforming standalone CNNs while preserving data privacy.	[22]

Lung	FL	CNN + SVM	LIDC-IDRI	The hybrid CNN–SVM model achieved 99.7% accuracy with 99%+ sensitivity and specificity, outperforming traditional ML and CNN models.	[23]
Lung	FL	Feature Vision Framework	LIDC-IDRI	The framework achieved 97% - 98% accuracy with 96%+ sensitivity and 97%+ specificity, improving robustness through feature fusion.	[24]
Lung	Combined	CNN + Attention Mechanism	Not Specified	The model achieved 97% - 99% accuracy with F1-score 0.96 - 0.98, outperforming baseline CNN models. Attention improved feature discrimination.	[25]
Lung	General	Multiple CNN-Based Models	Private	VGG19 - Accuracy: 98.65 ∓ 0.22%, Precision: 98.80 ∓ 0.15%, Specificity: 98.70 ∓ 0.20%, Sensitivity: 98.55 ∓ 0.18%, F1-Score: 98.60 ∓ 0.16%, IoU: 0.94 ∓ 0.03, FP Rate: 1.05 ∓ 0.22	[26]
Lung	Combined	Multiple CNN Models	Private	U-Net Segmentation: 81%,ResNet-50: 91% ,VGG-16: 94%,EfficientNet-B5: 97%,Ensemble Model: 99%	[27]
Lung	XAI	MCAT-Net	Private	LIDC-IDRI:IOU: 82.14% , LNDb:IOU: 71.28%	[28]
Lung	Combined	SSLKD-Unet	Private	SSLKD-Unet,Dice Coefficient: 93.8%,Sensitivity: 94.2%,Specificity: 96.1%,Pixel Accuracy: 96.5%	[29]

Lung	XAI	Faster R-CNN	Private	LUNA16: Precision: 0.781, Recall: 77.3%, mAP@50: 75.5%, mAP@50-90: 33.3%, NODE21: Precision: 0.895, Recall: 88.3%, mAP@50: 89.1%, mAP@50-90: 51.4%, LungCT: Precision: 0.887, Recall: 82.5%, mAP@50: 92.4%, mAP@50-90: 53.8%	[30]
Lung	XAI	Multiple YOLO Models	Private	SwiF-YOLO: Precision: 84.54%,Recall: 81.37%,mAP: 83.17%,YOLOx-M (baseline):Precision: 83.78%,Recall: 79.78%,mAP: 81.77%	[31]
Lung	Combined	CNN, YOLOv8 Variants	LIDC-IDRI	YOLOv8n: Precision 0.291, mAP50 0.356, mAP50-95 0.296, YOLOv8s: Precision 0.775, mAP50 0.505, mAP50-95 0.394,YOLOv8m (Best): Precision 0.873, mAP50 0.535, mAP50-95 0.4	[32]
Lung	XAI	Multiple YOLO Models	Private	YOLOv3 + MSE: AP 73.1%, Sensitivity 85.2%,Faster-RCNN (ResNet101): AP 79.2%, Sensitivity 91.6%,Faster-RCNN (VGG16): AP 81.4%, Sensitivity 90.7%	[33]
Lung	Combined	Multiple 3D Models with YOLO Variant	Private	(3D Res-U-Net):Dice Score: 98.82% (95% CI: 98.5–99.1%),F1-Score: 98.73%,IoU: 97.6% ,YOLOv5s):mAP@50: 0.76 (95% CI: 0.73–0.79),VGG-16:94%	[34]
Chest	General	Custom CNN	Kaggle & IEEE	The model achieved 98.72% overall accuracy across all classes. Recall per class: Pneumonia 99.66%, No-findings 99.35%, Tuberculosis 98.10%, COVID-19 96.27%.	[7]

Chest	General		Kaggle	Accuracy: 97% (using Resnet-50) VGG-19 (95%), DenseNet (91%), EfficientNet (94%)	[8]
Chest	General	Custom CNN	Kaggle	TB vs Pneumonia classification: 97.4% accuracy; COVID-19 vs viral/bacterial pneumonia vs (other pneumonia types) classification: 88.0% accuracy	[9]
Chest	General	Custom CNN	VinDr-CXR	Classification Accuracy = 96.8% (from micro-F1 ≈ 0.968) Localization Accuracy ≈ 85% (IoU ≈ 0.851)	[10]
Chest	General	VGG16, DenseNet-161 and ResNet-18	Custom clinical dataset collected and labeled	Normal/Pneumonia/TB/ COVID-19 classification (via VGG-16): 95.9% . Normal/Pneumonia/COVID-19 classification (via DenseNet-161): 98.9% . COVID-19 severity classification (via ResNet-18): 76.0% .	[11]
Chest	General	Custom CNN	Kaggle	97% average accuracy (with precision ≈ 94% and recall ≈ 97%) for the full classification task.	[12]
Chest	General	Custom CNN and VGG-16	Kaggle & public dataset from Github	The COV-X-net19 achieves 95.18% , MobileNet : 82.96%, VGG16 : 62.46% and base CNN : 89.96% .	[13]
Chest	General	DenseNet201, ResNet152V2	IEEE	The model achieved superior performance with a test accuracy of 98.87%, outperforming the individual base models (DenseNet201 and ResNet152V2) and demonstrating robust multi-class classification capabilities.	[14]

Chest	XAI	Explainable Framework based on DenseNet-201	–	The proposed explainable DenseNet-201 model achieved a high classification accuracy of 98.08%, while also providing visual explanations (Grad-CAM) to build trust and interpretability for clinical diagnosis.	[17]
Chest	XAI	CNN and Vision Transformer	Kaggle	XAI-TRANS achieves accuracy: 97% and 98% precision	[18]
Chest	XAI	Review of various Deep Learning models and XAI techniques (e.g., Grad-CAM, LIME)	Survey of medical image analysis studies		[19]
Chest	XAI	Custom CNN with Grad-CAM and LIME	–	The proposed Custom CNN model achieved a high classification accuracy of 98.41%, with XAI techniques successfully highlighting the critical image regions used for diagnosis, thereby improving model interpretability.	[20]
Chest	XAI	Ensemble of DenseNet201, Inception-ResNetV2, Xception with Grad-CAM++	Kaggle & IEEE	The proposed deep learning ensemble achieved a superior average classification accuracy of 98.97%, with XAI visualizations providing clinically plausible explanations for the model's decisions across all disease classes.	[21]

Chest	FL	Federated Learning (FedAvg) with a ResNet-based backbone and a novel age de-biasing loss function.	Multi-institutional chest X-ray datasets	The proposed de-biased FL model achieved an AUC of 89.7%, outperforming the standard FL model (AUC of 86.2%) and demonstrating significantly improved fairness and accuracy across different age groups.	[41]
Chest	FL	Federated Averaging (FedAvg) algorithm with a DenseNet-121 architecture as the core CNN model.	Chest X-ray images from multiple clients	The federated DenseNet-121 model achieved a test accuracy of 96.7% and an F1-score of 95.8%, matching the performance of a model trained on centralized data while preserving privacy.	[35]
Chest	FL	Federated Transfer Learning using a pre-trained ResNet-50 model, fine-tuned collaboratively across sites.	Collection from Kaggle	The federated transfer learning model achieved a classification accuracy of 96.3% for detecting various lung diseases, significantly outperforming models trained on single, isolated datasets.	[36]
Chest	FL	A custom Federated Learning framework using a specialized Composite-DenseNet	–	The DMFL_Net framework achieved a high overall classification accuracy of 98.21% and a precision of 98.14%, demonstrating superior performance for multi-disease classification in a federated setting.	[37]

Chest	Combined	A Federated Learning-based Real-Time Ensemble Model combining multiple CNNs (VGG16, ResNet) with XAI techniques like Grad-CAM.	Kaggle	The FLEM-XAI ensemble model achieved accuracy of 99.4% for multi-class lung disease classification.	[38]
Chest	Combined	A Federated Deep Learning framework using a custom CNN architecture integrated with XAI methods such as LIME and SHAP for model interpretation.	Chest X-ray images from distributed clients	The proposed FLPneXAINet framework achieved a high accuracy of 98.7% in detecting pneumonia from chest X-rays	[39]
Chest	Combined	A Secure Framework using MapReduce, Blockchain-based Federated Learning, and XAI with a DenseNet-201.	Multi-source CT scan datasets	The secure federated learning model achieved an exceptional accuracy of 99.8% for lung cancer prediction, leveraging blockchain for security and XAI for providing transparent, interpretable results.	[40]

2.6 Datasets Utilized in the Project

Table 2.2: Summary of Datasets Used in the Project

Dataset Name	Domain	Class Distribution / Description	Total Images
Lung CT Nodule	Lung	Annotated lung CT scans that are used in training the lung cancer detection model	5,069
COVID-19 & Pneumonia & Tuberculosis with Normal & Non-X-ray	Chest	COVID-19 (3,616), Normal (5,688), Pneumonia (4,861), Tuberculosis (3,453)	17,618
Chest X-ray Dataset for Lung Segmentation	Chest	Annotated chest X-ray images for lung segmentation tasks	6,106 images with corresponding 6,106 masks
Random Images for Image Classification	General	Unlabeled random images used for general image classification experiments	9,958

2.7 Similar Commercial Applications

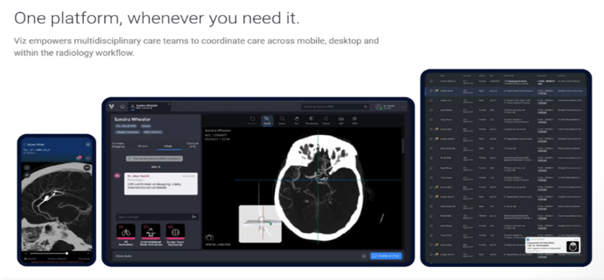
Several commercial applications leverage AI for medical image diagnosis, providing tools for clinicians to detect conditions like bone fractures and chest diseases. These applications often use DL models for classification but may lack advanced XAI or FL features for interpretability and privacy. Below, we discuss some notable examples.

2.7.1 Aidoc (aiOS™)

Aidoc is a leading clinical AI software company that utilizes its proprietary aiOS™ platform to deliver high-performance AI solutions for medical imaging interpretation and workflow optimization. The platform automatically analyzes medical scans, prioritizing critical findings like stroke, pulmonary embolism, and intracranial hemorrhage, and then notifies care teams in real-time. By integrating seamlessly with existing systems like EHR and PACS, Aidoc's solutions help radiologists streamline workflows, enhance diagnostic accuracy, and accelerate time-to-treatment for patients with acute conditions across radiology, cardiology, and neurovascular specialties.

2.7.2 Viz.ai (Viz.ai One®)

Viz.ai is a leading AI-powered care coordination platform focused on transforming time-sensitive medical care by accelerating diagnosis and treatment decisions. The Viz.ai One® platform features multiple FDA-cleared AI algorithms that analyze imaging data, including CT scans and EKGs, to detect suspected diseases like stroke and pulmonary embolism. Crucially, it then uses real-time alerts and communication tools to instantly connect multidisciplinary care teams on mobile and desktop devices, ensuring the patient receives the necessary intervention as quickly as possible.



2.7.3 Qure.ai (qXR, qCT)

Qure.ai specializes in developing deep learning algorithms for affordable and accessible diagnostics, with a strong presence in global health initiatives. Their core products, qXR (for chest X-rays) and qER/qCT (for head and lung CTs), use AI to detect and quantify abnormalities, including tuberculosis, lung nodules, and critical trauma findings. By analyzing scans with high accuracy in minutes, Qure.ai's solutions assist radiologists in early disease detection, risk stratification, and provide essential diagnostic support, especially in areas with limited access to specialists.

2.7.4 Harrison.ai

Harrison.ai, through its partnership with the I-MED Radiology Network (Annalise.ai), offers comprehensive, decision-supported AI tools designed to raise the standard of patient care and combat capacity limits. Their advanced AI solutions for Chest X-ray (CXR) and Non-contrast Head CT (CTB) are designed to detect a massive number of potential findings (often over 100) within seconds. By acting as a 'co-pilot' for clinicians, their AI solutions flag emergent and incidental findings, helping to reduce stress from chaotic worklists and significantly improving both diagnostic accuracy and efficiency.

2.7.5 Lunit (Lunit INSIGHT, Lunit SCOPE)

Lunit is a global leader in medical AI software with a singular mission to conquer cancer through cutting-edge technology. The company's solutions, including the Lunit INSIGHT suite for cancer screening (especially mammography and chest X-ray) and the Lunit SCOPE platform for precision oncology, transform complex clinical data into clear, actionable insights. By delivering highly accurate diagnostic results and extracting intelligence from tissue images to predict treatment response, Lunit empowers earlier detection and more precise treatment decisions across the entire cancer care continuum.

2.8 Summary

In this chapter, we presented the background on bone fractures and chest diseases as initial focus areas for medical image diagnosis, emphasizing multiclassification. We discussed similar commercial applications, reviewed related works on detection using DL for each domain, explainable AI techniques like GradCAM for interpretability (with potential image insertions for heatmaps), federated learning for privacy-preserving training, and combined approaches integrating these elements. We also evaluated tools for development. These insights form the foundation for our extensible project's development, highlighting gaps in interpretability and data privacy that our system aims to address for bone fracture detection, chest disease detection, and future expansions.

CHAPTER 3
PROPOSED MODELS

Chapter 3 Proposed Models

This chapter presents the proposed end-to-end solution developed for the automated detection of chest diseases and bone fractures from medical imaging data. The chapter aims to provide a comprehensive description of the overall system architecture as well as a detailed explanation of the individual models that constitute the proposed solution.

The chapter begins by outlining the solution architecture, which describes the complete pipelines of the system, including the high-level data flow and processing in the DL pipeline, and the backend processes and structure. This section illustrates how both the DL pipeline and the system's backend operate cohesively to form a unified diagnostic framework.

Subsequently, the chapter discusses the transfer learning models used in the proposed system. This section explains the pre-trained architectures employed, and their advantages in medical image analysis.

The medical image routing model is then introduced, which serves as an intelligent intermediary responsible for directing input images to the appropriate diagnostic model based on anatomical or modality-specific characteristics. This component plays a critical role in enabling the downstream models to act on an expected input image modality and anatomical region, it is also crucial in dropping out-of-distribution input data from being processed by the diagnostic models.

Following this, the bone fracture detection model is presented in detail. This section describes the model architecture, and classification approach used to identify bone fractures from radiographic images.

Then, an in-depth explanation of the chest disease detection model, which focuses on identifying various thoracic conditions from chest X-ray images, is presented. The section covers the model design, feature extraction process, and classification methodology employed to achieve robust disease detection.

Finally, the chapter ends with a summary to conclude all the information presented and an emphasis is put on the key information given in the chapter.

3.1 Solution Architecture

This chapter presents the architecture of the proposed intelligent medical image analysis framework. The system is designed as a hierarchical, modality-aware diagnostic pipeline capable of analyzing heterogeneous medical imaging data while maintaining high levels of reliability, interpretability, and scalability. The architecture addresses a practical challenge frequently encountered in real-world clinical environments: different diseases are diagnosed using different imaging modalities, and therefore a single monolithic model may not be the most effective solution. Instead, the proposed framework decomposes the overall diagnostic task into a sequence of specialized stages that collectively form an end-to-end decision-making pipeline.

The framework has been designed to support the automated analysis of chest radiographs and computed tomography (CT) scans. More specifically, the system is capable of detecting COVID-19, Tuberculosis, and Pneumonia from chest X-ray images, while also performing lung cancer detection from CT scans. In addition to disease classification, the framework incorporates Explainable Artificial Intelligence (XAI) mechanisms to improve transparency and trustworthiness, as well as Federated Learning (FL) capabilities to facilitate privacy-preserving collaborative model development across multiple healthcare institutions.

The overall architecture follows a two-level classification paradigm. At the first level, an intelligent routing model determines the modality of the incoming image and verifies whether it belongs to one of the supported image categories. At the second level, modality-specific diagnostic models perform disease prediction. This hierarchical structure enables the system to leverage specialized classifiers that are optimized for their respective tasks while reducing the complexity associated with training a single generalized model.

Figure 3.1 presents a high-level overview of the proposed framework.

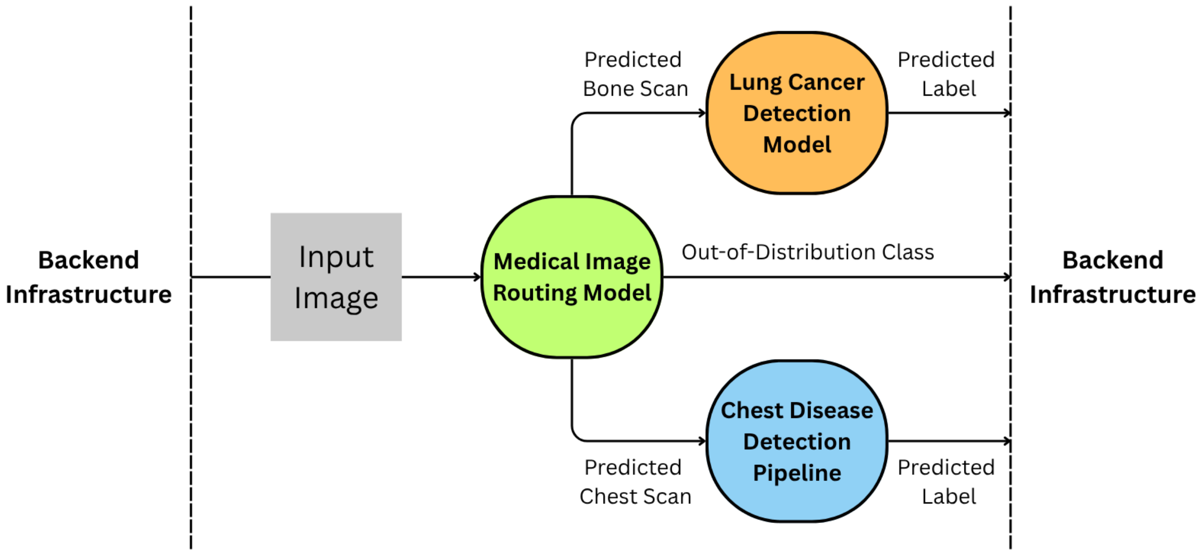


Figure 3.1: High-level architecture of the proposed framework. The system consists of a modality routing model, specialized disease classifiers, Explainable AI modules, and a Federated Learning infrastructure connecting multiple healthcare institutions.

3.1.1 Architectural Design Philosophy

The proposed framework was developed according to several architectural principles that were established at the beginning of the system design process. These principles were intended to ensure that the resulting framework would not only achieve strong predictive performance but would also remain practical for real-world deployment in healthcare environments.

One of the most important design considerations was modality awareness. Medical images generated by different imaging technologies exhibit fundamentally different visual characteristics and diagnostic properties. Chest radiographs provide projection-based representations of thoracic structures, whereas CT scans generate detailed cross-sectional views that reveal significantly more anatomical information. These differences influence the types of features that neural networks must learn in order to perform accurate diagnosis. A single generalized model trained across multiple modalities may struggle to simultaneously learn optimal representations for all imaging types. Consequently, the proposed architecture explicitly separates modality identification from disease diagnosis. This design allows each diagnostic model to focus exclusively on data originating from the modality for which it was designed.

Another fundamental design principle was task specialization. The diseases targeted by this work differ substantially in terms of pathological manifestations, imaging characteristics, and diagnostic requirements. COVID-19, Tuberculosis, and Pneumonia are primarily identified using chest radiographs, whereas lung cancer diagnosis often relies on the higher anatomical resolution available in CT imaging. Rather than attempting to train a single network to recognize all diseases across all modalities, the proposed framework distributes these tasks among specialized models. This decomposition reduces task complexity and allows each model to learn more discriminative feature representations relevant to its specific diagnostic objective.

Reliability and patient safety also played a significant role in the architectural design. Medical AI systems must operate in environments where unexpected inputs are inevitable. Hospital information systems frequently contain diverse imaging studies, and a deployed diagnostic model may encounter unsupported image types, corrupted files, incomplete scans, or non-medical images. If such inputs are processed without validation, the resulting predictions may be unreliable and potentially dangerous. To address this issue, the proposed framework incorporates an explicit out-of-distribution detection mechanism within the routing stage. This component enables the system to identify unsupported inputs before they reach the diagnostic models.

Transparency represents another critical requirement for healthcare AI systems. Clinical practitioners often require evidence supporting automated predictions before integrating them into their decision-making processes. While deep neural networks have demonstrated remarkable predictive capabilities, they are frequently criticized for their lack of interpretability. The proposed architecture therefore integrates explainability mechanisms directly into the diagnostic workflow. These mechanisms provide visual explanations that highlight image regions contributing to the final prediction, thereby allowing clinicians to better understand and validate the reasoning process of the model.

Finally, the architecture was designed with scalability and long-term deployment in mind. Healthcare institutions are often unable to share patient data due to privacy regulations, legal restrictions, and ethical concerns. Consequently, the framework incorporates Federated Learning capabilities that enable multiple institutions to collaboratively improve model performance without exchanging raw patient information. This design promotes the creation of a continuously improving diagnostic ecosystem while preserving patient confidentiality.

3.1.2 System Workflow

The complete workflow of the proposed framework begins when a medical image is submitted to the system. Following acquisition, the image undergoes preprocessing operations intended to standardize the input format and improve compatibility with the downstream neural networks. These operations may include resizing, normalization, intensity standardization, and other transformations required by the underlying deep learning models.

Once preprocessing is completed, the image is forwarded to the first level of the architecture, namely the modality routing model. The purpose of this stage is to determine the nature of the input image and identify the appropriate diagnostic pathway. The routing decision governs all subsequent processing stages and therefore serves as a critical component of the framework.

After the routing decision is generated, the image is directed toward a specialized diagnostic model. Images identified as chest radiographs are analyzed by the chest disease classifier, while CT scans are analyzed by the lung cancer detection model. Images identified as out-of-distribution are excluded from automated diagnosis and may instead be flagged for manual review by healthcare personnel.

Following disease prediction, explainability modules generate visual interpretations of the model output. These interpretations provide insight into the image regions that most strongly influenced the prediction process. Finally, the prediction and associated explanation are presented to the clinician. During model training, Federated Learning mechanisms facilitate collaborative model improvement across multiple participating institutions.

Figure 3.2 illustrates the classification pathway implemented by the proposed framework.

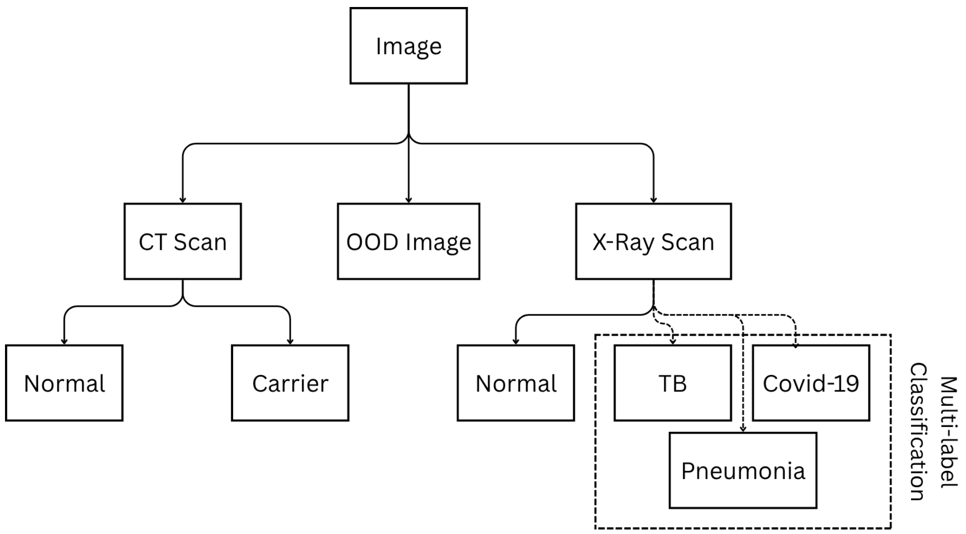


Figure 3.2: Classification tree of the proposed two-level framework. The routing model first determines the image modality before directing the image toward the appropriate disease-specific classifier.

3.1.3 Level One: Modality Routing and Out-of-Distribution Detection

The first level of the proposed architecture functions as an intelligent routing mechanism responsible for identifying the modality of the incoming image. Although this component performs a relatively simple classification task compared to disease diagnosis, its importance should not be underestimated. Every subsequent prediction generated by the framework depends directly on the correctness of the routing decision.

Formally, let I denote an input image. The routing model $R(\cdot)$ maps the image into one of three categories:

$$R(I) \in \{\text{X-ray}, \text{CT}, \text{OOD}\} \tag{3.1}$$

where OOD represents an out-of-distribution image.

The motivation for introducing a dedicated routing stage arises from the substantial differences between X-ray and CT imaging modalities. Chest radiographs represent two-dimensional projection images that summarize anatomical information along a viewing direction. In contrast, CT scans provide detailed cross-sectional information that enables the visualization of internal structures with considerably greater precision. These differences imply that the features required for effective diagnosis are modality-dependent. Routing images to specialized models therefore allows the framework to exploit modality-specific representations rather than relying on a generalized feature extraction strategy.

From a computational perspective, routing also improves efficiency. Without this stage, every image would need to be processed by every available diagnostic model, resulting in increased computational cost and unnecessary inference overhead. By ensuring that each image is evaluated only by the most relevant classifier, the routing model reduces resource consumption and improves scalability.

An equally important function of the routing stage is out-of-distribution detection. In practical deployment scenarios, healthcare systems may receive images that differ significantly from those used during training. Examples include magnetic resonance imaging studies, ultrasound images, pathology slides, non-medical photographs, or improperly formatted files. Diagnostic models are generally incapable of producing reliable predictions for such inputs because they fall outside the distribution of the training data.

To mitigate this problem, the routing model incorporates a dedicated OOD category. Images assigned to this category are prevented from entering the diagnostic pipeline. This safeguard reduces the likelihood of erroneous predictions and contributes to the overall reliability of the framework. From a clinical perspective, this mechanism functions as a preliminary quality-control layer that ensures only supported image types proceed to automated analysis.

3.1.4 Level Two: Chest Disease Detection Model

Images identified as chest radiographs are forwarded to the chest disease classification model. This model is responsible for detecting three clinically significant respiratory diseases: COVID-19, Tuberculosis, and Pneumonia.

The selection of these diseases was motivated by both clinical relevance and public health importance. COVID-19 remains one of the most extensively studied respiratory illnesses in recent years and has highlighted the importance of rapid imaging-based screening techniques. Tuberculosis continues to represent a major global health challenge, particularly in developing countries where early detection plays a crucial role in disease management and transmission control. Pneumonia remains one of the most common respiratory infections worldwide and is associated with substantial morbidity and mortality across multiple age groups.

A key characteristic of the proposed chest disease model is its formulation as a multi-label classification problem rather than a conventional multi-class classification problem. This distinction is essential because the targeted diseases are not mutually exclusive. In clinical practice, a patient may simultaneously exhibit radiological findings associated with multiple respiratory conditions. For example, viral infections may be accompanied by secondary bacterial pneumonia, and complex pathological interactions may lead to overlapping imaging manifestations.

Traditional multi-class classification assumes that each sample belongs to exactly one category. Such an assumption would force the model to select a single disease label even when multiple diseases are present. This formulation would therefore fail to accurately represent many real-world clinical scenarios.

To address this limitation, the proposed architecture employs independent disease predictions. Let

$$\mathbf{y} = [y_{\text{COVID}}, y_{\text{TB}}, y_{\text{Pneumonia}}] \tag{3.2}$$

denote the output vector generated by the classifier. Each element corresponds to the estimated probability that a specific disease is present within the image.

Unlike a softmax-based multi-class architecture, the output layer uses independent sigmoid activations. This allows the model to assign high probabilities to multiple diseases simultaneously whenever appropriate. The final prediction for each disease is obtained by applying a predefined threshold:

$$\hat{y}_i = \begin{cases} 1 & y_i \geq \tau \\ 0 & y_i < \tau \end{cases} \tag{3.3}$$

where τ denotes the classification threshold.

This formulation better reflects the realities of clinical diagnosis and provides greater flexibility when interpreting model outputs. Instead of receiving a single categorical prediction, clinicians are presented with independent disease probabilities that can be incorporated into broader diagnostic assessments.

3.1.5 Level Two: Lung Cancer Detection Model

Images classified as CT scans are forwarded to the lung cancer detection model. The decision to utilize CT imaging for lung cancer analysis is motivated by the superior anatomical detail provided by computed tomography compared to conventional radiography.

Lung cancer frequently manifests through subtle structural abnormalities such as pulmonary nodules, irregular masses, and localized tissue changes that may not be visible in chest radiographs. The high spatial resolution of CT imaging enables these features to be observed with significantly greater clarity, making CT the preferred modality for many lung cancer screening and diagnostic procedures.

The objective of the lung cancer model is to determine whether cancer-related findings are present within the input scan. The classifier can be represented mathematically as

$$P_{\text{Cancer}} = C(I) \tag{3.4}$$

where $C(\cdot)$ denotes the lung cancer detection network and P_{Cancer} represents the estimated probability of malignancy.

By focusing exclusively on CT-derived information, the model can dedicate its representational capacity to learning highly discriminative cancer-related features. This specialization improves the model's ability to identify subtle pathological patterns that may be difficult to detect using a generalized classifier.

3.1.6 Explainable Artificial Intelligence Integration

The increasing adoption of artificial intelligence within healthcare has generated significant interest in model interpretability. Although deep learning systems are capable of achieving remarkable predictive performance, their internal decision-making processes often remain difficult to understand. This lack of transparency can hinder clinical adoption, particularly when predictions influence critical healthcare decisions.

To address this challenge, Explainable Artificial Intelligence techniques are integrated directly into the proposed architecture. Rather than treating explainability as an optional post-processing step, it is considered a fundamental component of the diagnostic workflow.

Following prediction generation, the explainability module produces visual explanations illustrating the regions of the image that most strongly influenced the model's decision. These visualizations may take the form of saliency maps, activation maps, attention maps, or gradient-based explanations depending on the selected implementation.

The integration of explainability serves several purposes. First, it increases clinician confidence by providing visual evidence supporting the model's prediction. Second, it facilitates model validation by enabling researchers to verify that the network is focusing on medically meaningful regions rather than irrelevant artifacts. Third, it assists in identifying potential biases or failure modes that may otherwise remain hidden within the network. Ultimately, explainability strengthens the collaborative relationship between clinicians and AI systems by transforming predictions into interpretable decision-support information.

3.1.7 Federated Learning Infrastructure

A distinguishing feature of the proposed framework is its support for Federated Learning. This capability was incorporated to address one of the most significant challenges facing medical artificial intelligence: limited access to large and diverse datasets.

Traditional deep learning systems typically rely on centralized training approaches in which data from multiple institutions are collected and stored in a single repository. While effective from a machine learning perspective, this strategy introduces substantial privacy, security, and regulatory concerns. Healthcare organizations are often prohibited from sharing patient data due to legal and ethical restrictions.

Federated Learning offers an alternative paradigm in which training occurs locally at each institution. Rather than transmitting patient data, participating hospitals train local models using their own datasets and share only model parameters or parameter updates. These updates are subsequently aggregated to produce a global model.

Let w_i denote the model parameters learned by institution i, and let n_i denote the number of training samples available at that institution. Using the Federated Averaging algorithm, the global model is computed as

$$w_{\text{global}} = \sum_{i=1}^{K} \frac{n_i}{\sum_{j=1}^{K} n_j} w_i \tag{3.5}$$

where K represents the total number of participating institutions.

Beyond privacy preservation, Federated Learning provides several additional benefits. Because participating institutions often serve different patient populations, the resulting datasets contain greater demographic, geographic, and clinical diversity. Training across such heterogeneous data sources can improve model generalization and reduce susceptibility to institution-specific biases. Furthermore, the collaborative nature of Federated Learning enables continuous improvement of the framework as new institutions join the network and contribute additional knowledge.

3.2 Transfer-Learning Models Used

This section presents the transfer learning models employed in the proposed system and discusses the rationale behind their selection. Transfer learning is adopted to leverage pre-trained deep neural networks that have been trained on large-scale datasets, enabling effective feature extraction and improved generalization in medical imaging tasks with limited labeled data. The section provides an overview of the selected architectures and highlights their suitability for different components of the proposed diagnostic pipeline.

3.2.1 ResNet50 Model

Residual Networks (ResNets) were introduced to address the degradation problem observed in deep convolutional neural networks, where increasing network depth leads to saturation or decline in training accuracy. This issue emerges due to optimization difficulties such as vanishing gradients. ResNet addresses this limitation through the concept of residual learning, which allows the training of substantially deeper networks by enabling more effective gradient propagation.

The key idea of residual learning is to reformulate the learning objective of a neural network block. Instead of directly learning a desired underlying mapping $H(x)$, a residual block learns a residual function $F(x)$ such that:

$$H(x) = F(x) + x$$

where x represents the input to the residual block and $F(x)$ represents the output of the stacked convolutional layers within the block. The identity shortcut connection that performs the addition allows gradients to flow directly through the network during backpropagation, thereby minimizing vanishing gradient effects and improving training stability.

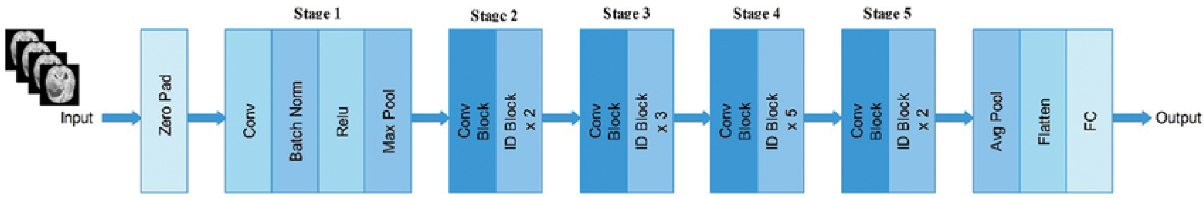


Figure 3.3: Architecture of the ResNet50 illustrating the successive stages of the model.

As illustrated in Figure 3.3, the ResNet50 architecture begins with an initial convolutional layer preceded by zero padding, followed by batch normalization, a ReLU activation function, and a max pooling operation. This initial stage is responsible for extracting low-level visual features from the input image. The network then proceeds through five sequential stages composed of convolutional blocks and identity blocks, each incorporating residual connections that bypass multiple convolutional layers.

ResNet50 employs a bottleneck design within its residual blocks, consisting of a sequence of 1×1, 3×3, and 1×1 convolutional layers. The 1×1 convolutions are used to reduce and subsequently restore feature dimensionality, thereby decreasing computational complexity while preserving representational power. This design enables ResNet50 to achieve substantial depth without incurring prohibitive computational costs.

Compared to shallower variants such as ResNet18 and ResNet34, ResNet50 offers increased representational capacity and improved ability to capture complex and abstract visual patterns. While deeper variants such as ResNet101 and ResNet152 further extend network depth, they introduce higher computational overhead and an increased risk of overfitting, particularly in medical imaging applications where annotated data is often limited. Consequently, ResNet50 represents a balanced trade-off between depth, efficiency, and performance within the ResNet family.

As shown in Figure 3.3, the final stages of ResNet50 consist of a global average pooling layer, followed by feature flattening and a fully connected layer. In transfer learning scenarios, this classification head can be replaced or fine-tuned to adapt the network to domain-specific medical tasks.

In the context of medical image analysis, ResNet50 is well-suited for extracting both low-level structural features and high-level semantic representations relevant to pathological patterns. The residual connections promote feature reuse and robust optimization, which is particularly important for detecting subtle abnormalities in medical images such as fractures or thoracic diseases. These characteristics make ResNet50 a reliable and widely adopted backbone for medical image classification tasks.

3.2.2 DenseNet121 Model

DenseNet121 is a deep convolutional neural network architecture belonging to the family of Densely Connected Convolutional Networks (DenseNets), which were introduced by Huang *et al.* in 2017 to address several limitations associated with traditional deep neural networks. DenseNet architectures were designed to improve feature propagation, encourage feature reuse, alleviate the vanishing gradient problem, and reduce the number of trainable parameters while maintaining high classification performance. Since its introduction, DenseNet121 has become one of the most widely adopted architectures in medical image analysis due to its ability to learn rich hierarchical representations from relatively limited datasets.

The fundamental idea behind DenseNet differs significantly from that of conventional convolutional neural networks. In traditional architectures, each layer receives information only from the immediately preceding layer and passes its output solely to the next layer. Although this sequential information flow is effective, it may lead to difficulties in gradient propagation as network depth increases. Furthermore, many learned feature maps become redundant because similar patterns are repeatedly relearned at different layers throughout the network.

To address these limitations, DenseNet introduces direct connections between layers. Instead of receiving input from only the previous layer, each layer receives the feature maps generated by all preceding layers within the same dense block. Consequently, information can flow directly from earlier layers to later layers without repeatedly passing through intermediate transformations. This dense connectivity pattern enables more efficient feature reuse and facilitates the training of very deep networks.

Figure 3.4 illustrates the overall architecture of DenseNet121.

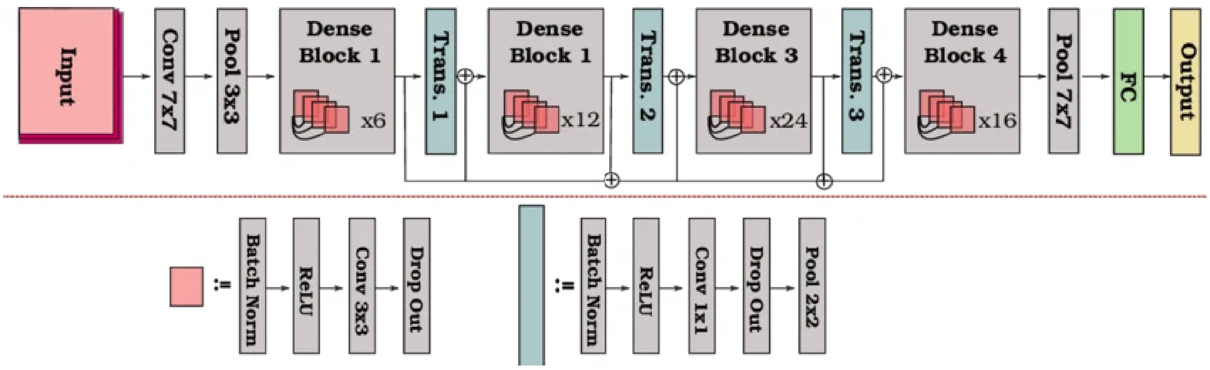


Figure 3.4: Overall architecture of DenseNet121 showing the arrangement of dense blocks, transition layers, and the final classification stage.

Motivation Behind Dense Connectivity

As neural networks become deeper, the flow of information and gradients through the network becomes increasingly difficult. One of the most common challenges encountered in deep architectures is the vanishing gradient problem, where gradients become progressively smaller as they are propagated backward through many layers. When gradients become extremely small, early layers receive little useful learning signal, slowing convergence and potentially reducing overall performance.

Residual Networks (ResNets) partially addressed this issue through skip connections that allow gradients to bypass certain transformations. DenseNet extends this concept further by establishing direct connections between every pair of layers within a dense block. Instead of merely adding outputs from previous layers, DenseNet concatenates them. This strategy preserves information from all preceding layers and makes it directly accessible to subsequent layers.

The dense connectivity mechanism provides several important benefits. First, it improves gradient flow during training, allowing deeper networks to be optimized more effectively. Second, it encourages feature reuse, reducing redundancy among learned representations. Third, it improves parameter efficiency because layers can leverage previously learned features rather than learning entirely new representations from scratch.

Dense Connectivity Mechanism

The defining characteristic of DenseNet is the dense connection pattern between layers. Let x_0 represent the input to a dense block and let x_l denote the output of the l^{th} layer. Unlike traditional convolutional networks, where the output of layer l depends only on the output of layer $l-1$, DenseNet computes each layer using the outputs of all preceding layers:

$$x_l = H_l([x_0, x_1, x_2, ..., x_{l-1}]) \tag{3.6}$$

where:

- $H_l(\cdot)$ denotes a composite transformation.
- $[\cdot]$ represents the concatenation operation.
- $x_0, x_1, ..., x_{l-1}$ are the feature maps generated by previous layers.

The concatenation operation differs fundamentally from the addition operation used in residual networks. Instead of merging features by summation, DenseNet preserves all previous feature maps and passes them directly to subsequent layers. As a result, each layer has access to both low-level and high-level information simultaneously.

This dense feature propagation mechanism allows later layers to exploit a richer collection of learned representations, improving feature diversity and reducing information loss.

Dense Blocks

The primary computational components of DenseNet121 are the dense blocks. A dense block consists of multiple densely connected layers arranged according to the connectivity pattern described previously.

Within each dense block, every layer receives as input the concatenated outputs of all preceding layers. Consequently, the number of feature maps grows progressively as additional layers are added. This growth is controlled through a parameter known as the *growth rate*.

The growth rate, typically denoted by k, determines the number of new feature maps generated by each layer. If the input to a dense block contains F_0 feature maps and the block contains L layers, the total number of feature maps at the end of the block can be expressed as:

$$F = F_0 + kL \tag{3.7}$$

where:

- F_0 is the number of initial feature maps.
- k is the growth rate.
- L is the number of layers within the block.

For DenseNet121, the growth rate is typically set to 32. This means that each layer contributes 32 new feature maps while simultaneously utilizing all previously generated feature maps.

The use of a relatively small growth rate contributes to the parameter efficiency of DenseNet architectures, allowing them to achieve competitive performance with fewer parameters than many alternative architectures.

Composite Function within Dense Layers

Each layer within a dense block consists of a sequence of operations designed to transform incoming feature maps into more informative representations.

The composite transformation $H_l(\cdot)$ generally consists of:

1. Batch Normalization (BN)
2. Rectified Linear Unit (ReLU)
3. Convolution

This sequence is commonly referred to as the BN-ReLU-Conv pattern.

Batch normalization stabilizes the learning process by normalizing intermediate activations. This reduces internal covariate shift and allows higher learning rates to be used during training.

The ReLU activation function introduces non-linearity into the network and is defined as:

$$f(x) = \max(0, x) \tag{3.8}$$

Following activation, convolutional filters extract spatial patterns from the input feature maps. Through repeated application of these operations, the network gradually learns increasingly abstract representations of the input image.

Bottleneck Layers

As the number of feature maps grows within dense blocks, computational complexity can increase significantly. To reduce this cost, DenseNet121 employs bottleneck layers.

A bottleneck layer consists of a 1×1 convolution applied before the standard 3×3 convolution. The purpose of this operation is to reduce the number of feature channels while preserving important information.

The bottleneck structure can be summarized as:

$$BN \rightarrow ReLU \rightarrow Conv(1 \times 1) \rightarrow BN \rightarrow ReLU \rightarrow Conv(3 \times 3) \tag{3.9}$$

The use of bottleneck layers substantially reduces computational requirements while maintaining representational capacity.

Transition Layers

Because feature map dimensions increase continuously within dense blocks, DenseNet introduces transition layers between consecutive blocks to control network complexity.

A transition layer performs two primary functions. First, it reduces the number of feature maps through a 1×1 convolution. Second, it decreases spatial resolution using average pooling.

The transition layer architecture typically follows:

$$BN \rightarrow ReLU \rightarrow Conv(1 \times 1) \rightarrow AvgPool(2 \times 2) \tag{3.10}$$

These operations prevent excessive growth in memory consumption while preserving the most informative features.

Transition layers also contribute to hierarchical feature extraction by gradually reducing spatial dimensions as network depth increases.

Architecture of DenseNet121

DenseNet121 consists of four dense blocks separated by three transition layers. The architecture begins with an initial convolution layer and concludes with global average pooling followed by a fully connected classification layer.

The distribution of layers across dense blocks is as follows:

Table 3.1: Layer distribution within DenseNet121.

Component	Number of Layers
Dense Block 1	6
Dense Block 2	12
Dense Block 3	24
Dense Block 4	16
Total Dense Layers	58

When all convolutional, pooling, and classification layers are included, the network contains 121 trainable layers, which gives the architecture its name.

The input image first passes through an initial 7×7 convolution layer followed by max pooling. Feature extraction then proceeds through the sequence of dense blocks and transition layers. Finally, global average pooling aggregates spatial information into a compact feature vector, which is passed to the final classification layer.

Advantages of DenseNet121 for Medical Image Analysis

DenseNet121 possesses several characteristics that make it particularly suitable for medical image classification tasks.

The dense connectivity pattern encourages extensive feature reuse, allowing the network to learn robust representations even when training datasets are relatively small. This characteristic is especially important in medical imaging, where acquiring large annotated datasets is often challenging.

The improved gradient flow resulting from dense connections facilitates stable optimization and reduces the risk of vanishing gradients. Consequently, deeper representations can be learned without the training difficulties commonly associated with very deep neural networks.

DenseNet121 is also highly parameter-efficient compared to many competing architectures. Despite its depth, the network requires fewer trainable parameters than several traditional convolutional architectures because previously learned features are reused rather than repeatedly regenerated.

Furthermore, the architecture has demonstrated strong performance across a wide range of medical imaging applications, including chest disease classification, pneumonia detection, COVID-19 diagnosis, tuberculosis screening, diabetic retinopathy analysis, and cancer detection. Its proven effectiveness in healthcare applications has made DenseNet121 a popular backbone network for medical image analysis research.

DenseNet121 in the Proposed Framework

In the context of this work, DenseNet121 was selected as the backbone architecture for the first-level routing model due to its strong balance between classification performance, computational efficiency, and feature extraction capability. The dense connectivity structure enables the model to capture both low-level radiological patterns and high-level pathological characteristics that are essential for accurate disease recognition.

Additionally, the feature reuse mechanism of DenseNet121 is particularly advantageous for medical datasets, where image variability can be substantial and labeled samples may be limited. By leveraging information from all preceding layers, the network can construct rich hierarchical representations that support robust classification performance.

Overall, DenseNet121 provides a powerful and efficient deep learning architecture that aligns well with the requirements of medical image analysis systems. Its dense connectivity, parameter efficiency, strong gradient propagation, and proven effectiveness in healthcare applications make it an appropriate choice for the disease classification tasks addressed in this project.

3.2.3 YOLOv8 Model

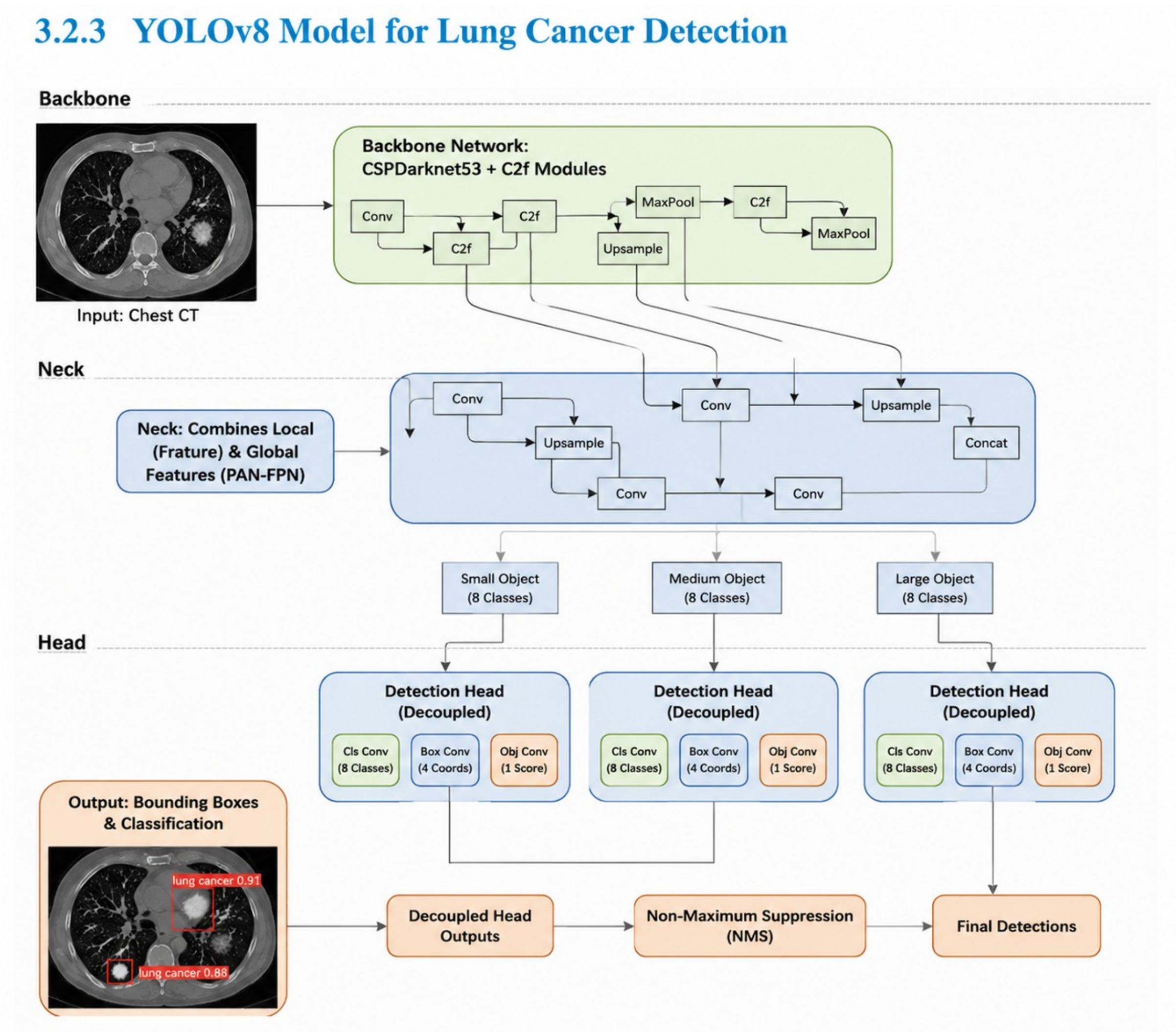


Figure 3.5: General architecture of YOLOv8 illustrating the backbone, neck, and detection head components.

Object detection has become one of the most important computer vision tasks in modern medical image analysis due to its ability to simultaneously identify and localize pathological findings within an image. Unlike conventional image classification models that only determine whether a disease is present, object detection models additionally provide spatial information regarding the location of abnormalities. This capability is particularly valuable in lung cancer diagnosis, where clinicians must not only determine the presence of a suspicious lesion but also identify its precise location within the lungs.

In the proposed framework, lung cancer detection is formulated as an object detection problem rather than a traditional image classification task. Instead of predicting a single label for an entire CT scan, the model identifies suspicious cancerous regions and generates bounding boxes surrounding them. To accomplish this task, the proposed system employs YOLOv8 (You Only Look Once Version 8), one of the most recent and advanced members of the YOLO family of object detection architectures.

YOLOv8 is a single-stage object detector, meaning that object localization and classification are performed within a single forward pass through the network. This design significantly reduces computational complexity compared to two-stage detectors such as Faster R-CNN, which first generate candidate regions and subsequently classify them. By combining both operations into a unified architecture, YOLOv8 achieves high detection accuracy while maintaining efficient inference speeds. These characteristics make it particularly suitable for deployment within real-time clinical decision support systems where rapid analysis of CT scans is desirable.

The complete processing pipeline employed by the proposed lung cancer detection model is illustrated in Figure 3.6.

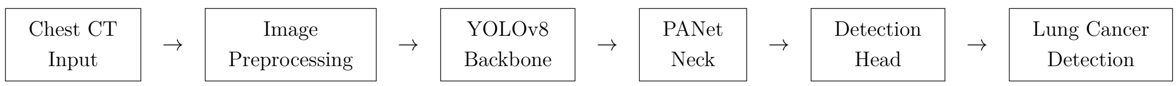


Figure 3.6: End-to-end processing pipeline of the proposed YOLOv8 lung cancer detection model.

As illustrated in Figure 3.5, YOLOv8 consists of three primary components: the backbone, the neck, and the detection head. Each component contributes a specific functionality to the overall detection process.

Backbone Network

The backbone serves as the feature extraction component of YOLOv8. Its primary objective is to transform raw CT scan images into hierarchical feature representations that can be used for lesion detection. YOLOv8 employs the C2f (Cross-Stage Partial Bottleneck with Two Convolutions) architecture, which is an evolution of the CSP (Cross Stage Partial) designs used in previous YOLO versions.

The backbone progressively applies convolutional operations to the input image, generating increasingly abstract feature maps. Early layers learn low-level visual characteristics such as edges, gradients, and texture variations, while deeper layers capture more complex semantic information associated with pulmonary nodules, masses, and abnormal tissue structures.

One of the key advantages of the C2f architecture is its ability to improve gradient propagation while reducing computational overhead. By reusing features across multiple layers, the network can learn richer representations without requiring excessive numbers of parameters. This characteristic is particularly beneficial in medical imaging applications where datasets are often smaller than those available in general computer vision tasks.

For lung cancer detection, the backbone learns to distinguish subtle visual differences between healthy lung tissue and suspicious lesions. These differences may include variations in density, texture, shape, and intensity patterns that are characteristic of malignant abnormalities.

Neck Network

The neck component is responsible for feature aggregation and multi-scale feature fusion. In YOLOv8, this functionality is implemented using a Path Aggregation Network (PANet).

Feature extraction layers at different depths of the backbone capture information at different scales. Shallow layers retain detailed spatial information, while deeper layers capture more abstract semantic features. Because lung cancer lesions can vary significantly in size, relying on a single feature scale would limit detection performance.

Small pulmonary nodules may occupy only a tiny portion of the CT image, whereas larger tumors may extend across substantial regions of the lung. The PANet structure addresses this challenge by combining feature maps generated at multiple scales. Through repeated top-down and bottom-up information flow, PANet enables the model to simultaneously utilize local texture details and broader anatomical context.

Consequently, the neck allows YOLOv8 to detect both small and large lesions while maintaining accurate localization performance across a wide range of tumor sizes.

Detection Head

The final component of YOLOv8 is the detection head. This module is responsible for generating the actual predictions produced by the network.

Unlike earlier object detectors that relied on predefined anchor boxes, YOLOv8 employs an anchor-free detection strategy. Rather than predicting offsets relative to predetermined anchors, the model directly predicts object centers and bounding box coordinates.

The detection head performs two independent tasks:

- Classification of detected regions.
- Localization of detected lesions.

This decoupled architecture allows each task to be optimized independently, improving overall detection performance. For each detected region, the model predicts:

- Bounding box coordinates.
- Object confidence score.
- Cancer detection probability.

The resulting predictions are then forwarded to the post-processing stage for refinement.

Multi-Scale Detection

A major challenge in lung cancer detection is the substantial variability in lesion size. Early-stage tumors may appear as extremely small nodules, whereas advanced-stage tumors can occupy large portions of the lung field.

YOLOv8 addresses this challenge through multi-scale detection. Detection heads operate on feature maps of different resolutions, enabling the model to identify abnormalities across multiple spatial scales.

This capability is particularly important in CT-based diagnosis because the early detection of small nodules is often critical for improving patient outcomes. Multi-scale processing allows the model to preserve fine-grained details while simultaneously leveraging high-level semantic information extracted from deeper network layers.

The full process of multi-scale feature extraction is shown in Figure 3.7

High-Resolution Features (Small Nodules)	Medium-Resolution Features (Intermediate Lesions)	Low-Resolution Features (Large Tumors)

$\downarrow$

Unified Multi-Scale Detection Output

Figure 3.7: Multi-scale feature extraction employed by YOLOv8 for detecting lung lesions of varying sizes.

Anchor-Free Detection Mechanism

Traditional object detection models rely on predefined anchor boxes that attempt to approximate the expected shapes and sizes of target objects. Although effective in many applications, anchor-based methods introduce additional hyperparameters and often struggle with irregularly shaped objects.

Lung tumors exhibit substantial variability in morphology and rarely conform to standardized geometric patterns. Consequently, anchor-based approaches may fail to adequately represent their boundaries.

YOLOv8 addresses this limitation by adopting an anchor-free detection strategy. Instead of matching objects to predefined anchor templates, the model directly predicts object centers and corresponding bounding box coordinates.

This design offers several advantages:

- Reduced hyperparameter complexity.
- Improved adaptability to irregular tumor shapes.
- Faster convergence during training.

- Enhanced localization performance.

These characteristics make anchor-free detection particularly suitable for medical imaging applications involving heterogeneous pathological structures.

Bounding Box Regression and Localization

Once suspicious regions have been identified, YOLOv8 must determine the exact boundaries of each lesion.

Bounding box regression is performed by predicting:

- Horizontal position.
- Vertical position.
- Width.
- Height.

These predictions are continuously refined throughout training using specialized localization loss functions. The resulting bounding boxes provide interpretable visual evidence regarding the location of suspicious findings and can assist clinicians during diagnostic review.

Non-Maximum Suppression

Object detectors frequently generate multiple overlapping predictions for the same lesion. To eliminate redundant detections, YOLOv8 employs Non-Maximum Suppression (NMS).

The process is illustrated in Figure 3.8.

Overlapping Bounding Boxes	→	IoU Calculation	→	Non-Maximum Suppression	→	Final Lung Cancer Detection

Figure 3.8: Non-Maximum Suppression procedure used to remove duplicate detections and retain the highest-confidence prediction.

The overlap between predicted and ground-truth boxes is measured using the Intersection over Union (IoU) metric:

$$IoU = \frac{Area(B_p \cap B_g)}{Area(B_p \cup B_g)}$$

where B_p denotes the predicted bounding box and B_g denotes the ground-truth bounding box.

Predictions with excessive overlap are suppressed, ensuring that each lesion is represented by a single final detection.

Training Objective and Loss Functions

YOLOv8 optimizes a composite loss function composed of multiple components.

Classification Loss (Binary Cross-Entropy) Binary Cross-Entropy loss guides the model in distinguishing cancerous lesions from background regions. This loss encourages accurate classification while minimizing false positive detections.

Box Localization Loss (CIoU) Complete Intersection over Union (CIoU) loss evaluates localization quality by considering overlap area, center distance, and aspect ratio differences between predicted and ground-truth boxes.

This loss can be expressed conceptually as:

$$L_{CIoU} = 1 - IoU + \text{Distance Penalty} + \text{Aspect Ratio Penalty}$$

where additional penalty terms improve localization precision.

Distribution Focal Loss (DFL) Distribution Focal Loss improves boundary estimation by modeling uncertainty in bounding box coordinates. This is particularly important in medical images where lesion boundaries may be ambiguous or partially obscured.

Together, these loss functions allow YOLOv8 to simultaneously optimize classification accuracy and localization quality.

Suitability of YOLOv8 for Lung Cancer Detection

Several characteristics make YOLOv8 particularly suitable for lung cancer detection.

First, its single-stage architecture provides rapid inference speeds while maintaining high detection accuracy. This allows the model to be integrated into real-time diagnostic support systems.

Second, multi-scale feature extraction enables the detection of lesions exhibiting substantial size variation, ranging from small pulmonary nodules to larger malignant masses.

Third, the anchor-free detection mechanism improves adaptability to irregular tumor morphologies commonly encountered in clinical practice.

Fourth, the generated bounding boxes provide intuitive visual explanations that can assist clinicians in understanding and verifying model predictions.

Finally, YOLOv8 benefits from transfer learning, allowing knowledge acquired from large-scale image datasets to be leveraged for medical imaging tasks. This capability is especially valuable when working with limited annotated medical datasets.

To evaluate the effectiveness of the proposed lung cancer detection model, Precision, Recall, and mAP50 are employed as the primary performance metrics. Precision quantifies the proportion of detected lesions that are truly cancerous, Recall measures the proportion of actual lesions successfully identified by the model, and mAP50 provides a comprehensive assessment of overall detection performance by evaluating both localization and classification accuracy at an Intersection over Union threshold of 0.50.

Collectively, these metrics provide a rigorous evaluation of the model's ability to accurately identify and localize lung cancer lesions within chest CT scans, ensuring that the resulting system can serve as a reliable component of the proposed intelligent diagnostic framework.

3.3 Medical Image Routing Model

This section presents the proposed medical image routing model employed as the first level of the hierarchical diagnostic framework and discusses the motivation behind its development, the formal problem formulation, and its architectural design. The routing model serves as the entry point of the proposed system, where the modality of each incoming image is identified and the image is subsequently forwarded to the most appropriate downstream diagnostic component. By introducing an explicit routing stage, the framework ensures that specialized diagnostic models receive only the image types for which they were designed, thereby improving overall system reliability, robustness, and scalability.

3.3.1 Motivation

The increasing adoption of deep learning within medical image analysis has enabled the development of highly specialized models for disease detection, diagnosis, and clinical decision support. However, most medical imaging models are designed for specific imaging modalities and diagnostic tasks. Consequently, their performance depends heavily on receiving inputs that match the data distribution encountered during training. Applying a modality-specific model to an inappropriate image type can significantly degrade performance and may result in unreliable predictions.

In the context of the proposed framework, two distinct diagnostic pathways are employed. The first pathway focuses on chest disease classification from chest X-ray images and is responsible for identifying the presence of COVID-19, Tuberculosis, Pneumonia, or the absence of disease. The second pathway focuses on lung cancer detection from chest CT scans using a dedicated object detection model. Since these diagnostic models operate on fundamentally different imaging modalities and address different clinical objectives, a mechanism is required to automatically determine which model should process each incoming image.

Real-world deployment scenarios further emphasize the importance of such a mechanism. Medical information systems may receive images originating from various imaging modalities, anatomical regions, acquisition devices, and clinical workflows. In addition, unsupported images may be inadvertently submitted to the system. Without a dedicated routing stage, these images would be forced through one of the diagnostic models regardless of their suitability, increasing the likelihood of incorrect predictions and reducing overall system reliability.

To address this challenge, a dedicated medical image routing model is introduced as the first stage of the proposed architecture. The model distinguishes between chest X-ray images, chest CT scan images, and an additional out-of-distribution category referred to as *Other*. By incorporating this third category, the framework explicitly recognizes unsupported images and prevents them from being processed by downstream diagnostic models. Images identified as chest X-rays are forwarded to the multi-label chest disease classification model, while CT scans are forwarded to the lung cancer detection model. Images classified as *Other* are rejected from further analysis. This design not only improves robustness and safety but also provides a modular foundation upon which additional imaging modalities and diagnostic models may be incorporated in future system extensions.

3.3.2 Problem Statement

The objective of the routing model is to automatically determine the modality of an incoming medical image and direct it toward the appropriate downstream diagnostic pathway. This task is formulated as a three-class image classification problem in which each input image must be assigned to one of three predefined categories.

Let the input image be represented by

$$X \in \mathbb{R}^{224\times224\times3},$$

where the image is resized to a fixed spatial resolution of 224×224 pixels and represented using three color channels. Given an input image X, the routing model predicts one of the following classes:

- Chest X-ray (0)
- Chest CT Scan (1)
- Other / Out-of-Distribution (2)

The model produces a probability distribution

$$\mathbf{p} = [p_0, p_1, p_2],$$

where

$$p_k = P(y = k \mid X)$$

and

$$\sum_{k=0}^{2} p_k = 1.$$

The predicted class label is obtained using

$$\hat{y} = \arg\max_{k} p_k.$$

Images classified as chest X-rays are forwarded to the multi-label chest disease classification model, while images classified as CT scans are directed to the lung cancer detection model. The out-of-distribution category serves as an additional safety mechanism that prevents unsupported images from being processed by downstream diagnostic components. By incorporating this category, the routing model improves the reliability and robustness of the overall framework and establishes a structured entry point for all incoming medical images.

The routing model therefore functions as a modality recognition system whose purpose extends beyond simple image classification. Its predictions directly determine the diagnostic pathway followed by each image and consequently influence all subsequent processing stages within the proposed architecture.

3.3.3 Model Architecture

The proposed medical image routing model is implemented using a transfer learning approach based on the DenseNet121 convolutional neural network architecture. The model constitutes the first level of the proposed hierarchical framework and is responsible for determining the appropriate diagnostic pathway for each incoming image. By accurately identifying image modalities and separating unsupported inputs from valid medical scans, the routing model ensures that downstream diagnostic models receive only the image types for which they were designed.

Several candidate architectures were investigated during development, including a custom convolutional neural network, ResNet50, DenseNet121, and EfficientNetB3. Although all evaluated models demonstrated the ability to learn modality-specific representations, DenseNet121 consistently provided the most favorable balance between classification performance, feature extraction capability, and computational efficiency. The dense connectivity mechanism employed by DenseNet121 promotes feature reuse throughout the network and improves gradient propagation, allowing the model to learn highly discriminative features while maintaining a relatively compact parameter count. Consequently, DenseNet121 was selected as the backbone architecture for the final routing model.

The architecture follows a structured pipeline consisting of image preprocessing, deep feature extraction using DenseNet121, feature aggregation through global average pooling, and final classification using fully connected layers. A high-level overview of the proposed architecture is illustrated in Figure 3.9.

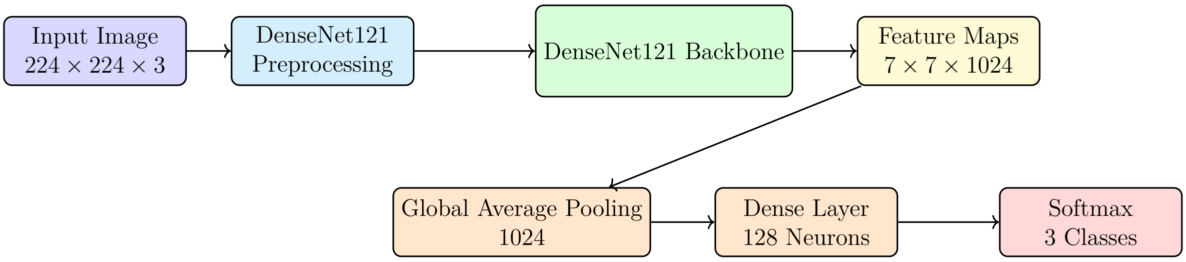


Figure 3.9: Two-row architecture of the DenseNet121-based medical image routing model.

The input to the model is a medical image represented by a tensor of dimensions $224 \times 224 \times 3$. Prior to feature extraction, the image undergoes the preprocessing procedure associated with DenseNet121. This preprocessing transforms the image into a format compatible with the ImageNet-pretrained weights used by the backbone network. By aligning the input distribution with the statistical properties encountered during pretraining, the model can effectively leverage the learned representations obtained from large-scale natural image datasets.

Following preprocessing, the image is passed through the DenseNet121 backbone with its original classification layers removed. DenseNet121 belongs to the family of densely connected convolutional neural networks, in which each layer receives feature maps from all preceding layers within a dense block. This dense connectivity pattern encourages feature reuse, facilitates gradient propagation during training, and improves parameter efficiency. As a result, DenseNet121 is capable of learning rich hierarchical representations while requiring fewer parameters than many conventional deep convolutional architectures.

For an input image of dimensions $(224, 224, 3)$, the DenseNet121 backbone produces a high-level feature tensor of dimensions $(7, 7, 1024)$. These feature maps contain abstract representations of visual patterns extracted from the image, including anatomical structures, texture information, intensity distributions, and modality-specific characteristics. Because information from earlier layers remains accessible throughout the network, DenseNet121 can effectively combine low-level and high-level features when constructing its final representation.

To transform these spatial feature maps into a compact feature representation suitable for classification, a Global Average Pooling (GAP) layer is applied. Unlike flattening operations that convert all feature map values into a large vector, GAP computes the average activation of each feature channel across all spatial locations. Given a feature tensor

$$F \in \mathbb{R}^{7\times7\times1024},$$

the pooled feature vector

$$z \in \mathbb{R}^{1024}$$

is computed as

$$z_k = \frac{1}{7 \times 7} \sum_{i=1}^{7} \sum_{j=1}^{7} F_{i,j,k},$$

where z_k represents the average activation of the k-th feature channel. This operation reduces the dimensionality of the representation while preserving the most important information learned by the backbone network. Furthermore, it significantly reduces the number of trainable parameters in the classification head and helps improve model generalization.

The resulting feature vector is subsequently processed by a fully connected layer consisting of 128 neurons with ReLU activation. This layer learns task-specific combinations of the extracted features and serves as an intermediate representation between the pretrained backbone and the final classification layer.

The final layer consists of three output neurons followed by a softmax activation function. This layer produces a probability distribution

$$\mathbf{p} = [p_{XRay}, p_{CT}, p_{Other}],$$

where each component corresponds to the predicted probability that the input image belongs to the chest X-ray, chest CT scan, or out-of-distribution category, respectively. The final prediction is obtained according to

$$\hat{y} = \arg\max_k p_k.$$

Images classified as chest X-rays are forwarded to the multi-label chest disease classification model, images classified as CT scans are directed to the lung cancer detection model, and images assigned to the *Other* category are rejected from further analysis. Consequently, the routing model serves as the gateway to the entire diagnostic framework and plays a critical role in ensuring that each image is processed by the most appropriate downstream component.

The successful operation of this routing stage is fundamental to the effectiveness of the proposed hierarchical architecture. Since all subsequent diagnostic decisions depend on accurate modality recognition, errors introduced at this stage may propagate throughout the remainder of the system. Therefore, the routing model acts not only as a classifier but also as a reliability and safety mechanism that enhances the robustness of the overall framework. The following sections describe the specialized diagnostic models that receive images from the routing component and perform disease-specific analysis.

3.3.4 Implementation of the First-Level Routing Model

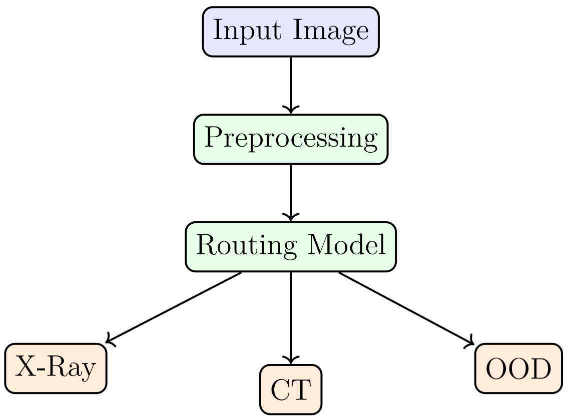
The first component implemented within the proposed framework is the modality routing model. As discussed in Chapter **??**, this model serves as the entry point of the entire classification pipeline and is responsible for determining the type of the incoming image before directing it toward the appropriate diagnostic model. Since the subsequent classification stages were specifically designed for particular imaging modalities, ensuring the accuracy of the routing stage was considered a critical requirement for the reliability of the overall framework.

The routing model was formulated as a three-class image classification problem in which each input image was assigned to one of three categories: chest X-ray images, chest CT scans, or out-of-distribution (OOD) images. The OOD category was intentionally incorporated to improve the robustness of the system during deployment. In practical healthcare environments, medical information systems may contain a wide variety of image types that fall outside the scope of the proposed framework. Without an explicit OOD category, unsupported images could be mistakenly forwarded to disease classification models, potentially resulting in unreliable predictions. The inclusion of this category therefore acts as a safety mechanism that prevents unsupported inputs from entering the diagnostic pipeline.

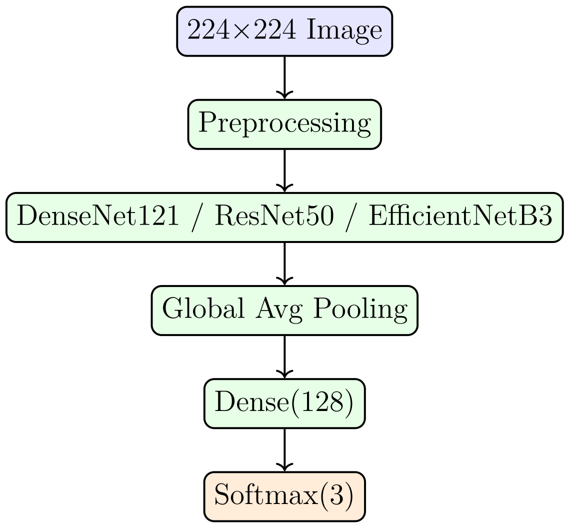
The final routing dataset was constructed by aggregating images originating from multiple sources. After dataset integration, the X-ray category contained 7,123 images, the CT category contained 2,534 images, and the OOD category contained 9,958 images. Although this aggregation process provided a sufficiently diverse collection of samples, it also introduced a noticeable class imbalance problem. The CT class represented a considerably smaller portion of the dataset compared to the other categories, creating a risk that the trained model would become biased toward the more frequently represented classes.

To mitigate this issue, several balancing strategies were incorporated into the training pipeline. Data augmentation techniques were applied to increase the effective diversity of minority-class samples while simultaneously improving model generalization. Augmentation operations included random rotations, horizontal flipping, zoom transformations, and intensity variations. These transformations were selected because they preserve medically relevant structures while exposing the model to realistic image variations that may occur across different acquisition settings. In addition to augmentation, class balancing procedures were employed during dataset preparation to reduce the influence of class distribution disparities on the learning process. These measures contributed to a more stable training process and improved the model's ability to recognize underrepresented classes.

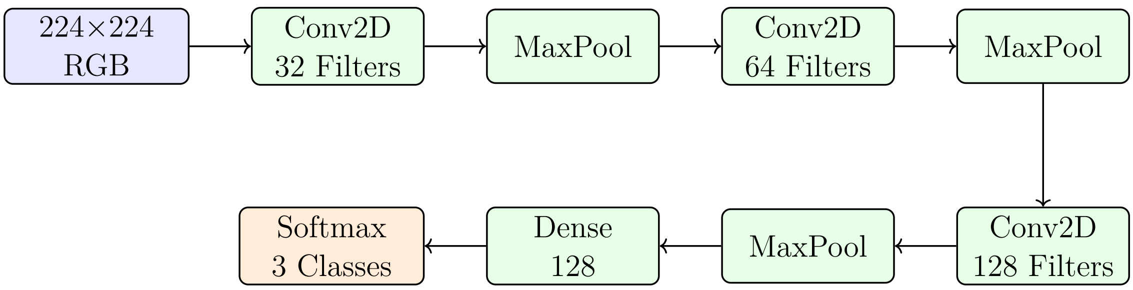
The complete routing workflow implemented within the framework is illustrated in Figure 3.10. The figure summarizes both the operational behavior of the routing model and the architectures investigated during the model selection process. Rather than committing immediately to a single architecture, several alternative models were implemented and evaluated in order to identify the most suitable solution for the routing task. This decision was motivated by the observation that image modality classification differs substantially from disease classification. While disease detection often requires learning subtle pathological patterns, modality recognition primarily depends on identifying broader structural and acquisition-specific characteristics. Consequently, it was necessary to experimentally determine whether a lightweight custom architecture would be sufficient or whether a deeper pretrained network would provide superior performance.



(a) Routing workflow.



(b) Transfer learning models.



(c) Custom CNN architecture used as a baseline model for modality routing.

Figure 3.10: Architectures and workflow investigated during the implementation of the routing model. Blue blocks represent inputs, green blocks denote processing stages, and orange blocks correspond to classification outputs.

The first architecture investigated was a custom convolutional neural network developed specifically for the routing task. The model begins with a rescaling layer that normalizes pixel intensities into the range $[0, 1]$. Feature extraction is then performed through three convolutional blocks containing 32, 64, and 128 filters respectively. Each convolutional layer uses a ReLU activation function and is followed by a max-pooling operation that progressively reduces the spatial dimensions of the feature maps while preserving the most salient visual information. After the final convolutional block, the extracted features are flattened and passed through a fully connected layer consisting of 128 neurons. The output layer employs a softmax activation function to generate probability estimates for the three target classes.

Although the custom CNN provided a relatively lightweight solution with low computational requirements, the project also explored the use of transfer learning techniques. Transfer learning has become a widely adopted strategy in medical image analysis because it enables models to leverage feature representations learned from large-scale image datasets. Instead of training a deep architecture entirely from scratch, a pretrained backbone can be used as a feature extractor while only a small classification head is trained on the target dataset.

Three state-of-the-art architectures were evaluated during experimentation: ResNet50, DenseNet121, and EfficientNetB3. For each architecture, the pretrained backbone was initialized using publicly available weights and subsequently frozen during training. Freezing the backbone prevented updates to the pretrained parameters and allowed the network to function as a fixed feature extractor. A Global Average Pooling layer was then applied to aggregate spatial information into a compact feature vector, followed by a fully connected layer containing 128 neurons and a final softmax classification layer producing probabilities for the three routing classes.

The implementation of transfer learning significantly reduced training complexity while benefiting from the rich visual representations learned by these networks during large-scale pretraining. However, preliminary experiments revealed that different architectures exhibited varying levels of sensitivity to optimization parameters. Consequently, additional effort was dedicated to hyperparameter tuning in order to maximize classification performance.

One of the most important optimization parameters investigated during experimentation was the learning rate. The learning rate directly controls the magnitude of weight updates during optimization and can substantially influence convergence behavior. A learning rate that is too large may cause unstable training, whereas an excessively small learning rate may prevent efficient convergence. Since each pretrained architecture possesses unique characteristics and parameter distributions, a single learning rate was not expected to perform optimally across all models.

To address this issue, architecture-specific learning rates were introduced. ResNet50 and DenseNet121 were trained using a learning rate of 10^{-2}, while EfficientNetB3 was trained using a learning rate of 10^{-4}. These values were selected based on empirical experimentation and repeated evaluation of training behavior. The custom CNN utilized the default Adam optimization configuration due to its comparatively smaller parameter space. This targeted hyperparameter tuning strategy improved convergence stability and ensured that each architecture was evaluated under favorable training conditions.

All routing models were trained using the Adam optimization algorithm and sparse categorical cross-entropy loss. The use of sparse categorical cross-entropy was appropriate because the routing task involves mutually exclusive classes, where each image belongs to exactly one category. Model performance was monitored using classification accuracy, which provided a direct measure of the model's ability to correctly identify image modalities.

To ensure a fair and consistent evaluation process, all experiments were conducted using identical dataset partitions. The aggregated dataset was divided into training, validation, and testing subsets according to a ratio of 70%, 15%, and 15%, respectively. Training was performed using a batch size of 8 and a total of 10 epochs. Maintaining a consistent experimental protocol allowed the performance differences observed between models to be attributed primarily to architectural characteristics rather than variations in the training procedure.

Following extensive experimentation, DenseNet121 was selected as the final routing model incorporated into the proposed framework. The architecture demonstrated the most favorable balance between classification performance, convergence behavior, and generalization capability. DenseNet121's dense connectivity pattern enables efficient feature reuse and improved gradient propagation, characteristics that proved particularly beneficial for distinguishing between imaging modalities exhibiting substantial structural differences. Furthermore, the architecture achieved strong performance while maintaining a relatively efficient parameter utilization strategy compared to alternative deep learning models.

The final routing model therefore consists of a DenseNet121 feature extraction backbone followed by a Global Average Pooling layer, a fully connected layer with 128 neurons, and a three-class softmax output layer. Once deployed within the complete framework, the model receives an input image, determines whether the image corresponds to a chest X-ray, a CT scan, or an out-of-distribution sample, and subsequently forwards the image to the appropriate downstream component. This routing decision constitutes the first stage of the proposed hierarchical classification pipeline and forms the foundation upon which all subsequent diagnostic analyses are performed.

3.4 Chest Disease Classification Model

This section presents one of the downstream diagnostic models developed within the project, focusing on automated multi-class classification of chest diseases from chest X-ray (CXR) images. The model is designed as a robust and interpretable pipeline that integrates standardized image preprocessing, lung region isolation, and deep learning–based classification using transfer learning. Emphasis is placed on architectural design choices and methodological foundations rather than performance evaluation, which is discussed in a later chapter.

3.4.1 Motivation

Deep learning has demonstrated strong potential in medical image analysis, yet several challenges remain, including variability in image acquisition, class imbalance, and the need for explainable decision-making. The proposed model addresses these issues through a multi-stage pipeline that enhances image quality, isolates clinically relevant lung regions, and applies a transfer learning–based classifier augmented with explainability techniques.

3.4.2 Problem Statement

The objective of this model is to perform multi-class classification of chest X-ray images into the following diagnostic categories:

$$\mathcal{C} = \{\text{NORMAL}, \text{PNEUMONIA}, \text{COVID-19}, \text{TUBERCULOSIS}\}.$$

Given an input chest X-ray image $I \in \mathbb{R}^{H \times W}$, the task is to learn a mapping

$$f_\theta : I \rightarrow \mathbf{p},$$

where $\mathbf{p} \in \mathbb{R}^{|\mathcal{C}|}$ is a probability distribution over disease classes and θ denotes the model parameters.

To improve robustness and clinical relevance, the classification task is constrained to lung parenchyma regions extracted via an explicit segmentation stage. Additionally, the model must provide interpretable outputs to support clinical trust, motivating the use of gradient-based visualization techniques.

3.4.3 Model Architecture

The diagnostic model is structured as a multi-stage pipeline, illustrated conceptually in Figure 3.11. The pipeline consists of three main components: image preprocessing, lung segmentation, and disease classification.

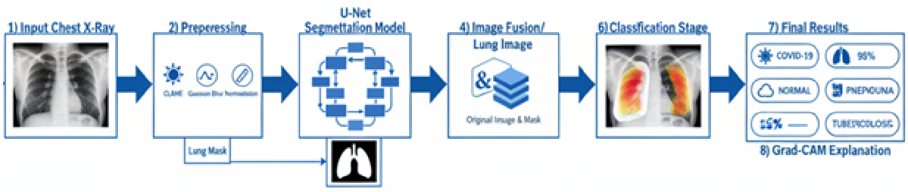


Figure 3.11: High-level overview of the multi-stage chest disease classification pipeline.

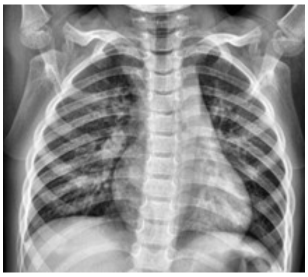
Each chest X-ray image undergoes a standardized preprocessing pipeline to enhance diagnostically relevant features. Images are first converted to grayscale and enhanced using Contrast Limited Adaptive Histogram Equalization (CLAHE). Noise introduced during enhancement is mitigated using Gaussian smoothing with a 5×5 kernel.

Pixel intensities are then normalized using Min–Max normalization: $I_{\text{norm}} = \frac{I - I_{\min}}{I_{\max} - I_{\min}}$ where $I_{\min}$ and $I_{\max}$ represent the minimum and maximum pixel values of the image.

Finally, images are converted to RGB format to ensure compatibility with the ResNet50 model. For a discussion of the ResNet50 model, refer to Subsection 3.2.1.

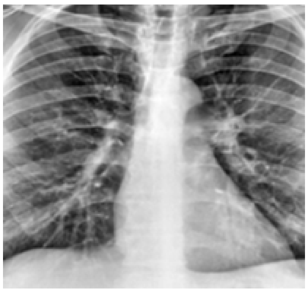
Figure 3.12 illustrates the effect of the preprocessing pipeline on representative CXR images.

(a) Original Pneumonia CXR



(b) Preprocessed Pneumonia CXR

(c) Original Pneumonia CXR



(d) Preprocessed Pneumonia CXR

Figure 3.12: Chest X-ray images before and after preprocessing, highlighting improved contrast and reduced noise.

To ensure that classification focuses exclusively on clinically relevant regions, a U-Net–based segmentation model is employed to extract lung masks. The U-Net architecture follows an encoder–decoder structure with skip connections that preserve spatial information, making it well suited for biomedical image segmentation. The U-Net architecture used is shown in Figure 3.13

The segmentation network outputs a binary lung mask $M \in \{0,1\}^{H\times W}$, which is applied to the preprocessed image using element-wise multiplication:

$$I_{\text{lung}} = I_{\text{norm}} \odot M.$$

Training of the segmentation model is guided by the Dice Loss, derived from the Dice–Sørensen Coefficient (DSC):

$$\text{DSC} = \frac{2|X \cap Y|}{|X| + |Y|},$$

where X and Y denote the predicted and ground-truth masks, respectively. The loss function is defined as

$$\mathcal{L}_{\text{Dice}} = 1 - \text{DSC},$$

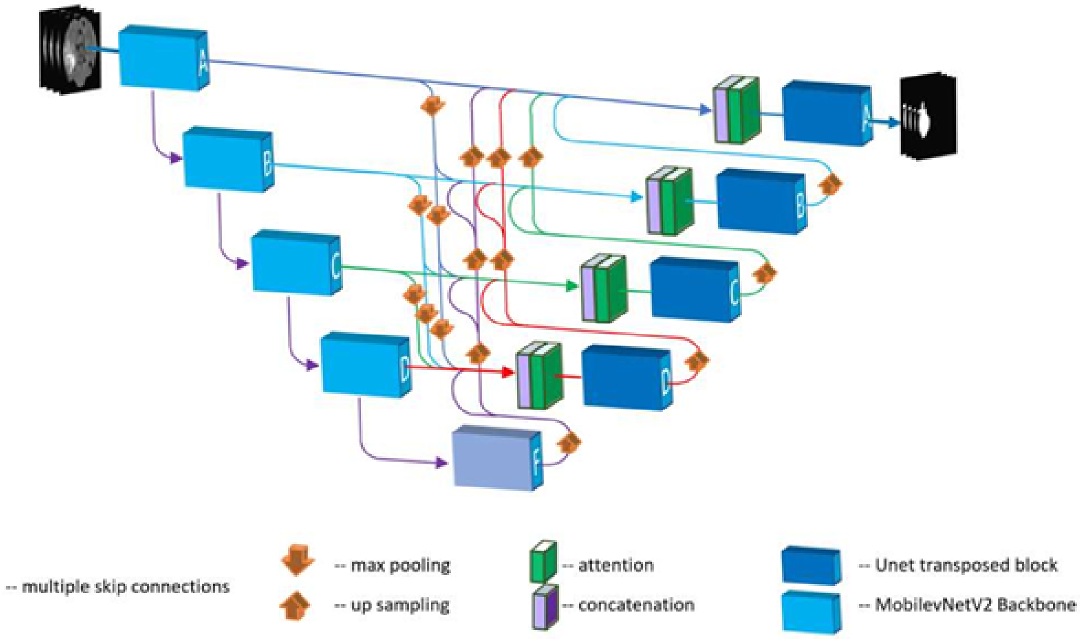


Figure 3.13: The U-Net architecture, illustrating the encoder-decoder structure with skip connections that merge high-resolution features from the contracting path with upsampled features in the expansive path. This design is highly effective for precise medical image segmentation.

which is particularly effective in handling class imbalance between lung and background pixels. The result of using segmentation on CXR images is illustrated in Figure 3.14

The final classification stage employs a ResNet50-based transfer learning model. ResNet architectures leverage residual connections to facilitate gradient flow and enable the training of deep networks. As the ResNet family is discussed in detail earlier in this book, only a brief overview is provided here.

A ResNet50 model pre-trained on ImageNet is used as a feature extractor, with early layers frozen to preserve general visual representations. A custom classification head is appended, consisting of global average pooling, fully connected layers with ReLU activation, dropout for regularization, and a final softmax layer producing class probabilities.

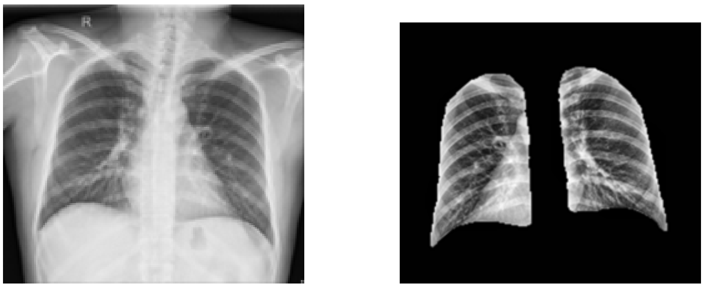
The model is trained using Sparse Categorical Cross-Entropy loss:

$$\mathcal{L}_{\text{CE}} = -\sum_{i=1}^{N} \log p_{i,c},$$

where $p_{i,c}$ denotes the predicted probability of the true class c for sample i. To address class imbalance, class-weighted loss scaling is applied during training.

To enhance interpretability, Gradient-weighted Class Activation Mapping (Grad-CAM) is employed to visualize the regions most influential to model predictions, an example is shown in Figure 3.15. Grad-CAM computes the gradients of the target class score with respect to the feature maps of the final convolutional layer and produces a coarse localization map highlighting salient regions.

These heatmaps are overlaid on the original chest X-ray images, enabling qualitative assessment of whether the model attends to clinically meaningful lung regions.



(a) Preprocessed CXR Image

(b) Segmented Lung Image after Mask Application

Figure 3.14: Application of the U-Net segmentation mask. The original preprocessed Figure 3.14a is processed to produce the final segmented lung Figure 3.14b, which serves as the input for the classification stage

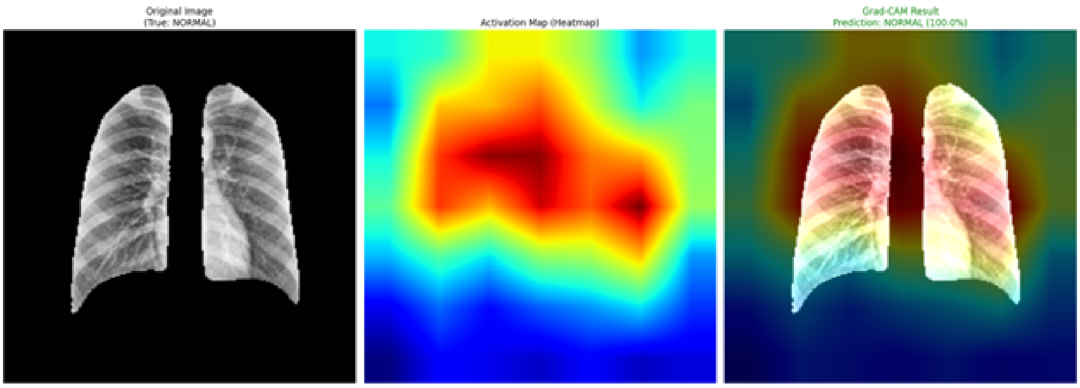


Figure 3.15: Grad-CAM visualizations highlighting regions contributing to model predictions.

3.5 Lung Cancer Detection Model

This section presents the lung cancer and tuberculosis detection model developed within the project, focusing on automated localization and classification of pulmonary abnormalities from chest X-ray (CXR) images[cite: 1, 2]. The model is designed as an end-to-end object detection pipeline that integrates binary mask conversion, bounding box generation, and deep learning–based detection using the YOLOv8 architecture[cite: 3]. Emphasis is placed on the data preparation strategy and architectural design choices that underpin the detection framework[cite: 4].

3.5.1 Motivation

Object detection frameworks offer distinct advantages over image-level classification for medical imaging tasks, as they provide explicit spatial localization of pathological regions rather than a single diagnostic label[cite: 5, 6]. For lung cancer and tuberculosis screening, identifying the precise location and extent of pulmonary lesions or infiltrates is clinically essential, as it directly informs radiological assessment and subsequent clinical decision-making[cite: 7]. The proposed model addresses this requirement by reformulating the diagnostic task as a bounding box regression and classification problem, enabling simultaneous localization and identification of disease regions within a single unified forward pass[cite: 8].

3.5.2 Problem Statement

The objective of this model is to perform multi-class detection of pulmonary abnormalities in chest X-ray images, identifying and localizing instances belonging to the following diagnostic categories[cite: 9, 10]:

$$C = \{\text{tumor}, \text{tuberculosis}\}[cite : 11] \tag{3.11}$$

Given an input chest X-ray image $I \in \mathbb{R}^{H \times W}$, the task is to learn a mapping[cite: 12]:

$$f_\theta : I \rightarrow \{(b_i, c_i, s_i)\}_{i=1\ldots N}[cite : 13] \tag{3.12}$$

where each tuple (b_i, c_i, s_i) denotes the predicted bounding box coordinates, class label, and confidence score for the i-th detected region, and θ represents the model parameters[cite: 14].

To bridge the gap between pixel-level segmentation annotations (binary masks) and the bounding box representation required by the detection framework, an explicit mask-to-box conversion stage is incorporated as the first step of the pipeline[cite: 15]. Each connected foreground component in the binary annotation mask is converted into a normalized YOLO-format bounding box, enabling the use of richly annotated mask datasets for detection training without manual re-labeling[cite: 16].

3.5.3 Model Architecture

The detection model is structured as a three-stage pipeline consisting of: (1) mask-to-bounding-box conversion and dataset preparation, (2) YOLO-format data organization and configuration, and (3) YOLOv8-based detection model training and evaluation[cite: 17, 18]. Each stage is described in detail below[cite: 19].

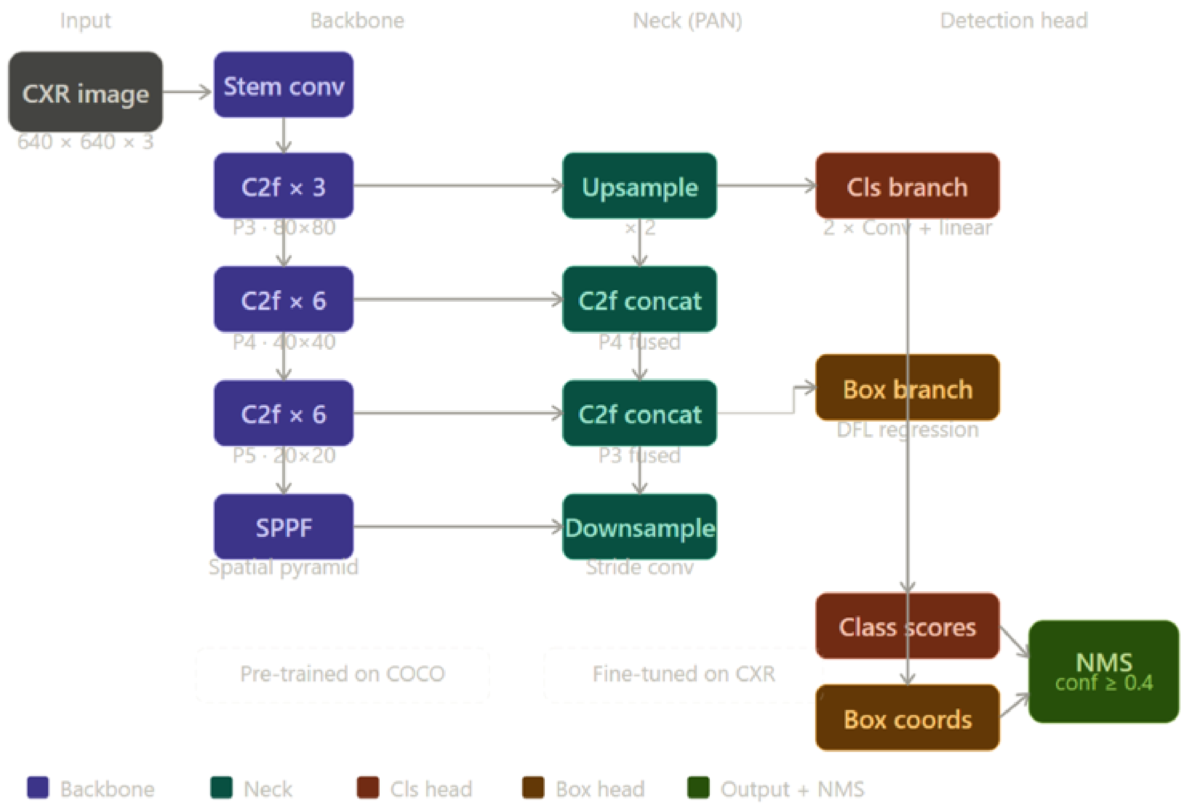


Figure 3.16: YOLOv8 architecture pipeline for automated localization and classification of pulmonary abnormalities.

Mask-to-Bounding-Box Conversion

Annotation masks for pulmonary lesions are provided as binary PNG images in which foreground pixels (intensity > 127) correspond to pathological regions[cite: 20, 21]. To convert these masks into YOLO-format bounding box annotations, connected component analysis is applied using external contour detection[cite: 22]. For each detected contour, an axis-aligned bounding box is computed and normalized with respect to the image dimensions, yielding center coordinates and extent values in the range $[0, 1]$[cite: 23].

Formally, for a contour with pixel-space bounding rectangle (x, y, w, h) within an image of width W and height H, the normalized YOLO representation is[cite: 24]:

$$c_x = \frac{x + w/2}{W}, \quad c_y = \frac{y + h/2}{H}, \quad n_w = \frac{w}{W}, \quad n_h = \frac{h}{H}[cite: 25] \tag{3.13}$$

Contours enclosing fewer than 5 pixels in either dimension are discarded as noise prior to annotation generation[cite: 26]. All foreground components within a given mask are assigned the class index corresponding to the image-level label (tumor or tuberculosis), since mask annotations are provided per diagnostic category[cite: 27].

Dataset Preparation and Splitting

Following bounding box conversion, the dataset is partitioned into training, validation, and test subsets using stratified random splitting[cite: 28, 29]. The default split proportions are 75% training, 15% validation, and 10% test, with random seed fixed at 42 to ensure reproducibility[cite: 30]. Image files and their corresponding YOLO label files are organized into the standard YOLO directory structure, with separate image and label folders for each split[cite: 31]. A YAML configuration file is generated automatically to specify dataset paths, the number of classes, and class names, serving as the interface between the prepared dataset and the YOLOv8 training framework[cite: 32].

YOLOv8 Detection Architecture

Object detection is performed using YOLOv8, the latest generation of the You Only Look Once (YOLO) family of single-stage detectors[cite: 33, 34]. YOLOv8 follows an anchor-free design and adopts a CSPDarknet backbone for feature extraction, a Path Aggregation Network (PAN) neck for multi-scale feature fusion, and a decoupled detection head that separately predicts classification scores and bounding box offsets[cite: 35]. This design enables real-time inference while maintaining competitive localization accuracy[cite: 36].

A YOLOv8n (nano) variant is employed as the base model, pre-trained on the COCO dataset, and fine-tuned on the prepared CXR detection dataset[cite: 37]. The model is trained for 50 epochs with an input resolution of 640×640 pixels and a batch size of 16[cite: 38]. Transfer learning is leveraged by initializing all network weights from the COCO pre-trained checkpoint, with all layers unfrozen to allow domain-specific adaptation[cite: 38]. The training objective combines binary cross-entropy loss for classification, distribution focal loss (DFL) for bounding box regression, and a complete intersection-over-union (CIoU) penalty term to improve localization precision[cite: 39]:

$$L = \lambda_{\text{cls}} \cdot L_{\text{cls}} + \lambda_{\text{box}} \cdot L_{\text{box}} + \lambda_{\text{dfl}} \cdot L_{\text{DFL}}[cite : 40] \tag{3.14}$$

where λ_{cls}, λ_{box}, and λ_{dfl} are scalar weighting coefficients that balance the contribution of each loss component during optimization[cite: 41].

After training, the best-performing checkpoint, selected based on validation mean Average Precision at IoU threshold 0.50 (mAP50), is retained for evaluation[cite: 42]. Model performance is assessed on both the held-out validation and test splits, reporting mAP50, mAP50–95, precision, and recall as primary evaluation metrics[cite: 43]. Qualitative evaluation is conducted by visually inspecting annotated predictions on test set images, examining the spatial correspondence between predicted bounding boxes and known pathological regions[cite: 44].

3.5.4 Interpretability Considerations

Unlike classification-only models that require post-hoc gradient visualization to identify diagnostically relevant regions, the detection framework provides explicit spatial outputs in the form of bounding boxes and associated confidence scores[cite: 45, 46]. Each prediction directly encodes the model's estimate of where a pathological region is located

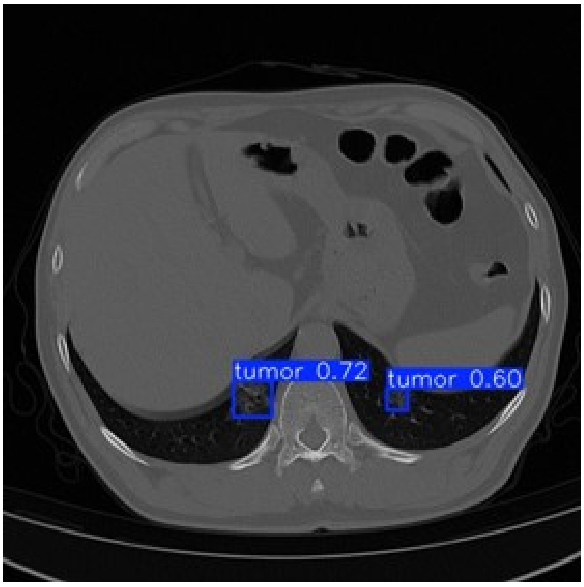


Figure 3.17: Qualitative evaluation showing tumor localization (confidence scores 0.72 and 0.60) on an axial CT slice.

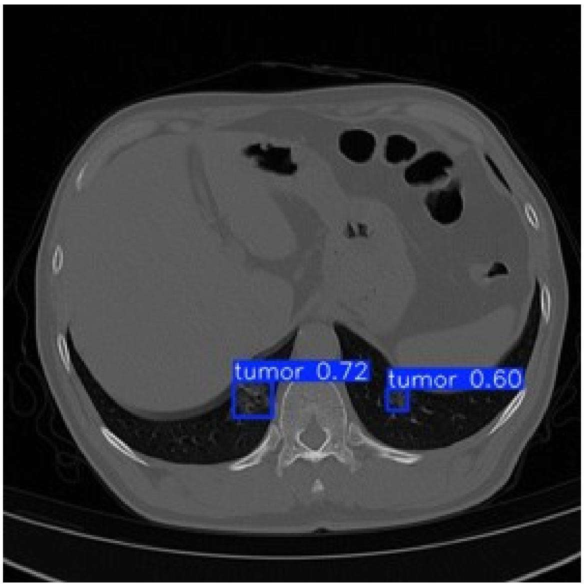


Figure 3.18: Qualitative evaluation showing tumor localization (confidence score 0.87) on an axial CT slice.

and how confident it is in that localization[cite: 47]. This built-in spatial explainability reduces the reliance on auxiliary interpretability methods and aligns more naturally with clinical reporting conventions, in which radiologists annotate specific anatomical regions rather than assigning image-level labels[cite: 48]. Confidence thresholding ($\tau = 0.4$) and non-maximum suppression with an IoU threshold of 0.6 are applied at inference time to filter low-quality or redundant detections, producing a concise set of high-confidence predictions per image[cite: 49].

3.5.5 Implementation of the Lung Cancer Detection Model

The second major component implemented within the proposed framework is the lung lesion detection model. Unlike conventional image classification systems that only determine whether an abnormality is present, the proposed approach performs object detection, enabling simultaneous identification and localization of suspicious pulmonary lesions within CT images. The model predicts both the spatial coordinates of lesion regions and the confidence associated with each prediction, thereby providing clinically meaningful visual outputs that can assist radiologists during image interpretation.

The detection pipeline was developed using the YOLOv8 architecture, which belongs to the family of single-stage object detectors. Single-stage detectors perform localization and classification simultaneously within a unified network, eliminating the need for separate region proposal and classification stages. This design significantly reduces computational complexity and inference time while maintaining competitive detection performance.

The complete implementation workflow consists of five primary stages:

1. Dataset preprocessing

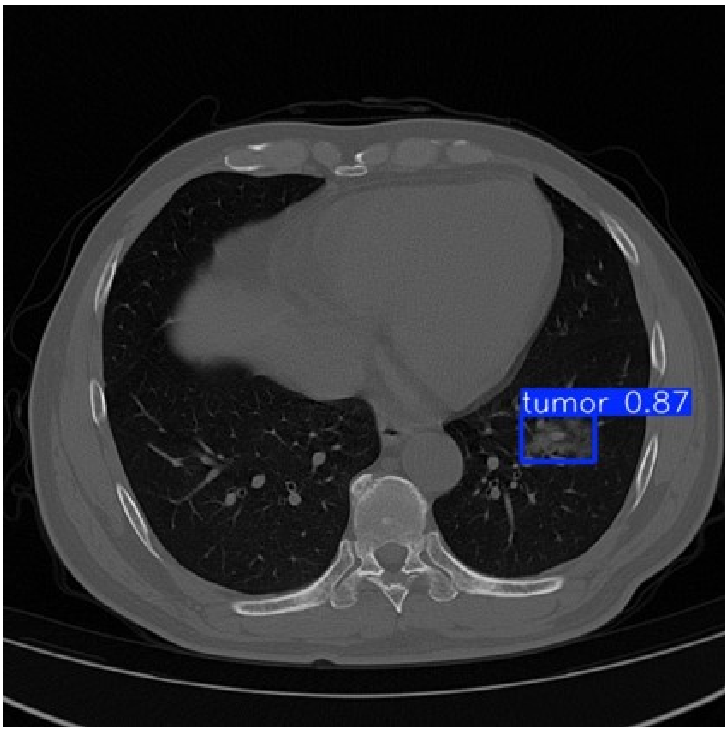


Figure 3.19: Example of an interpretable YOLO detection result on a chest CT image. The model localizes a suspected lesion using a bounding box and reports a confidence score of 0.87. Such spatial predictions provide inherent interpretability by explicitly identifying the anatomical region associated with the model's decision, reducing the need for post-hoc visualization techniques.

2. Annotation generation
3. Dataset partitioning
4. YOLOv8 model training
5. Performance evaluation and visualization

Dataset Preparation and Annotation Generation

The original dataset contains CT images together with binary segmentation masks that identify lesion regions. Although segmentation masks provide pixel-level information regarding lesion boundaries, the YOLO detection framework requires object-level annotations represented by bounding boxes. Consequently, a dedicated annotation generation pipeline was implemented to automatically convert segmentation masks into YOLO-compatible labels.

For every image-mask pair, the following operations were performed:

Step 1: Mask Loading Each binary mask was loaded as a grayscale image where:

- Pixel value 0 represented background tissue.
- Pixel value 255 represented lesion pixels.

The masks were subsequently thresholded to ensure binary separation between foreground and background regions.

Step 2: Contour Extraction The OpenCV contour detection algorithm was applied to identify connected lesion regions:

```
contours, _ = cv2.findContours(
mask,
cv2.RETR_EXTERNAL,
cv2.CHAIN_APPROX_SIMPLE
)
```

Only external contours were retained in order to avoid duplicate detections resulting from nested structures. The contour extraction stage transformed pixel-level lesion masks into geometrically defined lesion candidates.

Step 3: Bounding Box Generation For each detected contour, the minimum enclosing rectangle was computed:

```
x, y, w, h = cv2.boundingRect(contour)
```

where x, y represent the upper-left corner, w represents width, and h represents height. These coordinates define the lesion localization region used by the detector during training.

Step 4: YOLO Coordinate Conversion YOLO requires normalized coordinates expressed relative to image dimensions. The generated bounding boxes were therefore converted using:

$$x_{\text{center}} = \frac{x + w/2}{W} \tag{3.15}$$

$$y_{\text{center}} = \frac{y + h/2}{H} \tag{3.16}$$

$$w_{\text{norm}} = \frac{w}{W}, \quad h_{\text{norm}} = \frac{h}{H} \tag{3.17}$$

where W = image width and H = image height. The resulting annotation was stored using YOLO format:

```
class_id x_center y_center width height
```

Since only one lesion category was considered, all annotations were assigned `class_id` = 0.

Step 5: Label File Generation For each image (e.g., `image_001.png`), a corresponding annotation file was created (`image_001.txt`) containing all lesion bounding boxes present in the image. This automated procedure eliminated the need for manual annotation and ensured consistency throughout the dataset.

3.5.6 Dataset Organization

Following annotation generation, the dataset was reorganized according to the standard YOLO directory structure:

```
dataset/
|-- train/
|   |-- images/
|   `-- labels/
|-- valid/
|   |-- images/
|   `-- labels/
`-- test/
|-- images/
`-- labels/
```

Maintaining this structure allows direct integration with the Ultralytics YOLO training framework. A YAML configuration file was subsequently generated containing:

```
path: dataset
train: train/images
val: valid/images
test: test/images
nc: 1
names:
0: tumor
```

The YAML file serves as the interface between the dataset and the YOLO training engine.

YOLOv8 Detection Architecture

The proposed detection model employs YOLOv8 as the underlying deep learning architecture. The network consists of three major components:

- **Backbone:** The backbone is responsible for hierarchical feature extraction. Low-level layers capture edges, textures, and intensity gradients, while deeper layers learn lesion morphology, shape characteristics, and contextual lung structures.
- **Neck:** The neck aggregates information from multiple feature scales using Feature Pyramid Network (FPN) and Path Aggregation Network (PAN) mechanisms. This multi-scale representation is particularly important in lung lesion detection because lesions may vary considerably in size. Small nodules and large masses must both be represented effectively within the feature hierarchy.

- **Detection Head:** The detection head predicts bounding box coordinates, object confidence scores, and class probabilities for each candidate lesion. Unlike anchor-based detectors, YOLOv8 utilizes an anchor-free detection mechanism that simplifies prediction and improves localization accuracy.

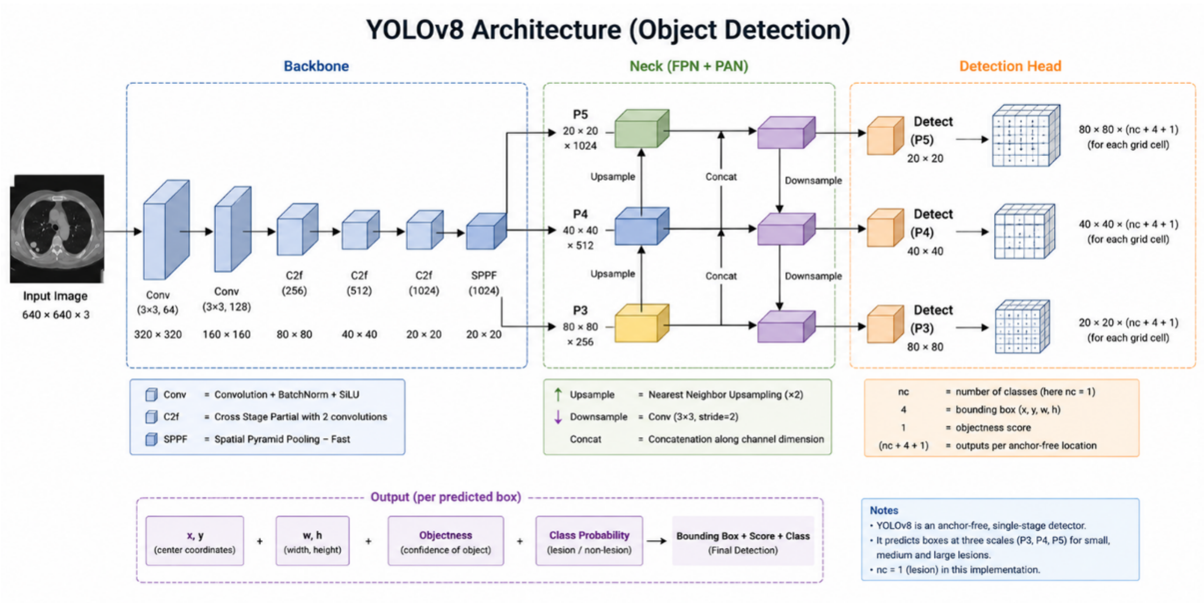


Figure 3.20: Detailed architectural diagram of the YOLOv8 object detection framework, illustrating the Backbone, Neck (FPN + PAN), and Detection Head stages for multi-scale prediction.

Transfer Learning Strategy

Training a deep object detector entirely from scratch typically requires extremely large medical imaging datasets. To overcome this limitation, transfer learning was employed. The detector was initialized using pretrained YOLOv8 weights trained on large-scale image datasets. The pretrained model already possesses strong low-level feature extraction capabilities, including detection of edges, contours, textures, and spatial patterns. Fine-tuning these representations on lung CT images significantly accelerates convergence while reducing the risk of overfitting.

Training Configuration

Model training was performed using the Ultralytics implementation of YOLOv8. The training pipeline automatically performs image loading, resizing, batching, label matching, loss computation, and weight optimization. Input images were resized to 640×640 pixels to ensure uniform tensor dimensions throughout training.

The optimization process simultaneously minimizes:

- **Localization Loss:** Measures bounding-box accuracy.
- **Classification Loss:** Measures lesion classification accuracy.
- **Distribution Focal Loss (DFL):** Improves bounding-box regression precision.

The total loss is defined as:

$$L_{\text{total}} = L_{\text{box}} + L_{\text{cls}} + L_{\text{DFL}} \tag{3.18}$$

During training, stochastic gradient optimization updates network weights through back-propagation until convergence.

Evaluation Methodology

After training, the best-performing model checkpoint was selected automatically according to validation performance. Performance was evaluated using standard object detection metrics:

Precision Measures the proportion of predicted lesions that are truly lesions.

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{3.19}$$

Recall Measures the proportion of true lesions successfully detected.

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{3.20}$$

Mean Average Precision (mAP) The primary detection metric used for evaluation. Two variants were reported:

- mAP@50
- mAP@50–95

mAP@50 evaluates detections using a single IoU threshold of 0.50, whereas mAP@50–95 provides a more stringent evaluation across multiple thresholds.

Prediction Visualization and Model Interpretability

A key advantage of the proposed detection framework is its inherent interpretability. Each prediction generated by YOLOv8 contains bounding box coordinates, a confidence score, and a class label. The confidence score reflects the model's certainty regarding the presence of a lesion within the predicted region.

Unlike classification-only approaches that require external explainability methods such as Grad-CAM or saliency maps, the detector directly highlights suspicious anatomical regions through bounding boxes. This spatial localization capability allows clinicians to visually inspect the exact image regions influencing the model's decision, thereby improving transparency and facilitating clinical validation of automated predictions.

3.6 Federated Learning

3.6.1 Concept and Motivation

Federated learning (FL) is a distributed machine learning paradigm in which a model is trained collaboratively across multiple independent nodes without any raw data leaving its source institution [42]. Rather than centralizing training samples on a shared server, FL brings the computation to the data: each participating node trains a local model on its private dataset and shares only the resulting model parameters with a central aggregation server, which combines the updates into an improved global model. This cycle repeats for a defined number of communication rounds until the global model converges.

The motivation for adopting FL in this work is rooted in the fundamental tension between data-driven model development and patient privacy. Healthcare institutions collect large volumes of annotated medical images, yet regulatory frameworks — including the Health Insurance Portability and Accountability Act (HIPAA) in the United States and the General Data Protection Regulation (GDPR) in the European Union — impose strict constraints on cross-institutional data sharing. Negotiating data transfer agreements is costly, time-consuming, and in many cases legally impractical. FL resolves this tension by enabling institutions to contribute to a shared model without relinquishing data custody, making large-scale collaborative medical AI both legally feasible and technically practical.

3.6.2 Federated Learning in Medical Imaging

The adoption of FL in the medical domain has grown substantially since its introduction, driven by the recognition that many clinical datasets are too small at any single institution to train robust deep learning models, while the combined data across institutions would be sufficient [43]. FL directly addresses this data insufficiency problem without requiring physical data consolidation.

In medical imaging specifically, FL has been applied across a broad range of modalities and anatomical targets. Early work by Sheller *et al.* demonstrated FL for brain tumor segmentation from MRI scans across multiple institutions, achieving performance comparable to centralized training without sharing any patient data [43]. Subsequent studies have extended FL to chest radiograph classification, diabetic retinopathy screening from fundus photography, histopathology slide analysis, and cardiac structure segmentation from echocardiography, among others [44].

This breadth of applicability is a key motivation for the design of the proposed system. Rather than building a solution tied to a specific model architecture or anatomical region, the FL framework developed in this work is designed to be **model-agnostic** and **organ-agnostic**: any classification or segmentation backbone can be substituted as the local model, and the training data can comprise images from any body region or imaging modality. The federated coordination layer — covering client management, weight aggregation, and communication — remains unchanged regardless of the underlying clinical task.

3.6.3 The Federated Averaging Algorithm

The aggregation algorithm that underpins the proposed system is Federated Averaging (FedAvg), introduced by McMahan *et al.* [42] and the most widely adopted FL optimization method to date. FedAvg operates in iterative communication rounds. Let K denote the number of participating clients, $\mathcal{D}_k$ the private dataset of client k, $n_k = |\mathcal{D}_k|$ its size, and $n = \sum_{k=1}^{K} n_k$ the total number of training samples across all clients. The global model at round t is parameterized by weight vector $\mathbf{w}_t$. Each round consists of three phases:

(1) Broadcast. The server distributes the current global model weights to all K clients:

$$\mathbf{w}_t \longrightarrow \{\text{client}_1, \dots, \text{client}_K\}. \tag{3.21}$$

(2) Local Training. Each client k independently minimizes its local empirical risk $\mathcal{L}_k$ over E local epochs using a gradient- based optimizer, producing updated local weights:

$$\mathbf{w}_t^k = \mathbf{w}_t - \eta \nabla \mathcal{L}_k(\mathbf{w}_t), \tag{3.22}$$

where η is the local learning rate and $\mathcal{L}_k$ is the task-specific loss function evaluated over the client's local dataset:

$$\mathcal{L}_k(\mathbf{w}) = \frac{1}{n_k} \sum_{(x_i, y_i) \in \mathcal{D}_k} \ell(f_{\mathbf{w}}(x_i), y_i). \tag{3.23}$$

(3) Aggregation. The server collects the locally updated weights from all clients and computes the new global model as a weighted average proportional to each client's dataset size:

$$\mathbf{w}_{t+1} = \sum_{k=1}^{K} \frac{n_k}{n} \mathbf{w}_t^k. \tag{3.24}$$

This process minimizes the global objective:

$$F(\mathbf{w}) = \sum_{k=1}^{K} \frac{n_k}{n} \mathcal{L}_k(\mathbf{w}). \tag{3.25}$$

Equation (3.24) assigns greater influence to clients with larger datasets, which is appropriate in the medical setting where institutional dataset sizes vary considerably. Clients contributing more annotated images should exert proportionally greater influence on the global model.

3.6.4 Key Parameters of a Federated Learning System

Configuring a federated learning system involves a distinct set of parameters beyond those of conventional deep learning. These parameters govern the coordination between server and clients and directly affect convergence speed, model quality, communication cost, and privacy. A clear understanding of each parameter is essential for adapting the system to different clinical tasks. Table 3.2 provides an overview of the critical parameters and their implications.

Table 3.2: Key Federated Learning Parameters and Their Clinical Implications

Parameter	**Definition**	**Implication**
Number of clients (K)	Total participating institutions	More clients improve data diversity and model generalization but increase coordination overhead.
Communication rounds (T)	Number of server–client training cycles	More rounds improve accuracy at the cost of higher communication volume. Convergence is typically monitored to select T.
Client fraction (C)	Proportion of clients selected per round ($C \in (0, 1]$)	$C = 1.0$ uses all clients every round (synchronous FL). $C < 1.0$ enables asynchronous, scalable training.
Local epochs (E)	Training epochs each client performs before returning weights	Higher E reduces communication rounds but risks *client drift* — divergence of local models due to non-IID data [45].
Local batch size (B)	Mini-batch size for local gradient computation	Affects local convergence speed, memory footprint, and gradient privacy (smaller batches increase reconstruction attack risk).
Local learning rate (η)	Step size for local gradient updates	Must be tuned per model architecture; too large a value exacerbates client drift in non-IID settings.
Aggregation strategy	Algorithm used to combine client updates (e.g., FedAvg, FedProx, FedNova)	Choice depends on data heterogeneity; FedAvg is the standard baseline; FedProx adds a proximal term to limit client drift.
Data partitioning scheme	How training data is divided across clients (IID vs. non-IID)	Non-IID partitions (realistic in healthcare) slow convergence and reduce final accuracy compared to IID partitions.
Minimum clients for aggregation	Minimum number of client updates required before the server aggregates	Prevents the global model from being updated on insufficient information; critical for stability.

Among these, the number of local epochs E and the aggregation strategy warrant particular attention in the medical imaging context. Increasing E reduces the number of costly communication rounds, which is beneficial in healthcare networks where bandwidth and uptime may be limited. However, when client datasets are statistically heterogeneous — a common condition across hospitals using different scanners, populations, or labeling protocols — high E causes the local models to overfit to their local distributions and diverge from the global optimum, a phenomenon known as *client drift* [45]. The selection of E therefore requires careful empirical tuning for each deployment scenario.

3.6.5 Privacy and Security

The primary privacy guarantee of federated learning is that raw data never leaves its source institution. The server observes only model weight tensors, which are orders of magnitude smaller than the original imaging data and contain no explicit patient identifiers. This stands in contrast to centralized approaches, where a single data breach exposes the entire training corpus. Table 3.3 summarizes the key differences.

Table 3.3: Privacy and Security: Centralized vs. Federated Learning

Property	**Centralized**	**Federated**
Raw data leaves institution	Yes	No
Single point of data breach	Yes	No
HIPAA / GDPR compliance	Difficult	Easier
Shared artifact	Full dataset	Model weights
Supports differential privacy	N/A	Yes
Resilient to server breach	No	Partial

It is important to acknowledge that gradient sharing is not unconditionally private. Gradient inversion attacks [46] have demonstrated that an adversarial server can reconstruct approximate training images from shared weight updates, particularly at small batch sizes. Two complementary mitigations are incorporated into the design of the proposed system:

Batch size selection. Using a sufficiently large mini-batch (e.g., $B \geq 32$) aggregates gradient information across many samples, significantly reducing the fidelity of any reconstruction attempt.

Differential Privacy (DP). For deployments requiring formal privacy guarantees, the system supports (ε, δ)-DP [47] by adding calibrated Gaussian noise to client updates before transmission:

$$\tilde{\mathbf{w}}_t^k = \mathbf{w}_t^k + \mathcal{N}\Big(0,\ \sigma^2 C^2 \mathbf{I}\Big), \tag{3.26}$$

where C is the gradient clipping threshold and σ is the noise multiplier. The privacy budget ε quantifies the worst-case privacy loss: smaller ε implies stronger privacy at the cost of model utility.

3.6.6 What Makes the Proposed System Distinctive

While federated learning has been applied in various medical imaging studies, the majority of existing implementations are designed around a fixed model architecture and a specific anatomical target, limiting their transferability to new clinical tasks. The proposed framework differentiates itself through the following design principles:

Model Agnosticism. The federated coordination layer — encompassing client management, weight serialization, aggregation, and communication — is fully decoupled from the local model definition. Any differentiable model that can be expressed as a parameter vector (e.g., CNNs for classification, U-Net variants for segmentation, Vision Transformers) can be substituted as the local model without modifying the federated infrastructure. This makes the system immediately applicable to new imaging tasks simply by swapping the model definition on the client side.

Staged Data Integration. The system incorporates a staged data feeding protocol that progressively increases each client's active training subset across federated rounds. This design reflects the clinical reality that expert-annotated medical imaging datasets grow incrementally over time as radiologists label new cases. Rather than requiring a fully annotated dataset before training begins, the system accommodates continuous data growth, enabling earlier model deployment and iterative improvement as annotation progresses.

Containerized Reproducibility. Each participating node is deployed as an isolated Docker container, ensuring that the federated environment is fully reproducible across different hardware configurations and operating systems. The container-based architecture also simplifies scaling: adding a new institution requires only the deployment of an additional client container with the appropriate data volume mount, with no changes to the server or existing clients.

Framework Choice. The system is implemented using the Flower (flwr) framework [48], selected for its support of multiple ML backends (PyTorch, TensorFlow, JAX), its clean abstraction of the client-server protocol, and its active development for healthcare FL research. The backend-agnostic design of Flower further reinforces the model-agnostic objective of the proposed system.

Collectively, these properties position the proposed FL framework as a general-purpose infrastructure for privacy-preserving collaborative medical image analysis, extensible to a wide range of clinical specialties and diagnostic tasks without architectural redesign.

3.7 Retrieval-Augmented Generation (RAG) Model

The proposed platform incorporates a Retrieval-Augmented Generation (RAG) model to provide an intelligent medical question-answering assistant capable of responding to user inquiries using trusted medical knowledge sources. Unlike conventional large language models that rely solely on their internal parametric knowledge, the RAG approach augments the language model with information retrieved from an external knowledge base at inference time. This enables the system to generate responses that are more accurate, context-aware, explainable, and grounded in domain-specific medical information.

The primary objective of integrating the RAG model within the proposed framework is to provide users with a reliable conversational assistant capable of answering questions related to chest diseases, lung cancer, medical imaging, diagnostic procedures, and other healthcare-related topics. By retrieving relevant information from curated medical documents before response generation, the system significantly reduces hallucinations and improves the factual consistency of generated answers.

3.7.1 Motivation

Although modern Large Language Models (LLMs) demonstrate remarkable natural language understanding and generation capabilities, they suffer from several limitations when deployed in healthcare environments. Most notably, they may generate inaccurate information, provide outdated medical recommendations, or produce responses unsupported by clinical evidence.

In medical applications, such limitations can negatively affect user trust and potentially lead to harmful outcomes. Consequently, relying solely on the internal knowledge of a pretrained language model is insufficient for healthcare-oriented systems.

Retrieval-Augmented Generation addresses these challenges by combining information retrieval techniques with generative language models. Instead of answering questions exclusively from learned parameters, the system first retrieves relevant medical knowledge from an external document repository and then uses this information to guide response generation. This approach ensures that answers remain grounded in verified sources while maintaining the conversational capabilities of modern language models.

3.7.2 Overview of the Proposed RAG Architecture

The implemented RAG system consists of four major components:

1. Medical Knowledge Base
2. FAISS Vector Database
3. Ollama Large Language Model
4. LangChain Orchestration Layer

The interaction between these components is illustrated in Figure 3.21.

The architecture begins when a user submits a natural language query through the platform interface. The query is forwarded to the FastAPI model backend, where LangChain orchestrates the retrieval and generation pipeline. Relevant document chunks

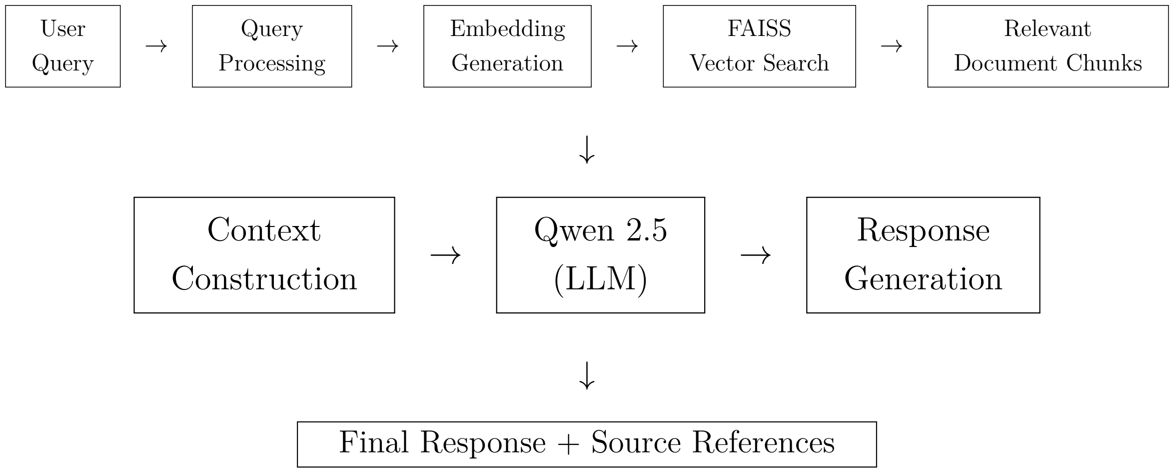


Figure 3.21: Overall architecture of the proposed Retrieval-Augmented Generation (RAG) model. The user query is transformed into vector embeddings, relevant medical knowledge is retrieved from the FAISS vector database, and the retrieved context is supplied to the Qwen 2.5 language model to generate a grounded and source-aware response.

are retrieved from the FAISS vector database and supplied as contextual information to the language model. The generated answer is then returned to the user along with references to the retrieved sources.

3.7.3 Knowledge Base Construction

The foundation of the RAG system is a curated collection of medical documents covering topics related to:

- Chest diseases
- Pneumonia
- Tuberculosis
- COVID-19
- Lung cancer
- Medical imaging
- Diagnostic procedures
- Healthcare guidelines

These documents are converted into machine-processable representations before being indexed. Since language models cannot efficiently search through large collections of raw documents during inference, each document is divided into smaller semantic chunks.

Chunking improves retrieval accuracy by allowing the system to retrieve only the most relevant portions of a document rather than entire files.

For each generated chunk, metadata is preserved, including:

- Document name
- Page number

- Source identifier
- Document category

This metadata later enables source attribution and transparency within generated responses.

3.7.4 Vector Embedding Generation

After document chunking, each text segment is transformed into a dense numerical representation known as an embedding vector.

Embeddings encode semantic meaning into a high-dimensional vector space, allowing documents with similar meanings to be located near one another mathematically.

Given a document chunk (d_i), an embedding function (E(·)) produces a vector representation $v_i = E(d_i)$ where $v_i \in \mathbb{R}^n$ and (n) represents the embedding dimensionality.

These embeddings capture semantic relationships between medical concepts and enable similarity-based retrieval during inference.

3.7.5 FAISS Vector Database

To support efficient retrieval over thousands of embedded document chunks, the system employs Facebook AI Similarity Search (FAISS).

FAISS is a high-performance vector database specifically designed for similarity search within large embedding collections. Instead of performing keyword matching, FAISS retrieves documents based on semantic similarity.

All generated embeddings are stored within the FAISS index.

During inference, a user query is embedded using the same embedding model: $q = E(x)$ where (x) denotes the user's query.

The FAISS index then searches for the (k) nearest vectors according to cosine similarity and returns the most relevant document chunks.

This retrieval process allows the system to locate contextually relevant medical information even when the exact wording differs from that found in the original documents.

3.7.6 Large Language Model Integration

The generative component of the RAG pipeline utilizes the Qwen 2.5 language model deployed locally through Ollama.

Running the model locally provides several advantages:

- Reduced operational cost
- Improved privacy

- Low latency
- Full deployment control
- Offline inference capability

The model configuration used in the implementation is summarized in Table 3.4.

Table 3.4: Configuration of the deployed language model.

Parameter	Value
LLM Engine	Ollama
Model	qwen2.5:3b
Temperature	0.5
Max Tokens	500
Keep Alive	Infinite (-1)

After retrieval, the selected document chunks are appended to the prompt supplied to the language model. Consequently, generated responses become grounded in retrieved medical knowledge rather than relying exclusively on pretrained parameters.

3.7.7 Query Processing Workflow

The complete RAG workflow implemented within the system is illustrated in Figure 3.22.

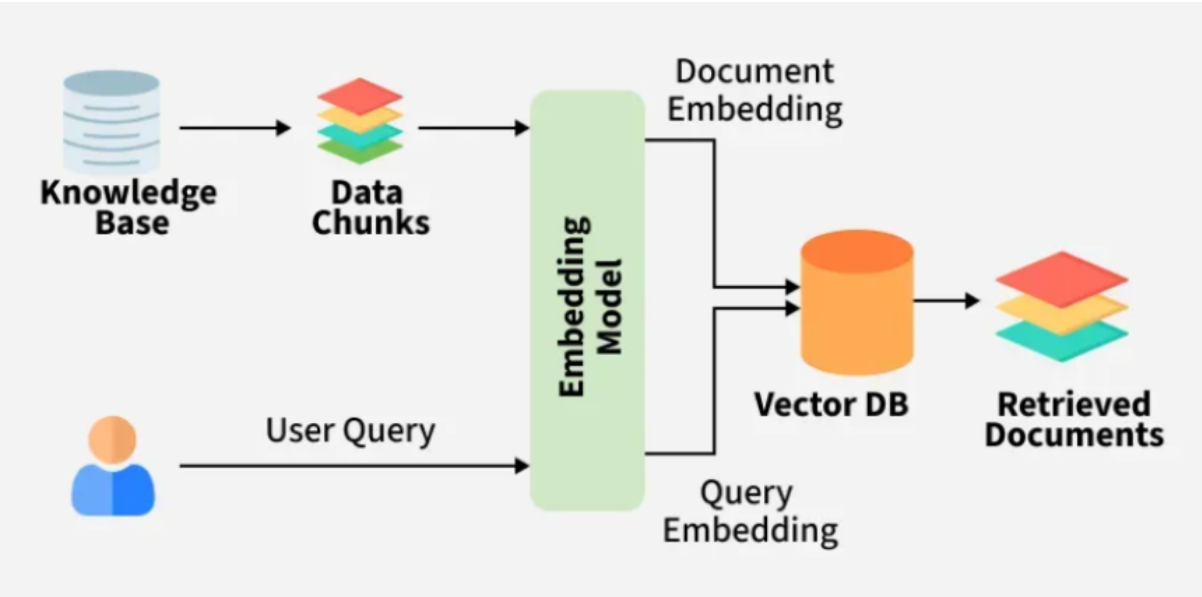


Figure 3.22: Query processing workflow of the proposed RAG system.

The workflow consists of the following steps:

1. User submits a medical question.
2. Query is forwarded to the FastAPI backend.

3. Query embedding is generated.
4. FAISS retrieves the most relevant document chunks.
5. Retrieved context is assembled.
6. Context and user query are passed to Qwen 2.5.
7. The language model generates a grounded response.
8. Source metadata is attached.
9. Final response is returned to the user.

3.7.8 Source Attribution and Explainability

One of the most important characteristics of the proposed RAG implementation is transparency.

Rather than generating answers without justification, the system preserves metadata associated with retrieved documents and returns source references alongside generated responses. This allows users to trace the origin of information and verify the supporting evidence used during answer generation. The underlying report emphasizes that retrieved documents and page references are preserved to improve transparency and user trust.

Providing source-aware responses is particularly important in healthcare applications where explainability and evidence-based recommendations are essential.

3.7.9 Advantages of the Proposed Approach

The implemented Retrieval-Augmented Generation model provides several advantages:

- Reduces hallucinations compared with standalone LLMs.
- Produces responses grounded in medical knowledge sources.
- Supports explainable and source-aware answers.
- Enables efficient retrieval through FAISS vector search.
- Preserves privacy through local deployment using Ollama.
- Can be continuously expanded by adding new medical documents.
- Integrates seamlessly with the overall X-Fed Neuroscan architecture.

Consequently, the RAG subsystem serves as an intelligent medical assistant that complements the image analysis capabilities of the platform by providing users with reliable, contextualized, and explainable healthcare information.

3.8 Platform Architecture

The system is a comprehensive web-based healthcare management system developed to streamline medical services and enhance diagnostic workflows through an integrated digital platform. The system provides a centralized environment where patients, doctors, administrators, and healthcare institutions can interact efficiently through role-based dashboards and specialized services. The platform allows patients to create accounts, manage their profiles, book appointments, track medical records, and communicate through an intelligent healthcare assistant. Doctors can manage appointments, analyze medical images, review patient information, and assign diagnostic results, while administrators oversee user management, system operations, and hospital coordination. The system also supports healthcare institutions participating in collaborative AI model development through federated learning mechanisms. Built using a modern multi-tier architecture, XFed NeuroScan combines a responsive web interface, secure backend services, cloud-based storage, and artificial intelligence modules to deliver a seamless user experience. The application emphasizes security, scalability, and usability by implementing role-based access control, secure authentication, real-time communication, and centralized healthcare workflows. By integrating healthcare management features with intelligent diagnostic services, XFed NeuroScan provides a unified platform that improves operational efficiency, supports medical decision-making, and enhances the overall experience for both healthcare providers and patients.

The platform is composed of four tightly integrated repositories:

- **Frontend:** React 18 single-pages application serving doctors, patients, and administrators.
- **Backend:** Spring Boot 3.5 REST and WebSocket API acting as the central system of record, handling authentication, business logic, user management, and communication between services.
- **Model Backend:** FastAPI-based AI service responsible for medical image analysis, chest disease classification, lung cancer detection, explainable AI heatmap generation, and a Retrieval-Augmented Generation (RAG) medical chatbot for intelligent question answering.
- **Federated-Learning Server:** Flower-based federated learning coordinator managing hospital registration, client monitoring, distributed training rounds, and global model aggregation.

Together, these components provide a complete healthcare platform that combines medical workflow management, AI-assisted diagnosis, intelligent medical assistance through RAG, and privacy-preserving federated learning across multiple healthcare institutions.

3.8.1 End-to-End System Architecture

The system follows a microservices architecture. Four independent services communicate over well-defined APIs. The frontend is the only entry point for human users; all backend services are internal.

Service Map

Table 3.6 lists all the different components and services of the proposed system.

Table 3.5: Summary of the proposed X-Fed NeuroScan framework.

Dimension	Detail
Primary Domain	Medical imaging and AI-based diagnostic systems
Key Innovation	Federated learning for privacy-preserving model improvement and collaborative model training across institutions
User Roles	Patient, Doctor, and Administrator
Imaging Modalities	Chest X-ray classification (COVID-19, Tuberculosis, Pneumonia, and Normal) and lung cancer detection from CT scans
AI Explainability	Grad-CAM heatmaps for chest disease predictions and lesion localization visualizations for lung cancer detection
Privacy Model	Raw medical images never leave hospital premises; only model parameters are exchanged during federated learning

Table 3.6: Technological components of the proposed system architecture.

Service	Technology	Port	Primary Role
Frontend	React 18, Vite, and TypeScript	3000	User interface for patients, doctors, and administrators
Backend	Spring Boot 3.5 and Java 21	8080	Authentication, business logic implementation, and API gateway services
Model Backend	FastAPI and Python 3	8000	Machine learning inference services and retrieval-augmented generation (RAG) chatbot functionality
Federated Learning Server	FastAPI, Flower, and Python 3	8001 / 50051	Federated learning coordination, client management, and hospital registry services

External Infrastructure

During system implementation, we have also relied on external components to not reinvent the wheel and leverage already existing powerful and free infrastructure, those components are listed in Table 3.7

Table 3.7: Supporting infrastructure and external services used by the proposed system.

Component	Role
Cloudflare R2 (S3-Compatible Storage)	Stores uploaded medical images, generated Grad-CAM heatmaps, lung cancer detection result overlays, and other imaging-related artifacts.
PostgreSQL	Primary relational database used to manage users, medical images, appointments, hospital information, federated learning participants, and other persistent application data.
Redis	Provides high-speed storage for refresh tokens and caching of presigned URLs to improve system performance and reduce unnecessary storage requests.
Ollama (qwen2.5:3b)	Local large language model (LLM) responsible for generating responses within the retrieval-augmented generation (RAG) chatbot component.
Gmail SMTP	Handles transactional email services including password reset requests, user invitations, account notifications, and hospital package communications.
FAISS	Vector similarity search engine used to index and retrieve relevant documents for the retrieval-augmented generation (RAG) pipeline.

Architecture Visualization

To provide a clearer understanding of the overall system structure, Figures 3.23 and 3.24 present visual representations of the proposed *X-Fed NeuroScan* framework. These diagrams complement the architectural descriptions presented throughout this chapter by illustrating the interaction between system components, data flow pathways, and deployment services. Together, they provide a comprehensive view of both the logical architecture of the framework and the technological infrastructure used to support its operation.

3.8.2 Frontend

The frontend is a React 18 single-page application built with Vite 6 and TypeScript. It serves three user roles — Patient, Doctor, and Admin — through role-gated dashboards,

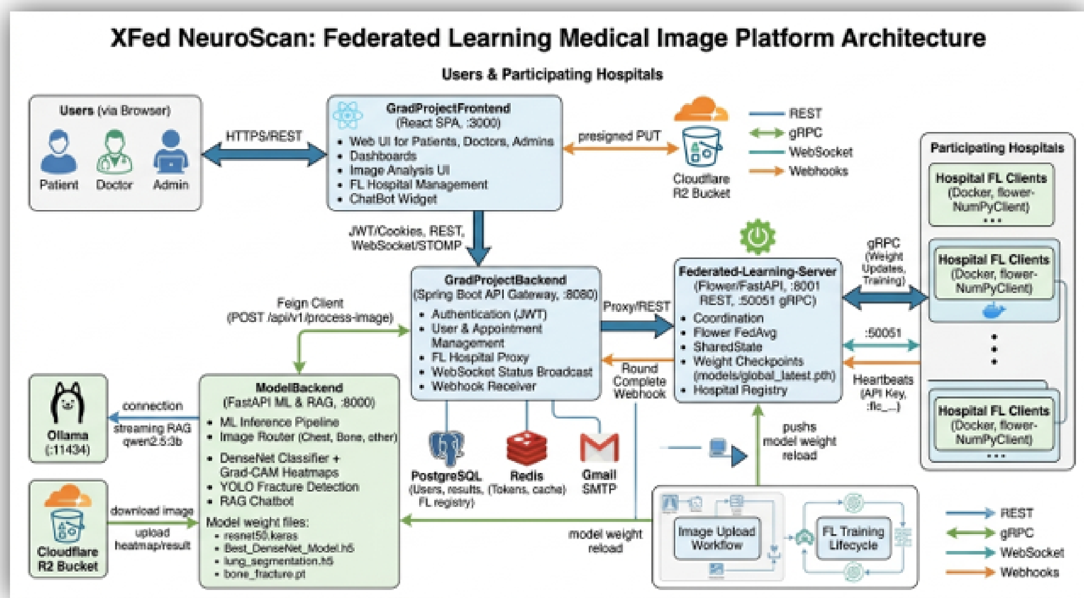


Figure 3.23: End-to-end logical architecture of the proposed X-Fed NeuroScan framework, illustrating the interaction between users, backend services, AI models, explainability components, federated learning infrastructure, and data storage systems.

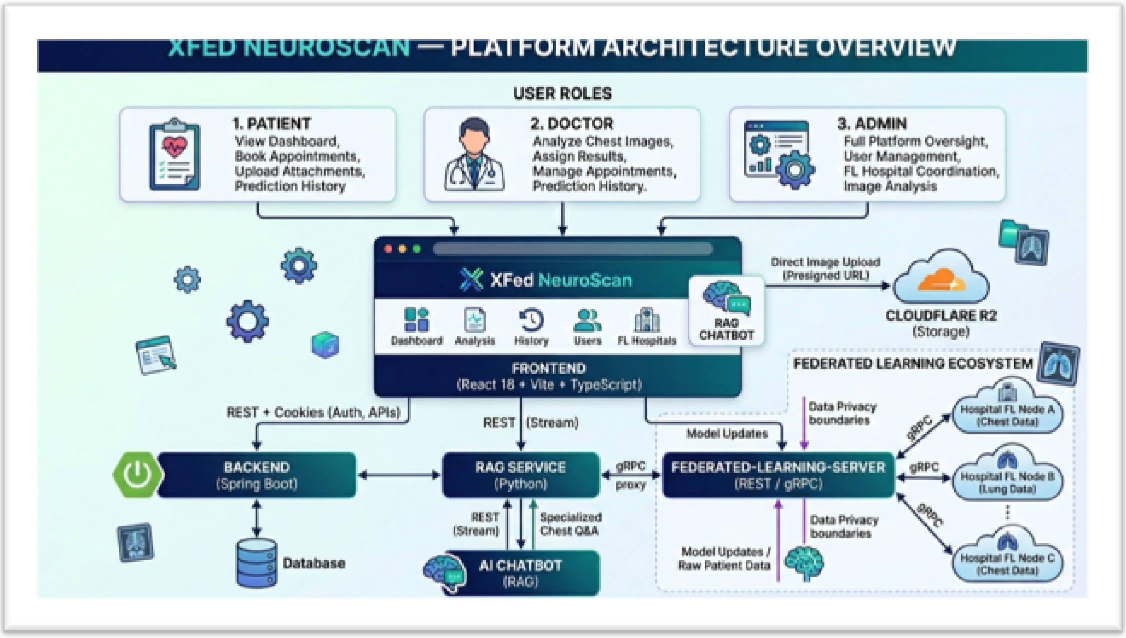


Figure 3.24: Deployment architecture of the proposed system showing the frontend application, backend services, model-serving infrastructure, federated learning server, databases, storage components, and external integrations.

and is the sole human interface into the platform.

Technology Stack

Different technologies were used during the development and implementation of the frontend components of the proposed system. They have been listed in Table 3.8.

Table 3.8: Frontend technologies used in the implementation of the proposed system.

Category	**Technology**
Framework	React 18
Build Tool	Vite 6 with `@vitejs/plugin-react-swc`
Programming Language	TypeScript 5 (Strict Mode)
Routing	React Router 7
Styling	Tailwind CSS v4
UI Primitives	Radix UI with shadcn-style wrapper components
Animations	Framer Motion (Motion)
Charts and Data Visualization	Recharts
Real-Time Communication	`@stomp/stompjs` and `sockjs-client`
Form Management	react-hook-form

User Roles & Capabilities

The system is designed as a role-based healthcare to ensure that each type of user interacts with the platform according to their responsibilities, permissions, and level of access. The system defines three main roles: Patient, Doctor, and Administrator, each serving a distinct purpose within the healthcare workflow.

Patients represent the end users of the system. Their role is limited to personal healthcare management, including registering accounts, booking appointments, viewing medical history, and accessing their own diagnostic results. This restricted access ensures that patients can interact with their healthcare data securely without modifying clinical or system-level information.

Doctors have a more advanced role focused on medical operations. They can upload and analyze medical images, review AI-generated predictions, interpret results, and assign diagnoses to patients. In addition, doctors manage appointments and clinical interactions with patients. This role requires broader access because it directly involves medical decision-making and the use of AI diagnostic tools. Administrators hold the highest level of control within the system. They are responsible for managing users, including creating and inviting doctors, monitoring system activity, and overseeing hospital and federated learning operations. Administrators also handle system configuration and ensure that all services operate securely and efficiently.

The separation of roles is essential for maintaining security, privacy, and operational clarity. By limiting access based on responsibilities, the system prevents unauthorized actions, reduces the risk of data misuse, and ensures that each user interacts only with the features relevant to their function. This role-based design also improves usability by providing a tailored interface for each type of user, making the system more efficient and easier to navigate.

Table 3.9: User roles and capabilities within the proposed X-Fed NeuroScan system.

Role	**Capabilities**
Patient	View personalized dashboard statistics, book appointments with doctors, upload medical image attachments, access diagnostic results, and review prediction history.
Doctor	Analyze chest medical images using the integrated AI models, assign diagnostic results to patients, manage appointment schedules, review patient records, and access prediction history.
Administrator	Perform full platform administration including user management, federated learning hospital registration, hospital selection for federated training rounds, monitoring system activity, and conducting image analysis through the integrated diagnostic framework.

Key Pages and Routing

Frontend is structured as a single-page application with role-based routing to ensure secure and organized access to system features. Each user type is directed to specific pages depending on their permissions and responsibilities within the platform.

Public users can access the landing page and authentication-related pages, including login, signup, password recovery, and invite-based account activation. These pages are only available to guests who are not yet authenticated.

Authenticated users are redirected to role-specific dashboards after login. Patients have access to their personal dashboard and medical history, where they can view appointments and diagnostic results. Doctors are provided with a dedicated dashboard along with access to image analysis tools and patient-related medical data. Administrators have the highest level of access, allowing them to manage users, monitor system activity, and control federated learning operations.

In addition to dashboards, administrators also access specialized modules for user management and federated learning configuration, including hospital registration and training round setup. The image analysis page is restricted to doctors and administrators due to its clinical importance and use of AI-based diagnostic tools.

Overall, the routing structure enforces strict role-based access control, ensuring that each user interacts only with the features relevant to their responsibilities while maintaining system security and usability.

Table 3.10: Access control policy for system pages and administrative modules.

Page	**Access**
Landing Page	Publicly accessible to all users, including unauthenticated visitors.
Authentication Pages	Accessible only to guest users who have not yet authenticated.
Password Recovery	Accessible only to guest users initiating the password reset process.
Invite Activation	Accessible only through a valid invitation token provided to the invited user.
Image Analysis	Accessible to users with Administrator or Doctor privileges.
Prediction History	Accessible to Administrators, Doctors, and Patients according to their authorization level.
Administrator Dashboard	Accessible exclusively to users with the Administrator role.
Doctor Dashboard	Accessible exclusively to users with the Doctor role.
Patient Dashboard	Accessible exclusively to users with the Patient role.
User Management	Accessible exclusively to Administrators for managing user accounts and permissions.
Federated Learning Hospital Management	Accessible exclusively to Administrators for registering, updating, and monitoring participating hospitals.
Federated Learning Training Round Setup	Accessible exclusively to Administrators for configuring and launching federated learning training rounds.

Core Features

Authentication & Session Management The frontend implements a full authentication lifecycle using JWT access tokens are stored in memory (not localStorage); refresh tokens are stored in HttpOnly cookies. On app load, the context attempts a silent token refresh to restore sessions without requiring re-login. GuestRoute prevents authenticated users from visiting auth. pages; Require Dashboard Role guards all dashboard routes.

Medical Image Analysis The image analysis flow uses a presigned-URL pattern to avoid routing large files through the backend:

- Step 1: Frontend requests a presigned R2 PUT URL from Spring Backend
- Step 2: Frontend uploads the image directly to Cloudflare R2
- Step 3: Frontend calls the api who is responsible for image-processing with the R2 key
- Step 4: Backend triggers Model inference and returns prediction, confidence, and heatmap URL

Federated Learning Management Admins manage FL hospitals through two dedicated pages. FLHospitalsPage shows all registered hospital nodes with live ONLINE/OFFLINE status via WebSocket. FL Select Hospitals Page lets admins choose which hospitals participate in the next training round. Registration generates an API key displayed once via ApiKeySuccessDialog.

AI Chatbot (RAG) A floating chat widget is available on every page. Responses are streamed from the RAG service. The chatbot gracefully degrades to keyword-based mock answers if the RAG API is unreachable.

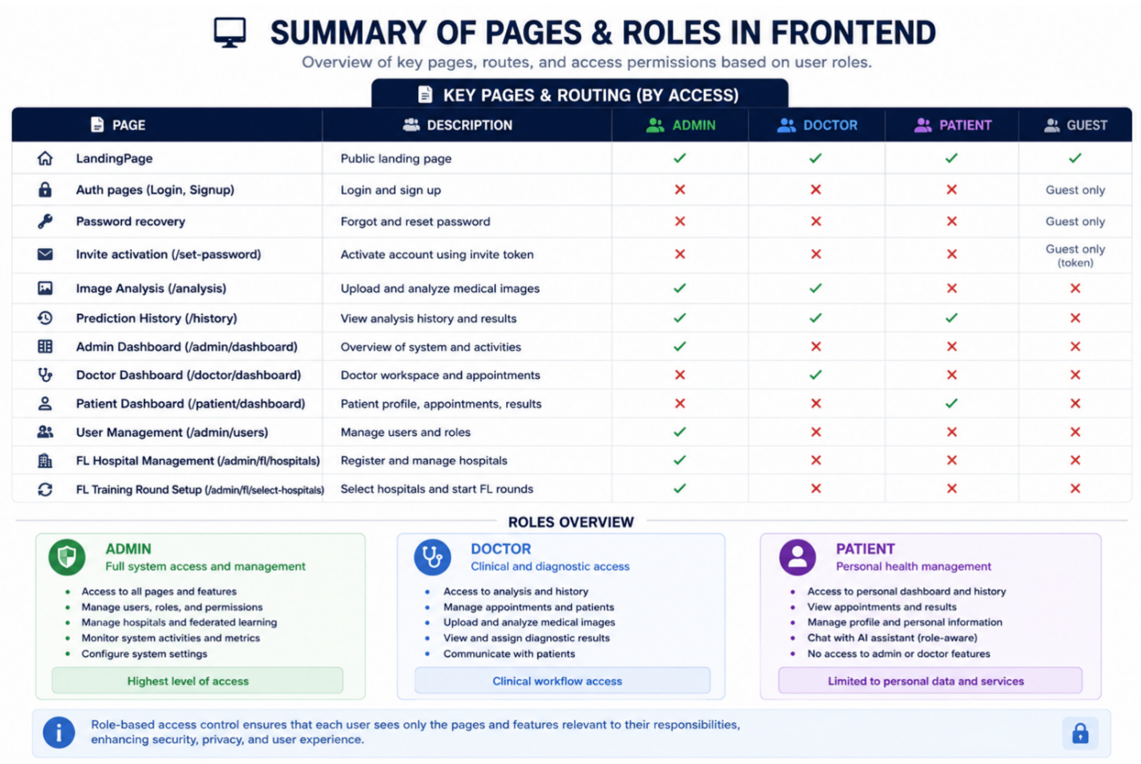


Figure 3.25: A visualization of the roles and pages implemented in the web application.

3.8.3 Centralized Backend (Spring Boot)

Centralized backend is a Spring Boot 3.5 application written in Java 21. It serves as the central API gateway and system of record for authentication, user management, medical image orchestration, appointments, federated learning coordination, and real-time WebSocket broadcasts.

Technology Stack

Various technologies were utilized in the implementation of the backend of the proposed end-to-end system. All of these technologies are listed in Table 3.11.

Table 3.11: Backend technologies used in the implementation of the proposed system.

Category	Technology
Framework	Spring Boot 3.5.4
Programming Language	Java 21
Build Tool	Maven
Security	Spring Security 6 with JSON Web Token (JWT) authentication using JJWT 0.12.3
Persistence Layer	Spring Data JPA, Hibernate, and PostgreSQL
Cache and Session Management	Spring Data Redis
HTTP Client	Spring Cloud OpenFeign
Real-Time Communication	Spring WebSocket, STOMP, and SockJS
Object Storage Integration	AWS SDK v2 for S3-compatible storage (Cloudflare R2)
Email Services	Spring Mail with Gmail SMTP
API Documentation	SpringDoc OpenAPI (Swagger UI)
Object Mapping and Code Generation	MapStruct and Lombok

Security & Authentication

Security is implemented with JWT access tokens and HttpOnly refresh-token cookies stored in Redis:

- Access Tokens: short-lived JWTs returned in the login response body
- Refresh Tokens: longer-lived tokens stored in Redis (key: email:deviceId) and set as HttpOnly cookies
- Logout: access tokens blacklisted when still valid; refresh token deleted from Redis
- FL API keys: encrypted at rest using AES via EncryptionUtils
- FL webhook endpoints: authenticated with a shared secret (not JWT)
- Soft delete: deleted users cannot log in

In the current system design, **login is restricted to patients only**. Doctor and Admin accounts are not self-authenticated through public signup; instead, they are created and managed exclusively by an administrator and activated through an invitation-based workflow. This ensures strict control over privileged roles in the healthcare system.

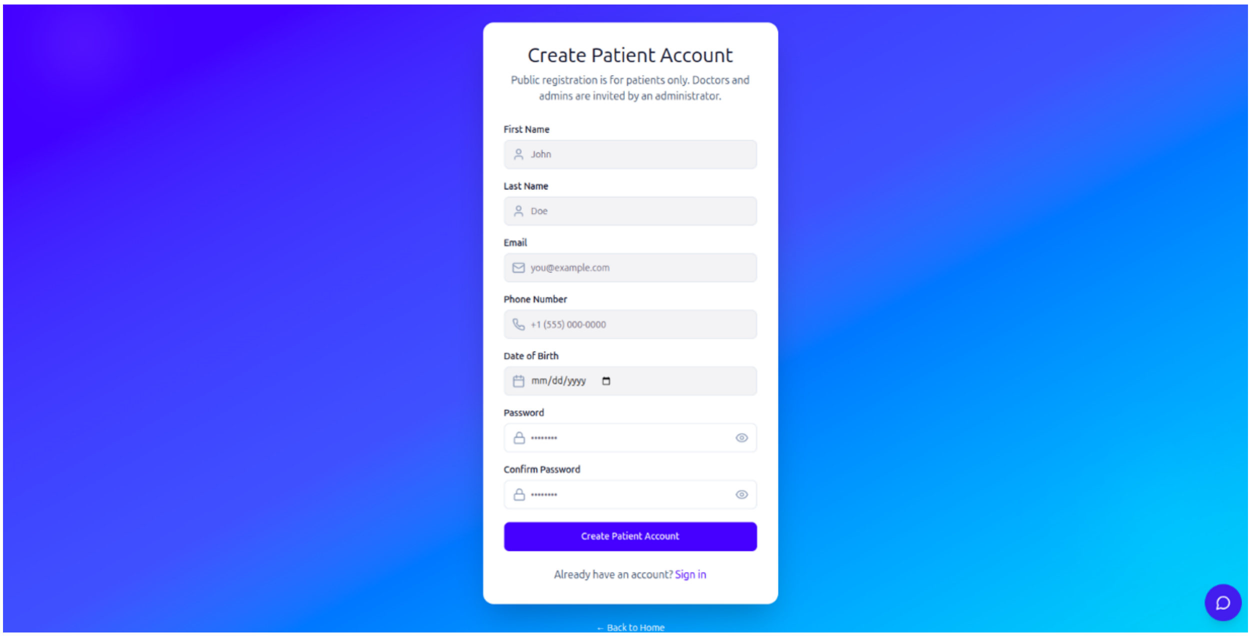


Figure 3.26: An image of the Create Account interaction from the web application.

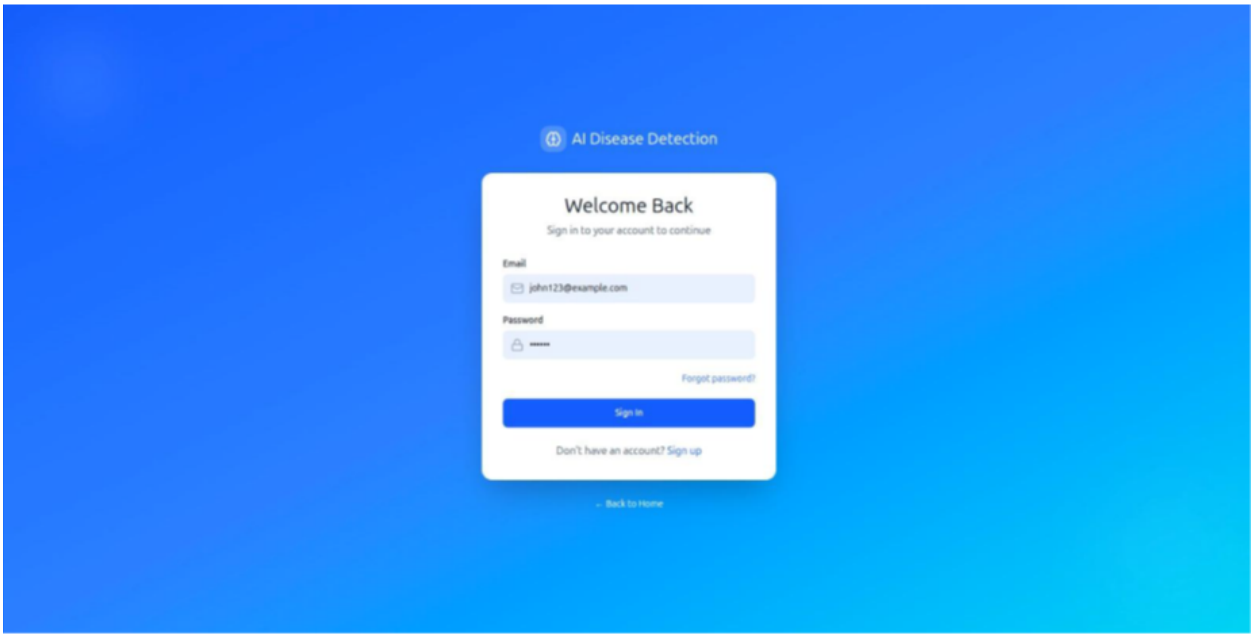


Figure 3.27: An image of the Login interaction from the web application.

After login, user information is stored in PostgreSQL, but sensitive session data is handled separately for security. Passwords are never stored in plain text; instead, they are hashed using **BCrypt** before being saved in the database, making them irreversible and protected even if the database is compromised.

During authentication, the input password is hashed and compared with the stored hash. If valid, the system generates a short-lived **JWT access token** and a long-lived **refresh token**. The access token is used for API requests, while the refresh token is stored securely in **Redis** and also set as an **HttpOnly cookie**.

When the user logs out, the access token is blacklisted and the refresh token is removed from Redis, fully terminating the session. This design ensures secure password storage, stateless authentication, and strong session protection.

In the system, only Admin users have the authority to create or manage Doctor and other Admin accounts, ensuring strict role control and preventing unauthorized privilege escalation.

New privileged users (Doctors or Admins) cannot register through the public signup process. Instead, an Admin must explicitly create or invite them through the administrative interface. This invitation-based flow allows the system to maintain tight control over sensitive roles that have access to clinical data, AI analysis tools, and system-wide management features.

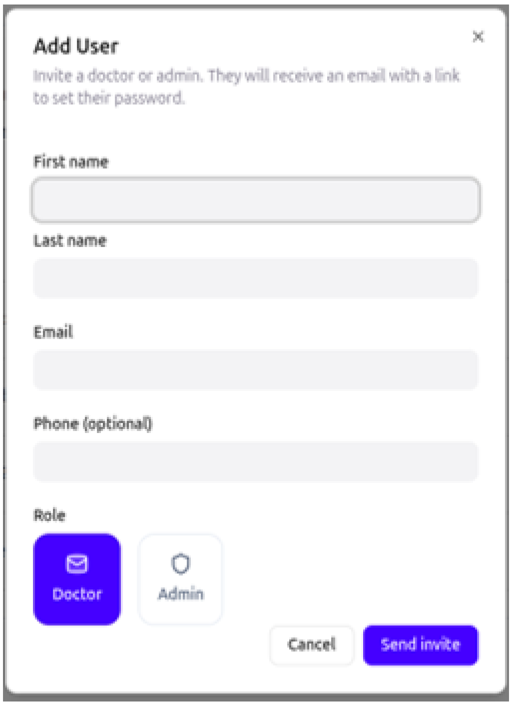


Figure 3.28: An image of the Add User interaction from the web application.

After an Admin creates a new Doctor or Admin account, the account does not become active immediately. Instead, it enters a pending state until it is explicitly approved. This approval step ensures that no privileged user gains access to the system without proper verification and administrative confirmation.

During this pending phase, the user record is already stored in the database, but the account remains inactive, meaning the user cannot log in or access any system features. Once the Admin approves the account, its status is updated to active, and the user is then able to authenticate and use the system according to their assigned role.

In the user management interface, the Admin has full control over viewing and organizing all system users through a role-based filtering system. This allows the Admin to efficiently manage large numbers of accounts by categorizing users according to their current role and status.

In addition to account creation and approval, the Admin also has full control over existing privileged accounts, as shown in 3.29. This includes the ability to disable or delete accounts at any time. Disabling an account temporarily prevents the user from logging in without removing their data, while deletion marks the account as removed (soft delete), ensuring it is excluded from authentication and system operations but still preserved for audit or historical purposes.

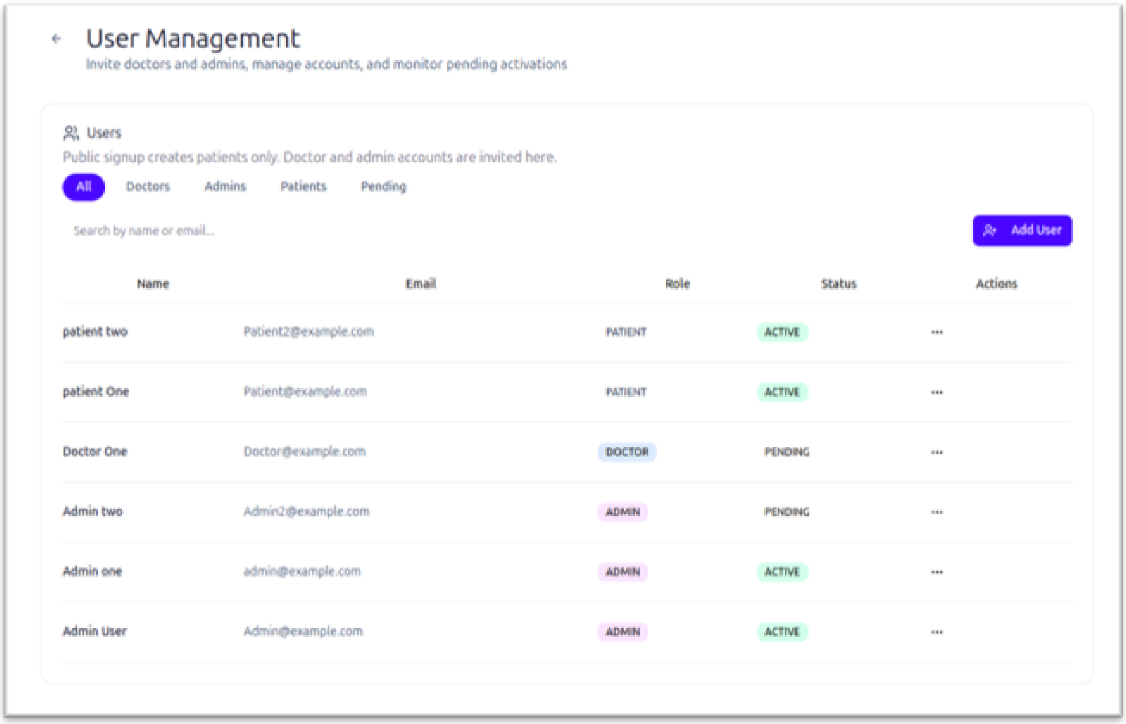


Figure 3.29: An image of the User Management tab from the web application.

When a user selects “Forgot Password”, the system follows a secure multi-step flow to ensure that only the legitimate account owner can reset the password.

First, the user enters their registered email address in the frontend. This request is sent to the Spring Boot backend, which validates whether the email exists in the system. If the user is found, Spring Boot generates a secure password reset token, typically a short-lived, random token linked to the user account.

This token is then stored temporarily (or associated with the user record) and sent to the user via email using Gmail SMTP. The email contains a secure reset link that includes the token as a parameter.

When the user clicks the link, they are redirected to the password reset page in the frontend. The new password is submitted along with the token, and Spring Boot validates the token to ensure it is still valid and has not expired or been used before.

If the token is valid, the system hashes the new password using BCrypt and updates it in the PostgreSQL database, replacing the old password hash. After successful reset, the token is invalidated to prevent reuse.

Overall, the forgot password process ensures security by combining email verification, time-limited tokens, secure transmission, and encrypted password storage.

So in summary User authentication in system uses a secure JWT-based system where only patients can self-register and log in, while doctors and admins are created by admins. Passwords are stored in the database using BCrypt hashing.

After successful login, the system issues a short-lived JWT access token and a long-lived refresh token stored in Redis as an HttpOnly cookie. The access token is used for requests, and the refresh token allows session renewal.

The system also supports a secure forgot password workflow, where a password reset token is generated, sent to the user via email, and validated before allowing password updates. The new password is then hashed using BCrypt and stored in the database.

On logout, tokens are invalidated by blacklisting the access token and removing the refresh token. Role-based access control and soft delete ensure only authorized and active users can access the system.

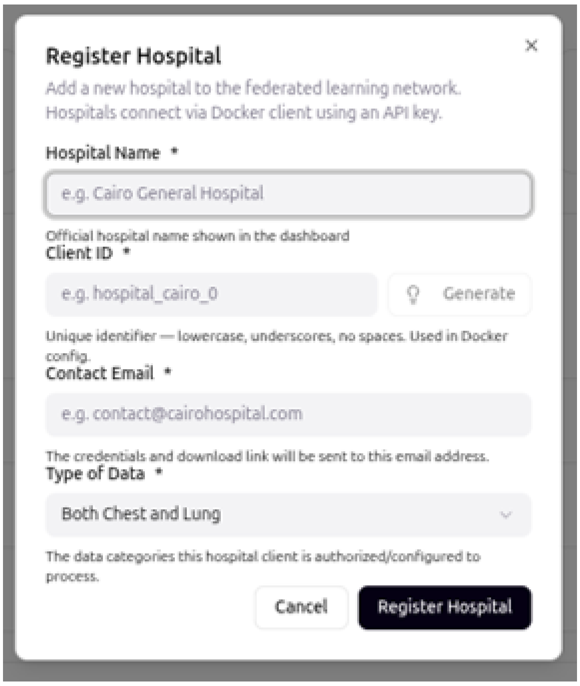


Figure 3.30: An image of the Add Hospital interaction from the web application.

Backend handles the full lifecycle of registering new hospitals in the federated learning system. When an Admin submits a hospital, Spring Boot validates the request, ensures

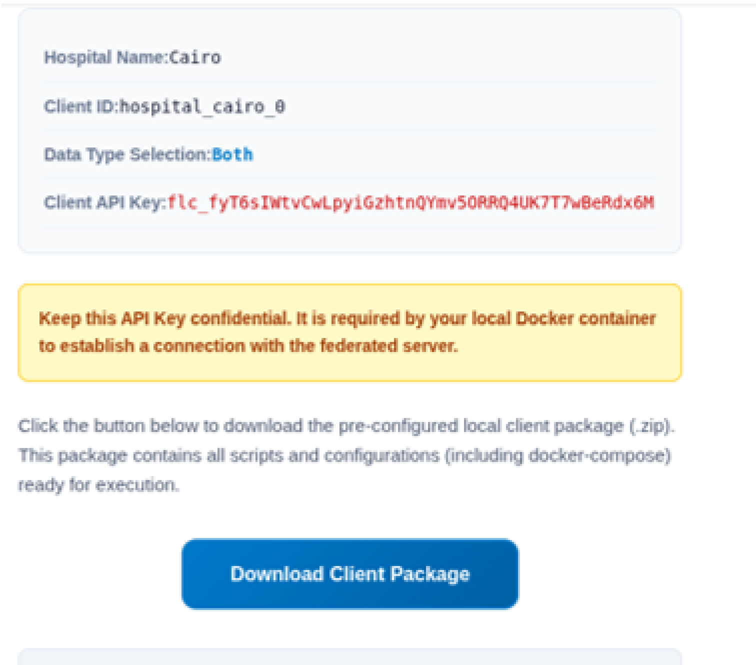


Figure 3.31: An image of the client hospital getting a confidential API key interaction from the web application.

only Admins can perform the action, and then generates a unique FL client ID and encrypted API key using AES before storing them in PostgreSQL. The hospital is initially saved with a PENDING or INACTIVE status.

After that, Spring Boot registers the hospital with the Federated Learning Server by sending its credentials and metadata, allowing it to join the federated learning network. It may also provide a Docker-based client package for hospital deployment.

Spring Boot receives and synchronizes key data from the Federated Learning (FL) Server, especially hospital online/offline status, which is determined by heartbeat signals sent from hospital clients to the FL Server. Spring Boot does not calculate this status itself; instead, it periodically fetches updated client information from the FL Server.

Overall, Spring Boot acts as a bridge that collects, stores, and distributes FL Server status updates, giving administrators a live view of the federated learning network.

When a new hospital is registered in the federated learning system, Spring Boot not only creates and stores the hospital record, but also sends an automated email notification via Gmail SMTP as part of the onboarding process.

After the Admin submits and approves the hospital registration, Spring Boot generates the hospital's federated learning credentials (such as the FL client ID and encrypted API key) and securely stores them in the database. Once the registration is successfully completed, Spring Boot triggers the email service to notify the hospital.

This email is sent using Gmail SMTP integration and typically contains important onboarding information such as the hospital's activation status, instructions for joining the federated learning network, and in some cases, a link or attachment for downloading the Docker-based FL client package required to connect to the FL server.

This step ensures that hospitals are immediately informed after registration and can proceed with setup and participation in federated learning without manual coordination. It also improves system automation and reduces administrative overhead while maintaining secure delivery of sensitive setup information.

Medical Image Pipeline

The backend orchestrates image analysis without storing raw images locally:

- Presigned PUT URL is generated for direct R2 upload (5-minute expiry).
- After upload, backend saves image metadata to PostgreSQL
- ModelBackend is called via OpenFeign (Feign client) with a presigned GET URL
- ML results — prediction label, confidence score, heatmap S3 key — are persisted
- Presigned GET URLs for original and heatmap images are cached in Redis (59 min TTL)

Small parts of the pipeline can be shown in **??**, and 3.33 where the user is asked to upload a medical image to the system so that they are directed to the right models and diagnosed.

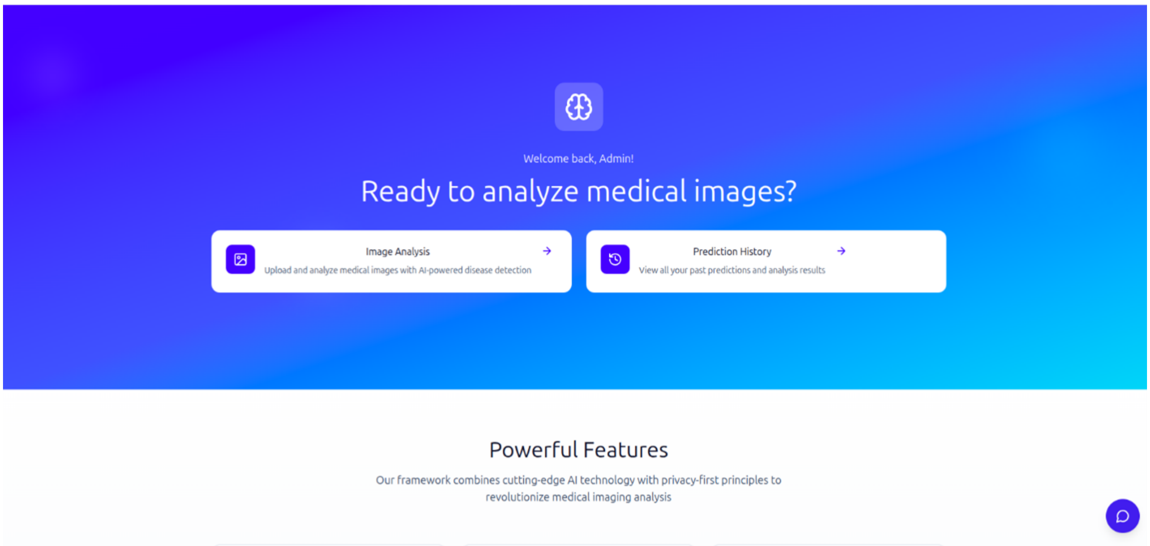


Figure 3.32: An image greeting page of the web application.

Prediction history can be viewed and also has statistics like (total scans , AVG. confidence) , and also every prediction's details can be viewed or deleted, as shown in Figure 3.34, and the details of each scan can be reviewed as well, as shown in Figure 3.35.

Some examples of the Lung Cancer model in action are shown in Figure 3.36, and Figure 3.37. The system also makes sure to show the confidence as ordinary in medical applications of computer vision systems, as well as the processing time and the resulting heatmap, if applicable.

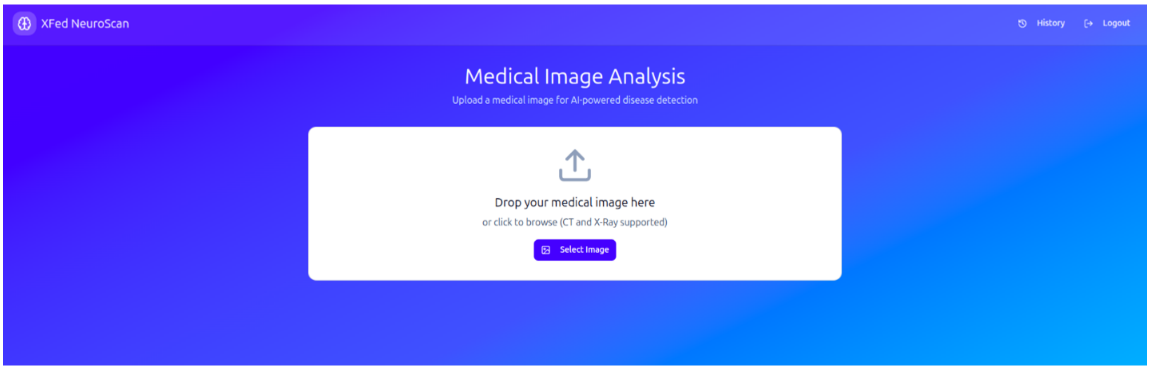


Figure 3.33: An image of the system asking the user to upload a scan to diagnose.

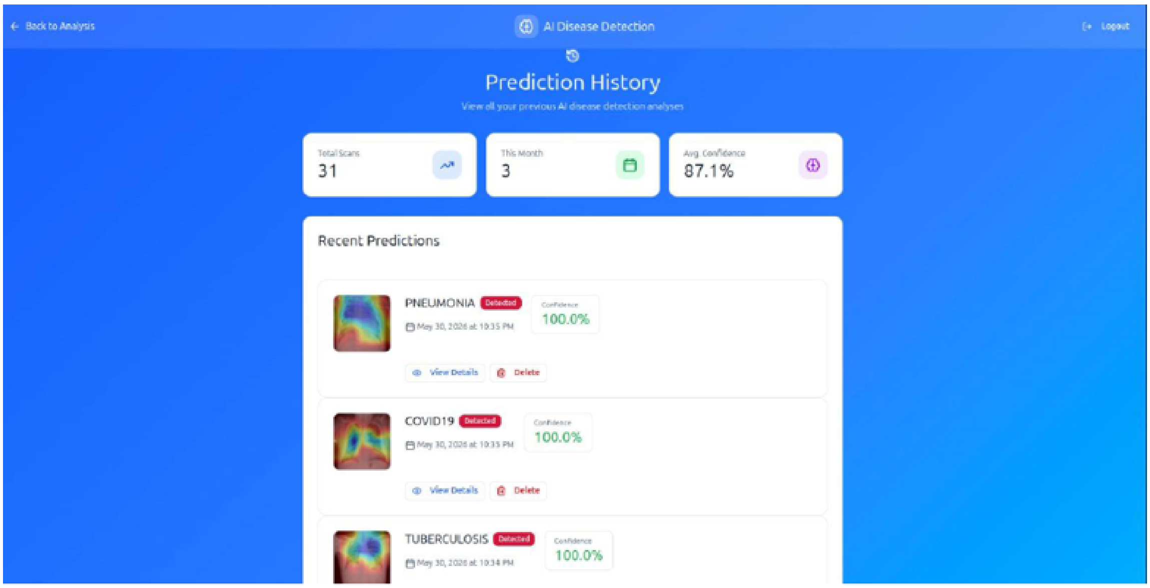


Figure 3.34: An image of the system showing the prediction history of the logged-in user.

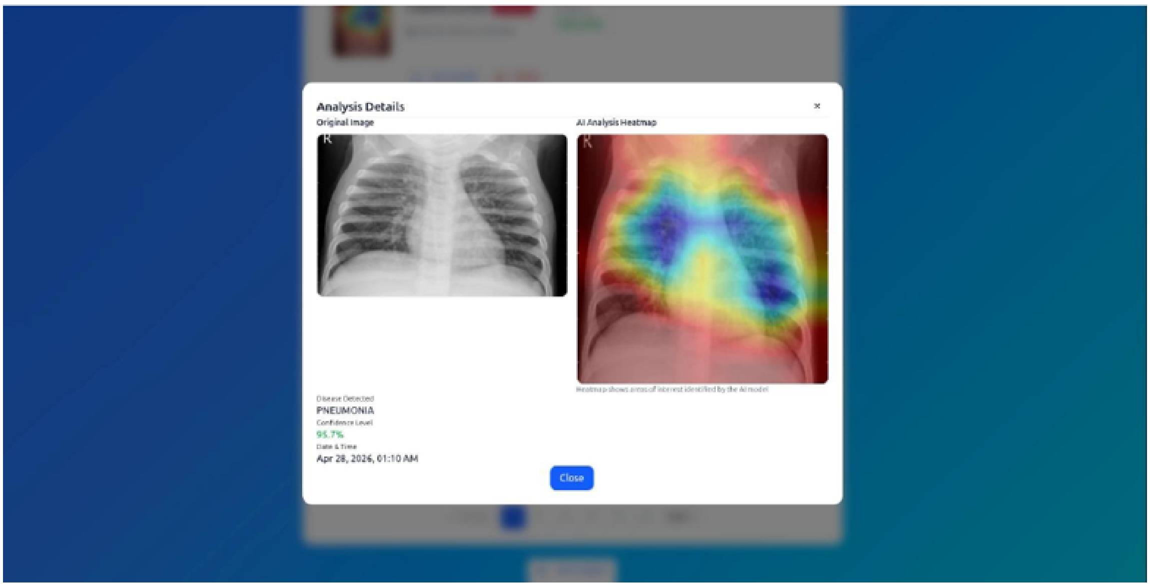


Figure 3.35: An image of the system showing the details of one of the user's historical predictions.

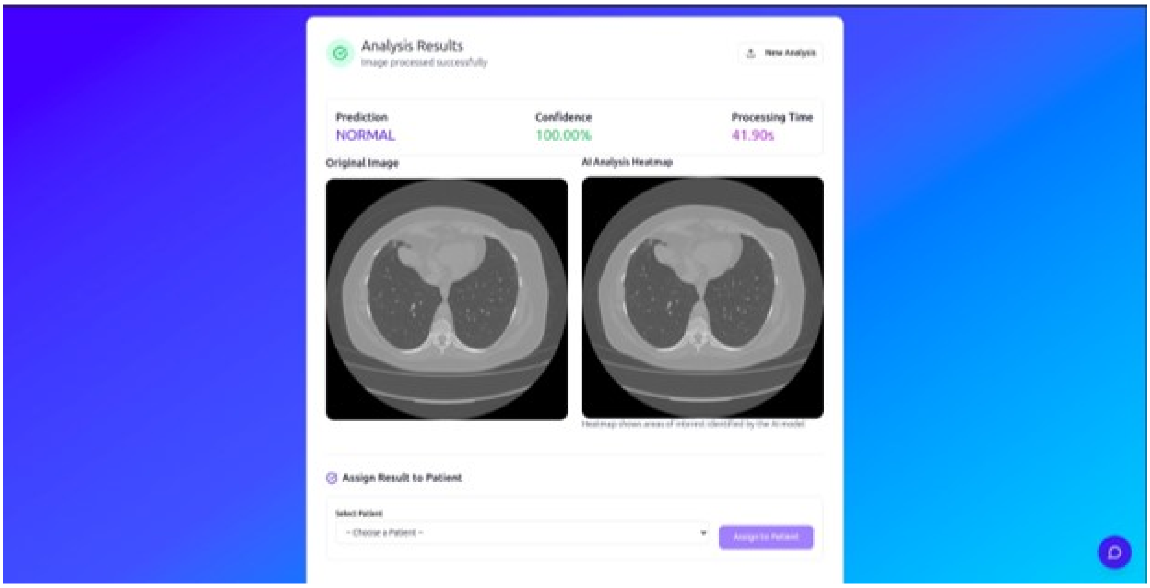


Figure 3.36: One of the of the examples of using the system on the a lung cancer image.

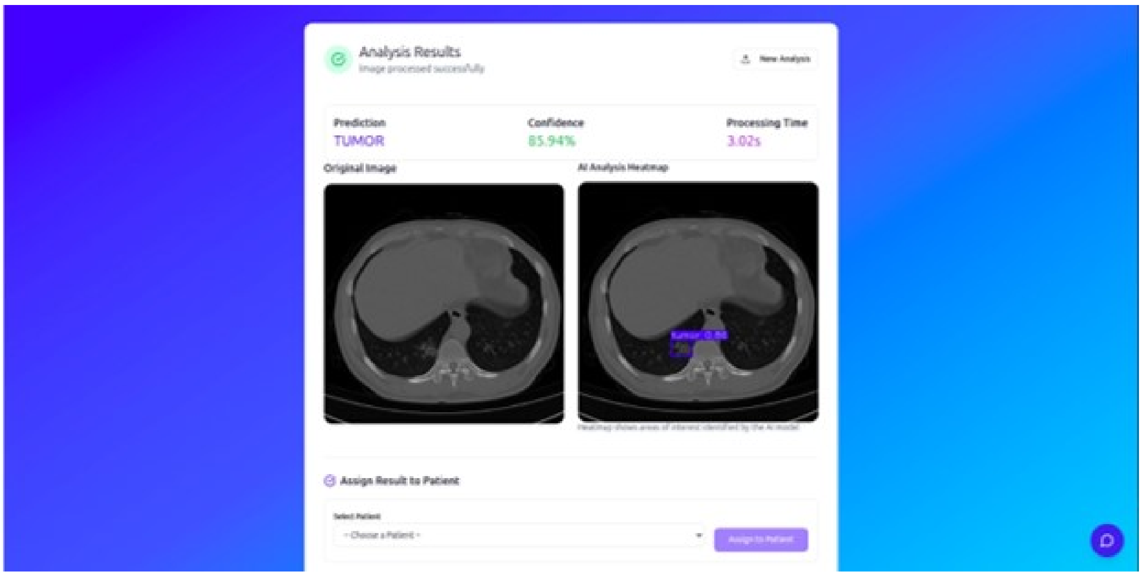


Figure 3.37: Another example of using the system on the a lung cancer image.

Appointment Management

The appointment management module enables patients and doctors to coordinate medical consultations through a structured workflow. Patients can create appointments by selecting a doctor, choosing a date and time, specifying the appointment type, and providing optional notes or supporting attachments. Once submitted, the appointment information is stored in the database and becomes available to the assigned doctor.

Doctors can view their scheduled appointments, review patient details, and update the appointment status as needed. They can mark appointments as completed, add consultation notes, or cancel appointments when necessary. Patients can also view the status of their appointments and track any updates made by the doctor.

All appointment data is managed through the Spring Boot backend and stored in PostgreSQL, ensuring persistence and secure access. The module provides an organized way to manage patient-doctor interactions while maintaining a complete record of appointment history within the platform.

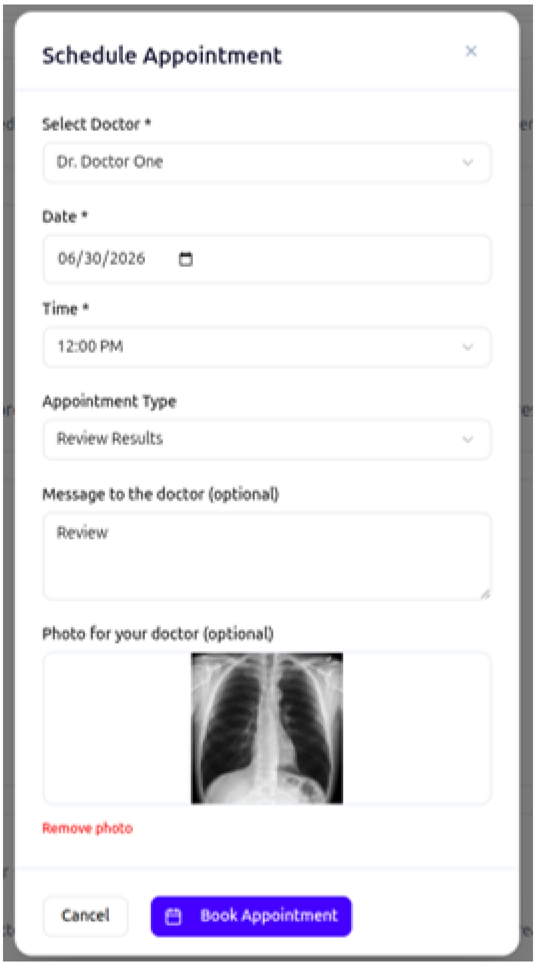


Figure 3.38: An image of the Schedule Appointment interaction from the system.

Real-Time WebSocket Spring WebSocket with STOMP over SockJS provides live FL hospital status to the frontend. A scheduled poller runs every 10 seconds, fetches client status from the FL Server, syncs to PostgreSQL, and broadcasts to /topic/fl/clients. Hospital nodes that miss heartbeats are marked OFFLINE automatically.

Email Service Spring Mail with Gmail SMTP sends three types of emails:

- Password Reset: link to `/reset-password?token=...`
- Admin Invite: link to `/set-password?token=...` for new doctor/admin accounts
- FL hospital onboarding: download link for the hospital Docker client package plus connection details.

Core API Modules The core modules exposed by the API and implemented in it a re listed in Table 3.12

Table 3.12: Backend service modules and their responsibilities within the system.

Module	Description
Authentication	Handles the full authentication lifecycle including user registration, login, token issuance, refresh token management, and logout operations.
Image Upload	Generates and returns Cloudflare R2 presigned PUT URLs to enable secure direct client-side uploads of medical images.
Image Processing	Orchestrates the machine learning inference pipeline, triggers model execution, and stores as well as retrieves diagnostic results from the persistence layer.
Appointments	Manages the full appointment workflow, allowing patients to book appointments and doctors to approve, complete, or cancel scheduled sessions.
Admin User Management	Provides administrative capabilities for managing user accounts, including creation, modification, role assignment, and deactivation.
Federated Learning Hospitals	Handles hospital registration and management within the federated learning ecosystem, acting as a proxy interface to the federated learning server.
Federated Learning Webhooks	Processes callback events from the federated learning server after training rounds, updating system state and storing training metadata accordingly.

3.8.4 ML Service - ModelBackend

ModelBackend is a FastAPI microservice that hosts all AI capabilities. At startup it loads four ML components into memory and exposes a unified image-processing endpoint plus a RAG chatbot. It runs on port 8000 and is called by centralized Backend for image analysis and directly by the frontend for chatbot interactions.

ML Models

Different backbone models were used for the different tasks the proposed system performs. All the AI models and architectures used are listed in Table 3.13.

Table 3.13: Machine learning components and their corresponding frameworks and purposes within the proposed system.

Component	Framework	Purpose
Router Model	Keras (DenseNet121)	Classifies input images into X-ray, CT scan, or out-of-distribution categories using 224×224 image resolution as the first stage of the diagnostic pipeline.
Chest Classification	TensorFlow/Keras (DenseNet121)	Performs four-class chest X-ray diagnosis including COVID-19, Normal, Pneumonia, and Tuberculosis classification in a multi-label-ready setup.
Lung Segmentation	TensorFlow/Keras (U-Net)	Extracts lung regions from medical images prior to classification to enhance focus on relevant anatomical structures and improve diagnostic accuracy.
Lung Cancer Detection	YOLOv8	Detects and localizes lung cancer regions in CT scans using bounding box predictions for object-level medical diagnosis.
RAG Stack	LangChain, Ollama, and FAISS	Implements a retrieval-augmented generation chatbot for medical question answering using vector-based document retrieval and local LLM inference.

FastAPI serves as the AI engine of XFed NeuroScan and relies on Spring Boot to integrate its models into the complete healthcare platform. While FastAPI is responsible for running AI models and generating predictions, Spring Boot manages the workflow and communicates with the AI service whenever analysis is requested.

Separating Spring Boot and FastAPI into two services improves scalability and maintainability, allowing the healthcare system and AI models to be developed, updated, and deployed independently without affecting each other.

Image Processing Pipeline

The image processing module is designed to automatically analyze uploaded medical images and route them to the appropriate AI model based on their content. This architecture allows the system to support multiple diagnostic pipelines while maintaining a single entry point for image analysis.

Unified Routing Every uploaded image first passes through the Router Model, which serves as the initial classification layer. The router determines whether the image is a chest X-ray, lung cancer image, or an unsupported image type. A confidence threshold of 0.5 is used to ensure reliable routing decisions. Images classified as other are immediately rejected to prevent invalid data from being processed by diagnostic models. Once classified, chest X-rays are forwarded to the DenseNet-based chest disease pipeline, while lung cancer images are sent to the YOLOv8 detection pipeline.

Chest Pipeline The chest disease analysis pipeline is optimized for detecting four classes: COVID-19, Pneumonia, Tuberculosis, and Normal. Before classification, several preprocessing steps are applied to improve image quality and model performance.

First, a U-Net lung segmentation model isolates the lung region from the surrounding anatomy, allowing the classifier to focus only on clinically relevant areas. Next, Contrast Limited Adaptive Histogram Equalization (CLAHE) enhances local contrast within the segmented lungs, making subtle abnormalities easier to detect. A Gaussian Blur (5×5) filter is then applied to reduce image noise while preserving important structures.

The processed image is normalized to the range [0,1], converted to RGB format, and resized to 224×224 pixels, which matches the input requirements of the DenseNet model. The DenseNet classifier then predicts one of the four disease categories and returns a confidence score indicating prediction certainty.

To improve explainability, the system generates a Grad-CAM heatmap from the final convolutional layers of the network. This heatmap highlights the image regions that most influenced the model's decision, providing visual evidence to support the prediction. Both the original image and the generated heatmap are uploaded to Cloudflare R2 for storage and later retrieval by the frontend.

Lung Cancer Pipeline The lung cancer pipeline is based on YOLOv8, an object detection model capable of identifying suspicious cancerous regions within medical images. After the router classifies an image as a lung cancer case, it is forwarded directly to the YOLOv8 model for inference.

The model performs real-time detection using a confidence threshold of 0.3, identifying potential cancer regions and generating bounding boxes around detected abnormalities. For each detected region, the model provides confidence scores indicating the likelihood of cancer presence.

The detected regions are then visually annotated on the image by drawing bounding boxes and labels, creating an interpretable result that can be reviewed by medical professionals. The final annotated image is uploaded to Cloudflare R2 under the lung cancer results directory, allowing it to be displayed in the frontend and stored as part of the patient's analysis history.

This dual-pipeline architecture combines classification-based diagnosis for chest diseases with object-detection-based localization for lung cancer, enabling the platform to provide both diagnostic predictions and visual explanations while maintaining a unified processing workflow.

An overall visualization of the image processing pipeline is shown in Figure 3.39

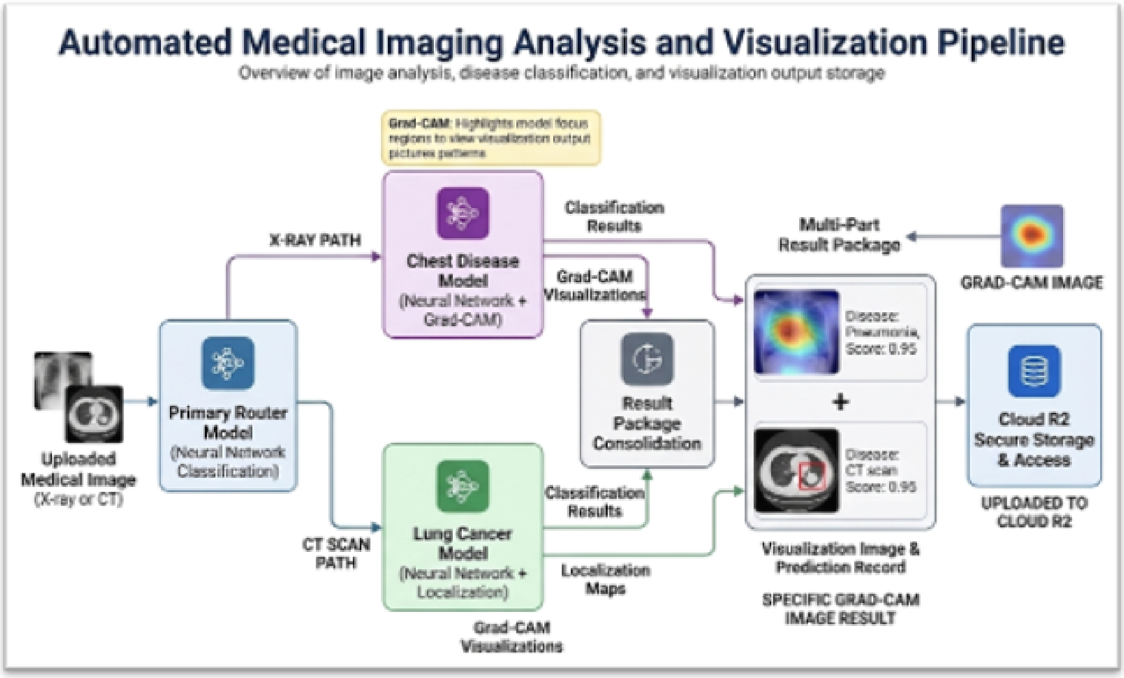


Figure 3.39: A simple visualization of the overall image processing pipeline from the backend's perspective.

Grad-CAM Explainability and Localization

For every chest X-ray prediction, the system generates a Grad-CAM (Gradient-weighted Class Activation Mapping) heatmap to provide interpretability for the AI decision. This mechanism highlights the regions of the image that most influenced the model's prediction by tracing gradients flowing into the final convolutional layers of the DenseNet network.

The generated heatmap is visualized using a JET colormap and overlaid on the original X-ray image , allowing both the anatomical structure and the activated regions to remain clearly visible. This overlay helps radiologists and clinicians visually verify whether the model is focusing on medically relevant areas such as lung opacities, consolidations, or abnormal tissue patterns rather than irrelevant regions.

To support persistence and later retrieval, both the original Grad-CAM output and the processed overlay image are stored in Cloudflare R2 under the `heatmaps/` directory. These images are served to the frontend using secure presigned URLs, ensuring controlled and time-limited access to sensitive medical data.

Localization Capability

Beyond explainability, Grad-CAM also provides a form of weak localization for disease regions within the chest X-ray. Although the DenseNet model is primarily designed for classification, Grad-CAM transforms it into a pseudo-localization system by identifying spatial regions that contribute most strongly to the predicted class.

This allows the system to:

- Approximate the location of abnormal lung regions.
- Highlight areas associated with diseases.
- Provide spatial context without requiring a dedicated detection model

In practice, this means the system not only predicts the presence of a disease but also indicates where in the lungs the abnormality is likely located. This improves clinical trust and makes the AI output more interpretable, bridging the gap between pure classification models and spatially aware diagnostic tools.

Overall, Grad-CAM serves as both an explainability mechanism and a lightweight localization method, enhancing transparency and clinical usability of the AI predictions.

RAG Chatbot Module

Retrieval-Augmented Generation (RAG) is an AI technique that enhances LLM responses by grounding them in a private document corpus rather than relying solely on the model's pre-trained knowledge. This is critical for domain-specific applications — such as medical systems, where accuracy, traceability, and up-to-date information are essential.

Standard RAG Pipeline

- Ingest: Load source documents (PDF, TXT, etc.)
- Chunk: Split long documents into overlapping smaller pieces.
- Embed: Convert each chunk to a dense numerical vector using a sentence embedding model.
- Store: Persist vectors in a FAISS vector database, organized by category.
- Retrieve: On a user query, embed the question and find the top-k most similar chunks.
- Generate: Pass the retrieved chunks and the question to the LLM to produce a grounded answer.

The classifier is the key differentiator of this RAG system. Rather than searching all documents indiscriminately, each query is first categorized, enabling targeted retrieval from the correct FAISS index.

Table 3.14: Categories of input handled by the RAG-based medical chatbot.

Category	**Typical Content**
Medical	Queries related to chest diseases and lung cancer, including symptoms, diagnosis, interpretation of medical findings, and general medical imaging questions.
General	Greetings, small talk, and off-topic or non-medical conversational queries that do not require clinical or diagnostic knowledge.

Categories

Model Architecture The classifier uses a lightweight 1D Convolutional Neural Network (CNN) designed for speed and bilingual text support, the full visualization of the architecture is shown in Figure 3.40.

1. Input: Token IDs (max 50 tokens)
2. Embedding Layer (vocab → 64 dims)
3. Conv1d (64 → 128 channels) + ReLU + Dropout(0.3)
4. Conv1d (128 → 128 channels) + ReLU
5. AdaptiveMaxPool1d(1)
6. Fully Connected: 128 → 64 → 3
7. Output.

Bilingual Tokenizer

- Tokenizes Arabic , English, and numeric tokens
- Vocabulary built from training data (300+ labeled question pairs)
- Fixed-length encoding: max 50 tokens, padded with <PAD>, unknown words as <UNK>
- Outputs: classifier.pth, tokenizer.json, config.json

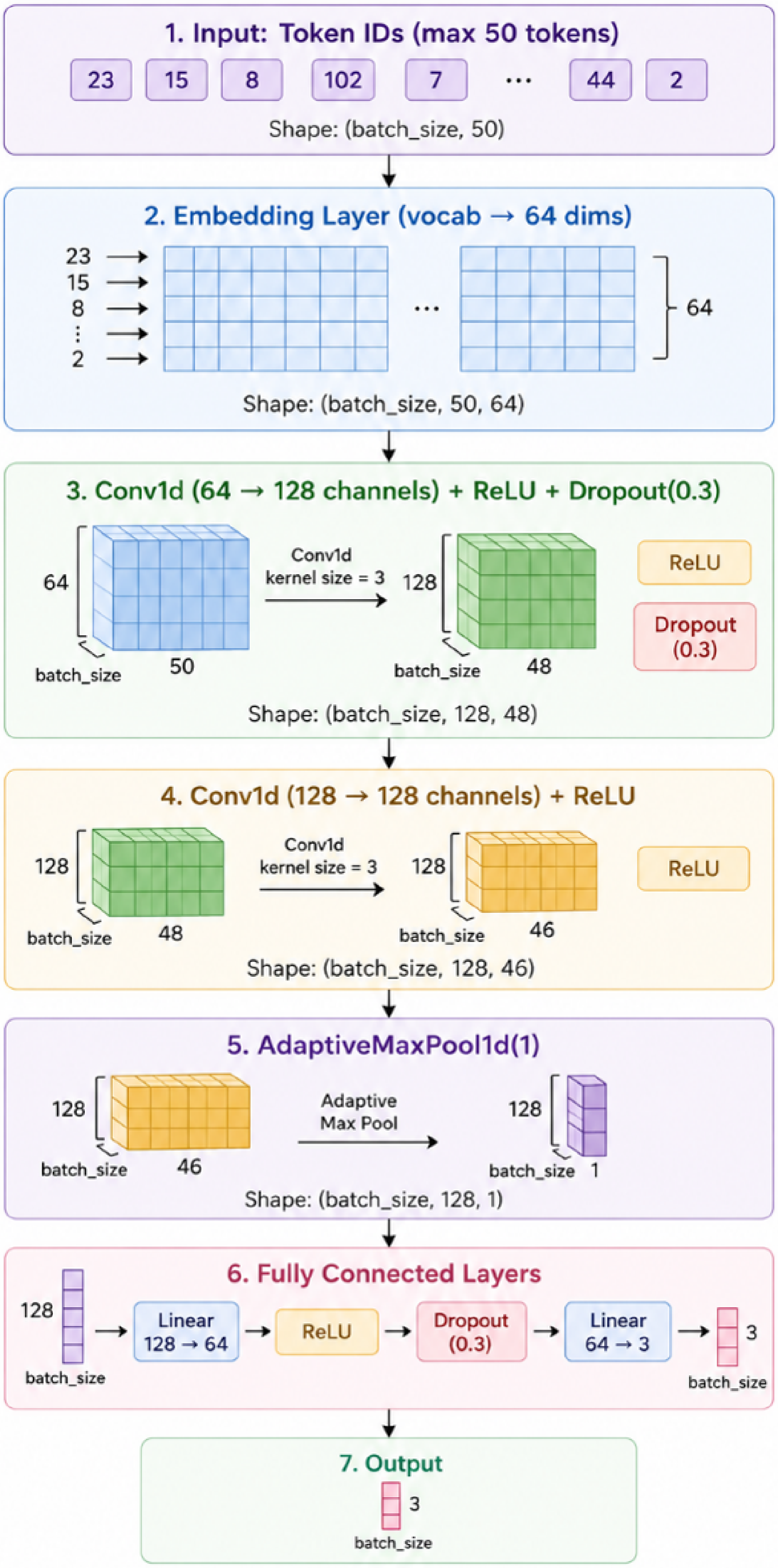


Figure 3.40: A simple visualization of the overall RAG architecture.

Training Configuration The model was trained for 100 epochs using the Adam optimizer with a learning rate of 0.001 and a batch size of 16. The vocabulary size is 352 tokens, and each input token is mapped into a 64-dimensional embedding space. The hidden layer dimension is set to 128, Each input sequence is limited to a maximum length of 50 tokens to ensure consistent processing and performance.

LLM Configuration The configurations used for the RAG engine are shown in Table 3.15 for reproduce-ability purposes.

Table 3.15: Configuration parameters of the RAG-based LLM engine.

Parameter	**Standalone Value**
LLM Engine	Ollama
Model	qwen2.5:3b
Temperature	0.5
Max Tokens	500
Keep Alive	Infinite (-1)

Prompt Design The LLM is strictly instructed to answer only from the retrieved context. If the context does not contain the answer, the model is instructed to say "I don't know." Responses are capped at three sentences and must match the language of the question.

The text is split using a RecursiveCharacterTextSplitter with a chunk size of 1000 characters and an overlap of 150 characters between consecutive chunks. This ensures that the text is broken into manageable segments while preserving context between adjacent chunks for better retrieval accuracy.

RAG Independence from Image models RAG does not receive chest X-ray predictions, GradCAM heatmaps, lung cancer results, or Spring Boot patient records. It is a standalone medical Q&A chatbot co-located in every page purely for deployment convenience.

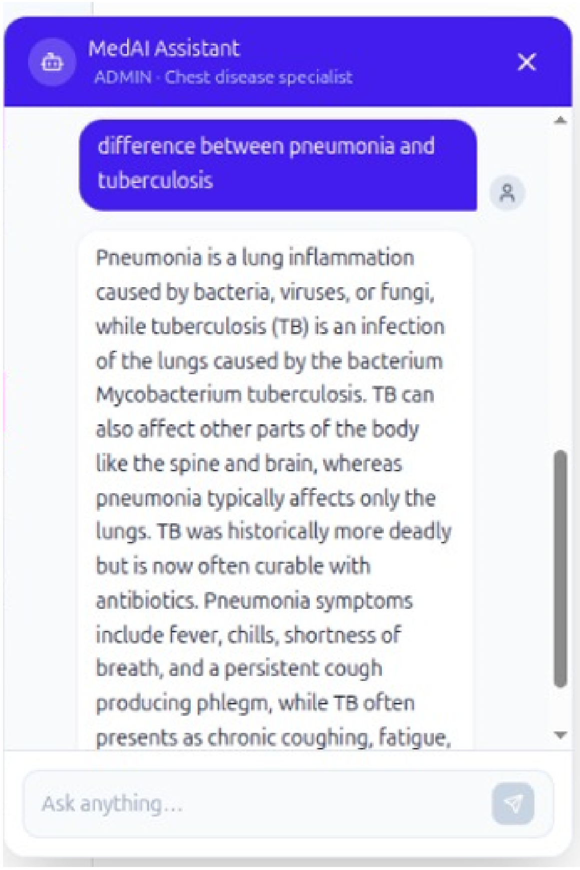
Examples of using the RAG agent implemented in the system are shown in Figure 3.41

Performance Profile

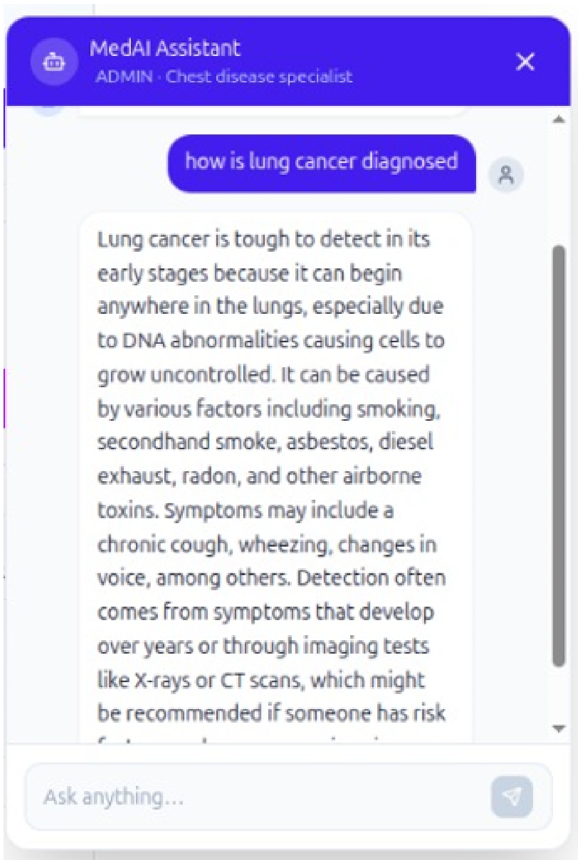
The performance profile of the proposed system is discussed in Table 3.16.

Running the ModelBackend requires sufficient computational resources because it hosts multiple AI components simultaneously, including image classification models (DenseNet, YOLO, U-Net), the router model, and the RAG pipeline with embedding models and an LLM (Ollama). For stable performance, a GPU is strongly recommended for accelerating deep learning inference, especially for chest X-ray and lung cancer detection tasks.

In addition, the system requires adequate CPU resources to handle concurrent API



(a) Example of a medical query processed by the RAG agent, showing retrieval of relevant documents and generated response.



(b) Example of a general/off-topic query handled by the RAG agent with a conversational response without medical retrieval.

Figure 3.41: Illustrative examples of the behavior of the retrieval-augmented generation (RAG) agent in different query scenarios. The system adapts its response strategy depending on whether the input is medical or general in nature.

Table 3.16: Computational performance and execution environment of the proposed system components.

Component	**Approximate CPU Time**	**GPU**
Router	50–100 ms	Automatically enabled if CUDA is available
Chest Pipeline (Segmentation + Prediction + Grad-CAM)	1–5 s	CUDA auto-enabled
Lung Cancer YOLO Inference	1–5 s	CUDA auto-enabled
R2 Download / Upload	Network-dependent	N/A

requests, image preprocessing, and routing logic, while RAM (at least 4–6 GB minimum, higher recommended) is needed to keep multiple models, FAISS indexes, and embeddings loaded in memory without performance degradation. Overall, a balanced setup with a capable GPU, multi-core CPU, and sufficient RAM ensures smooth and responsive AI processing across all services in the ModelBackend

3.8.5 Federated Learning Server

The Federated-Learning-Server is responsible for enabling privacy-preserving collaborative training across multiple hospitals without requiring any centralization of medical data. It allows hospitals to jointly train a shared model while ensuring that raw patient images never leave their local environments, maintaining strict data privacy and compliance with healthcare constraints.

The system is built using a hybrid architecture that combines a FastAPI REST service and a Flower-based gRPC server within a single Python process. The FastAPI layer acts as the control interface for registration, monitoring, and coordination tasks, while the Flower server handles the actual federated learning workflow, including distributed training and weight aggregation.

Both components are synchronized through an in-memory SharedState singleton, which maintains the current training status, active clients, and round coordination signals. This design allows seamless communication between REST-based administrative actions and gRPC-based training execution, ensuring that federated learning rounds are executed in a coordinated and efficient manner across all participating hospitals.

Design Principles

The design principles of the federated learning server are listed in Table 3.17.

The Federated-Learning-Server follows a dual-process architecture where two interconnected services run simultaneously within the same system. The main process runs a Uvicorn-based FastAPI server on port 8001, which acts as the REST bridge responsible for hospital registration, client management, monitoring, and handling webhooks from the main backend system. In parallel, a daemon process runs a Flower gRPC server on port 50051, which is responsible for coordinating the actual federated learning workflow, including distributed training and model weight aggregation across participating hospitals.

This separation allows the system to clearly divide responsibilities between control-plane operations (handled by FastAPI) and training-plane execution (handled by Flower). Both processes share a common in-memory SharedState, ensuring synchronization between REST actions and federated learning events while maintaining efficient and real-time coordination.

Training Round Lifecycle

A complete federated training round follows these steps:

Table 3.17: Core design principles and their implementation in the federated learning framework.

Principle	**Implementation**
Privacy first	Only model weight updates are transmitted between clients and the server; raw medical images never leave the hospital premises under any circumstance.
On-demand training	The Flower server remains idle until explicitly triggered by an administrator via the `POST /start-round` endpoint, ensuring that training rounds are fully controlled and non-automatic.
In-process coupling	FastAPI and Flower operate within a shared runtime environment using a `SharedState` singleton, eliminating the need for internal HTTP communication and reducing overhead.
Weight hot-reload	After each federated aggregation round, the updated global model weights are immediately pushed to the Model Backend, enabling real-time deployment of improved models.

- Step 1: Admin selects online hospitals and initiates a round
- Step 2: Spring calls start-round with round id, min clients, deadline minutes
- Step 3: FastAPI sets SharedState `trigger_event`, unblocking the Flower server thread
- Step 4: Hospital clients receive global weights via gRPC, train locally on their data
- Step 5: CustomFedAvg aggregates weight updates using the FedAvg algorithm
- Step 6: Aggregated weights saved and pushed to ModelBackend
- Step 7: Webhook notifies spring of round completion or failure
- Step 8: trigger event cleared; Flower waits for the next admin trigger

After a hospital is successfully registered in the federated learning system, Spring Boot finalizes the onboarding process by generating the necessary FL credentials and preparing a deployment package for that hospital. Once the registration is approved, the system automatically sends an email notification to the hospital using Gmail SMTP.

This email contains all essential onboarding information, including the hospital's federated learning client ID, encrypted API key, and instructions for connecting to the Federated Learning Server. It also provides a Docker-based hospital client package, which is the only component the hospital needs to run locally.

From the hospital side, the setup process is intentionally simple. The hospital team only needs to:

1. Put the data in the required folder

2. Run the container provided in the package

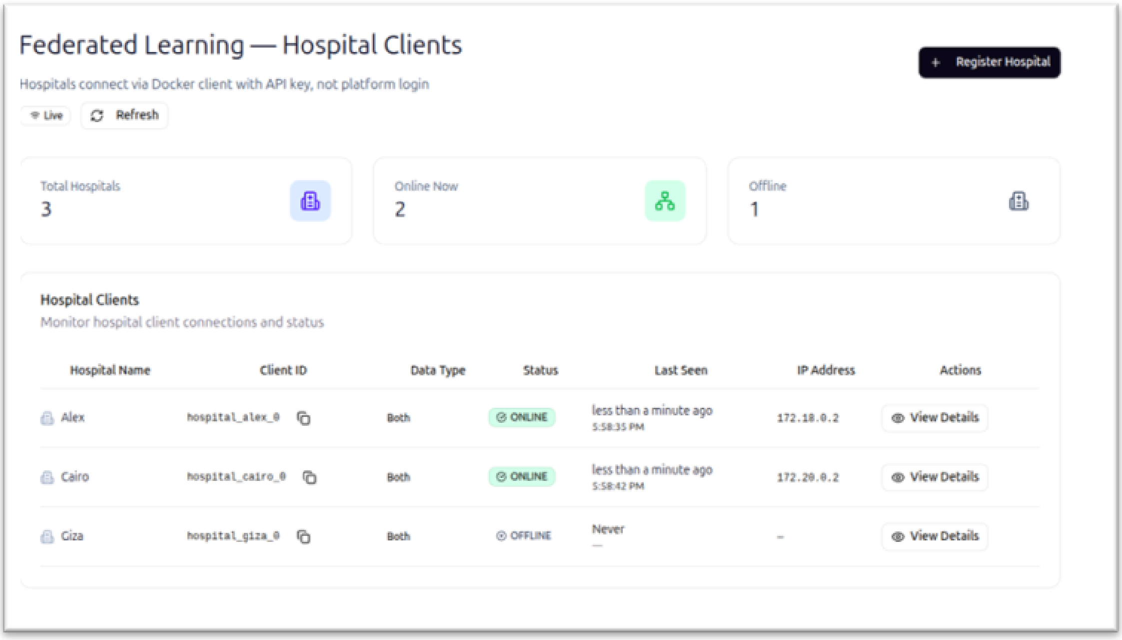


Figure 3.42: The federated learning dashboard of the web application.

Once the container is started, the hospital client automatically connects to the Federated Learning Server, begins sending heartbeat signals, and becomes eligible for participation in training rounds. After this step, no further manual integration is required, as all communication and training coordination are handled automatically by the system.

This design ensures a smooth onboarding process where hospitals do not need to manage complex machine learning setups, and can join the federated learning network by simply configuring and running a pre-built container

Once the container is running and sending heartbeat signals, the hospital is marked as ONLINE and becomes eligible to participate in federated training rounds.

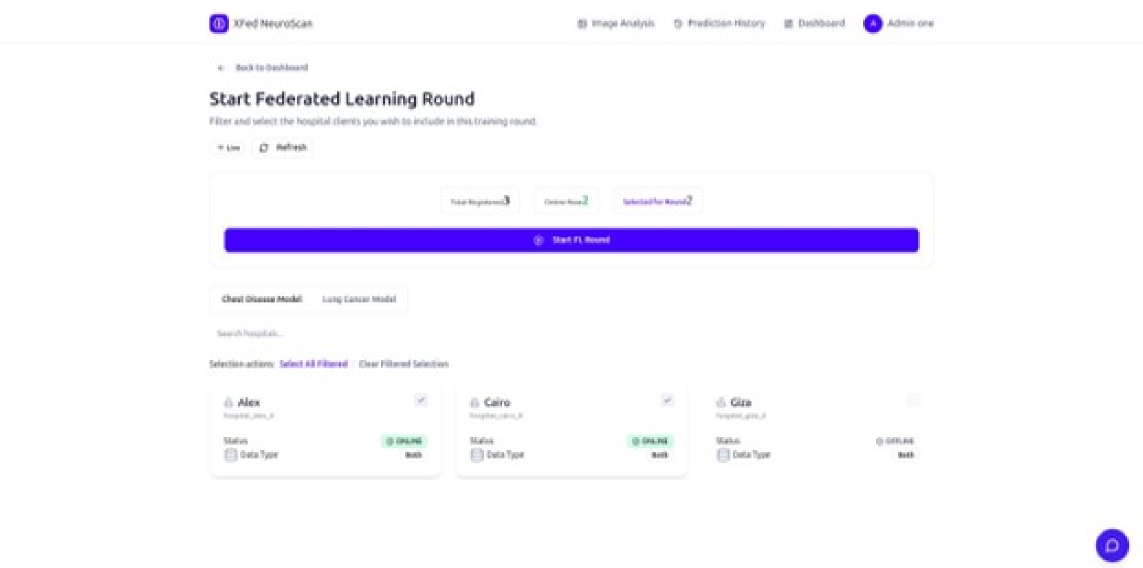


Figure 3.43: The federated learning management tab.

Round Status States

The full states and status code for the federated learning server are listed and explained in Table 3.18

Table 3.18: Federated learning server execution states and their meanings.

Status	**Meaning**
IDLE	No federated learning round is currently running; the Flower server is blocked and waiting for an external trigger event.
STARTED	A training round has been triggered and participating hospital clients are actively performing local model training.
AGGREGATING	The server is performing FedAvg aggregation over received client updates to compute the new global model.
COMPLETED	The federated learning round has successfully finished, and the updated global model weights have been saved and pushed to the Model Backend for deployment.
FAILED	The federated learning round has failed due to issues such as timeouts, client-side errors, or model update push failures.

Security

The Federated Learning security model is designed to ensure strict isolation between hospitals, the platform backend, and the training infrastructure. A Platform API key is used exclusively by the centralized Backend to securely register new hospital clients, retrieve client lists, and initiate federated learning rounds, ensuring that only the trusted backend can control system-level operations.

Each hospital node is assigned a unique Client API key, which is used solely for heartbeat authentication to confirm that the hospital is online and eligible to participate in training. In addition, a Webhook secret is shared between the FL Server and centralized Backend to verify and secure all outbound callbacks related to training round completion and status updates.

Hospitals communicate with the system only through outbound connections to port 8001 for REST heartbeats and port 50051 for gRPC-based training, eliminating the need for direct inbound access. Importantly, admin users never interact directly with the FL Server; instead, all operations are securely routed through centralized Backend using JWT authentication, maintaining a centralized and controlled security layer.

The various key types and their roles are shown and explained in great detail in Table **??**.

Table 3.19: Authentication key types and their roles within the federated learning system.

Key Type	Holder	Scope
Platform API Key	Centralized Backend	Used by the backend system to register clients, initiate federated learning rounds, and retrieve the list of participating clients.
Client API Key	Each hospital node	Used exclusively for heartbeat authentication to verify active participation and connectivity of hospital clients.
Webhook Secret	Federated Learning Server and Centralized Backend	Used to authenticate outbound callback requests from the federated learning server after training rounds to ensure secure communication.

3.8.6 End-to-End Data Flows

Medical Image Analysis Flow

1. The frontend user selects an image and requests a presigned URL from the Spring Backend.
2. The frontend uploads the image directly to Cloudflare R2 using the provided presigned URL.
3. After upload, the frontend calls the image processing service with the stored R2 key.
4. The Spring Backend saves image metadata in PostgreSQL, generates a presigned URL, and forwards the request to the FastAPI service via Feign client communication.
5. The FastAPI service downloads the image from R2, routes it using the trained routing model, and executes the appropriate specialist pipeline (either chest disease classification or lung cancer detection).
6. The FastAPI service uploads the generated outputs (such as Grad-CAM heatmaps or annotated detection images) back to R2 and returns the prediction, confidence score, and storage key.
7. The Spring Backend stores the final inference results in the database and generates presigned URLs for secure result visualization.
8. The frontend displays the prediction results, confidence scores, and visual explanations such as Grad-CAM overlays or bounding box annotations to the user.

User Authentication Flow

1. The user logs in via the login page, where credentials are verified by Spring Security.

2. If authentication succeeds, an access JWT is returned in the response body, and a refresh token is set as an HttpOnly cookie.
3. The refresh token is also stored in Redis with a defined key and a 30-day expiration period.
4. The frontend stores the access token in memory, while the authentication context initializes and loads the user role and profile information.
5. When the access token expires, the frontend calls the refresh token endpoint, which issues a new access token and rotates the refresh cookie.
6. During logout, the access token is blacklisted and the refresh token is removed from Redis, fully invalidating the user session.

Federated Learning Hospital Status Flow

1. The administrator registers a hospital through the frontend interface, and the Spring forwards the registration request to the Federated Learning (FL) Server.
2. The FL Server generates a unique API key, which is encrypted and stored by the backend. Subsequently, an onboarding email containing the Docker-based client package is sent to the hospital.
3. The hospital deploys the provided Docker client, which establishes a connection to the Flower server via gRPC for federated learning participation.
4. The hospital client initiates a heartbeat process that continuously sends periodic signals to the FL Server every 30 seconds to indicate active status.
5. The Spring Backend executes a polling mechanism every 10 seconds to retrieve the latest client status updates from the FL Server.
6. The retrieved client statuses are synchronized with the PostgreSQL database and broadcast in real time using STOMP WebSocket communication.
7. The frontend subscribes to the WebSocket channel and dynamically updates hospital ONLINE/OFFLINE indicators in real time based on received status updates.

3.9 Summary

This chapter presented the proposed end-to-end model for automated detection of chest diseases and bone fractures from medical images. The overall solution architecture was introduced, highlighting the integration of a deep learning–based diagnostic pipeline with supporting backend processes to enable efficient and scalable medical image analysis.

The deep learning pipeline was described in detail, illustrating the sequential flow from image preprocessing and feature extraction to intelligent routing and task-specific inference. Transfer learning was adopted as a core strategy to leverage pre-trained deep neural networks, enabling effective feature representation and improved generalization in the presence of limited labeled medical data.

The chapter also introduced the medical image routing model, which enables dynamic task allocation by directing input images to specialized diagnostic models. This design enhances modularity and allows the system to support multiple medical imaging tasks within a unified framework.

Then, the architecture of the chest disease classification model was discussed in detail. The design was illustrated, and the use of ResNet50 as the main feature extractor was justified. The formulation of the problem and the motivation behind such a model were presented while detailed architectural aspects of the transfer-learning model were omitted to avoid redundancy.

Finally, the bone fracture detection model was presented as a task-specific component of the system, implemented using an unmodified YOLOv8 architecture. The formulation of the fracture detection problem and the motivation for adopting a one-stage object detection approach were discussed, while detailed architectural aspects were referenced to earlier sections to avoid redundancy.

Overall, this chapter established the structural and methodological foundation of the proposed system, providing a clear understanding of its components and design choices. The described models form the basis for the experimental evaluation and performance analysis presented in the subsequent chapter.

CHAPTER 4
RESULTS & EVALUATION

Chapter 4 Results and Evaluation

This chapter presents the experimental results obtained from the implementation and evaluation of the proposed framework, *X-Fed NeuroScan: Federated Learning with Explainable AI for Medical Image Classification*. The primary objective of this chapter is to assess the effectiveness of the developed models, validate the design decisions discussed in previous chapters, and demonstrate the capability of the proposed system to accurately analyze medical images within a hierarchical diagnostic pipeline.

The evaluation process was designed to investigate both the individual performance of each model and the overall effectiveness of the integrated framework. Since the proposed architecture consists of multiple interconnected components, a comprehensive assessment is necessary to ensure that each stage performs its intended function reliably. Particular emphasis is placed on the routing model, which serves as the first level of the classification pipeline and is responsible for identifying the modality of incoming medical images before directing them to the appropriate downstream diagnostic model. The performance of the disease classification and lung cancer detection models is subsequently analyzed to determine their ability to perform accurate medical image diagnosis within their respective domains.

To provide a rigorous and objective evaluation, multiple performance metrics are employed throughout this chapter. These metrics include accuracy, precision, recall, F1-score, loss values, confusion matrices, and training convergence behavior. In addition, comparative experiments are conducted to evaluate different model architectures and hyperparameter configurations, allowing the selection of the most suitable models for inclusion within the final framework. Training and validation curves are also examined to assess optimization stability, convergence characteristics, and generalization performance.

Beyond predictive performance, this chapter also evaluates the practical aspects of the proposed framework. The effectiveness of transfer learning, data augmentation, class balancing strategies, and other implementation choices is analyzed to determine their contribution to overall model performance. Furthermore, the results obtained from the explainability component are discussed to demonstrate how visual explanations can enhance model transparency and improve the interpretability of diagnostic decisions. The chapter also examines the federated learning implementation and its ability to support collaborative model training while preserving data privacy.

The results presented throughout this chapter provide quantitative and qualitative evidence regarding the feasibility of the proposed approach for medical image analysis. By systematically evaluating each component of the framework and comparing alternative design choices, this chapter establishes the effectiveness of the developed system and provides a foundation for the discussion, conclusions, and future research directions presented in the subsequent chapters.

4.1 Evaluation of the Medical Image Routing Model

The medical image routing model represents the first level of the proposed hierarchical diagnostic framework and therefore plays a critical role in determining the overall effectiveness of the system. Since all incoming images must first pass through the routing stage before being analyzed by specialized diagnostic models, the reliability of this component directly impacts the performance of the entire framework. Consequently, a comprehensive evaluation was conducted to assess the ability of the routing model to accurately distinguish between chest X-ray images, chest CT scan images, and out-of-distribution images categorized as *Other*.

The evaluation process was designed not only to measure the classification performance of the final selected model but also to compare multiple candidate architectures and investigate the effect of hyperparameter selection on model performance. In particular, several deep learning architectures were trained and evaluated under identical experimental conditions, including a custom convolutional neural network, ResNet50, DenseNet121, and EfficientNetB3. The objective was to identify the architecture that offered the best combination of classification accuracy, generalization capability, and computational efficiency for the image routing task.

4.1.1 Implementation Environment and Development Tools

The routing model was implemented using the Python programming language and developed within a Jupyter Notebook environment. The deep learning pipeline was built using TensorFlow version 2.20.0 and Keras version 3.12.0, which provided the necessary functionality for constructing, training, and evaluating convolutional neural networks. Data preprocessing, dataset partitioning, performance evaluation, and metric computation were performed using Scikit-learn version 1.8.0.

Model training and experimentation were conducted on a workstation equipped with an NVIDIA RTX 2060 graphics processing unit (GPU). The availability of GPU acceleration significantly reduced training times and enabled the efficient evaluation of multiple candidate architectures and hyperparameter configurations. Table 4.1 summarizes the software and hardware environment used during the implementation and evaluation process.

Table 4.1: Software and hardware environment used for implementing the routing model.

Component	**Specification**
Programming Language	Python 3.12
Deep Learning Framework	TensorFlow 2.20.0
Neural Network API	Keras 3.12.0
Machine Learning Library	Scikit-learn 1.8.0
Development Environment	Jupyter Notebook
GPU	NVIDIA RTX 2060

4.1.2 Dataset Preparation and Experimental Setup

The aggregated routing dataset consisted of three image categories corresponding to the target classes of the routing problem. The first category contained chest X-ray images, the second category contained chest CT scan images, and the third category consisted of out-of-distribution images grouped under the *Other* class. The final dataset contained a total of 19,615 images distributed across the three classes as shown in Table 4.2.

Table 4.2: Class distribution of the routing dataset.

Class	**Number of Images**
Chest X-Ray	7,123
Chest CT Scan	2,534
Other	9,958
Total	19,615

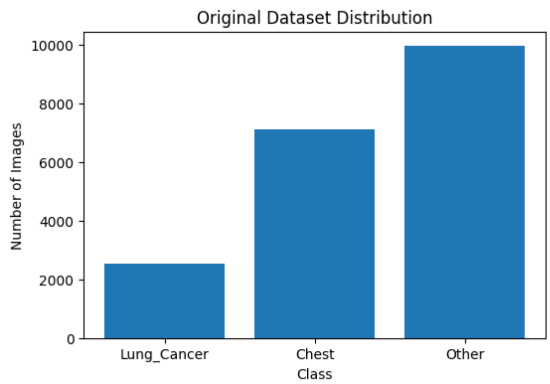


Figure 4.1: Distribution of the routing dataset classes.

Figure 4.2: Summary of the routing dataset distribution.

To ensure a fair evaluation procedure, the dataset was partitioned using a stratified sampling strategy. This approach preserved the original class distribution across all subsets and prevented sampling bias from influencing the evaluation results. The dataset was divided into training, validation, and testing subsets using a ratio of 70%, 15%, and 15%, respectively.

A number of techniques were employed to mitigate the effects of class imbalance. First, class weights were incorporated during training to increase the contribution of minority classes to the optimization process. Second, stratified sampling was utilized to preserve representative class distributions across all dataset splits. Third, extensive data augmentation was applied to improve generalization and increase sample diversity. The augmentation pipeline included common image transformation techniques such as rotation, translation, zooming, flipping, brightness adjustment, and contrast variation. Geometric transformations that could significantly distort medical structures, such as excessive shearing, were intentionally avoided in order to preserve clinically relevant image features.

All models were trained using a batch size of 8 for a total of 10 epochs. The same training, validation, and testing partitions were used across all experiments to ensure consistency when comparing model performance.

4.1.3 Learning Rate Optimization

Since learning rate selection plays a crucial role in determining convergence behavior and final model performance, multiple learning rate values were investigated for each architecture. The evaluated learning rates included 10^{-5}, 10^{-4}, 10^{-3}, and 10^{-2}.

The experiments revealed that different architectures exhibited different optimization characteristics. DenseNet121 and ResNet50 achieved their highest performance when trained using a learning rate of 10^{-2}, whereas EfficientNetB3 performed best with a learning rate of 10^{-4}. Due to its relatively shallow architecture and limited feature extraction capacity, the custom convolutional neural network achieved its best results with a learning rate of 10^{-5}.

Table 4.3 summarizes the optimal learning rate selected for each evaluated architecture.

Table 4.3: Optimal learning rates identified during hyperparameter tuning.

Model	**Optimal Learning Rate**
Custom CNN	10^{-5}
ResNet50	10^{-2}
DenseNet121	10^{-2}
EfficientNetB3	10^{-4}

These findings highlight the importance of architecture-specific hyperparameter tuning. Applying a common learning rate across all models would have resulted in suboptimal performance and potentially misleading comparisons between candidate architectures.

4.1.4 Comparative Evaluation of Candidate Architectures

To identify the most suitable backbone architecture for the routing model, four candidate architectures were trained and evaluated under identical conditions. The evaluated architectures included a custom CNN developed specifically for this task, ResNet50, DenseNet121, and EfficientNetB3.

The comparison focused on commonly used classification metrics, including accuracy, precision, recall, F1-score, and loss. Table 4.4 summarizes the performance of all evaluated architectures.

Table 4.4: Performance comparison of candidate routing architectures.

Model	Accuracy	Precision	Recall	F1-Score	Loss
Custom CNN	1.0000	1.0000	1.0000	1.0000	0.0011
ResNet50	1.0000	1.0000	1.0000	1.0000	1.782×10^{-6}
EfficientNetB3	1.0000	1.0000	1.0000	1.0000	1.9816×10^{-5}
DenseNet121	1.0000	1.0000	1.0000	1.0000	1.5126×10^{-7}

The results demonstrate the superiority of DenseNet121 for the routing task. DenseNet121 achieved perfect classification performance on the test dataset, obtaining an accuracy, precision, recall, and F1-score of 100%. Furthermore, the model achieved an extremely low loss value of 1.5126×10^{-7}, indicating highly confident predictions and excellent convergence behavior.

The dense connectivity mechanism employed by DenseNet121 enables efficient feature reuse throughout the network and facilitates gradient propagation across deep layers. As a result, the model was able to learn highly discriminative representations that effectively separated the three image categories. Based on these findings, DenseNet121 was selected as the backbone architecture of the final routing model.

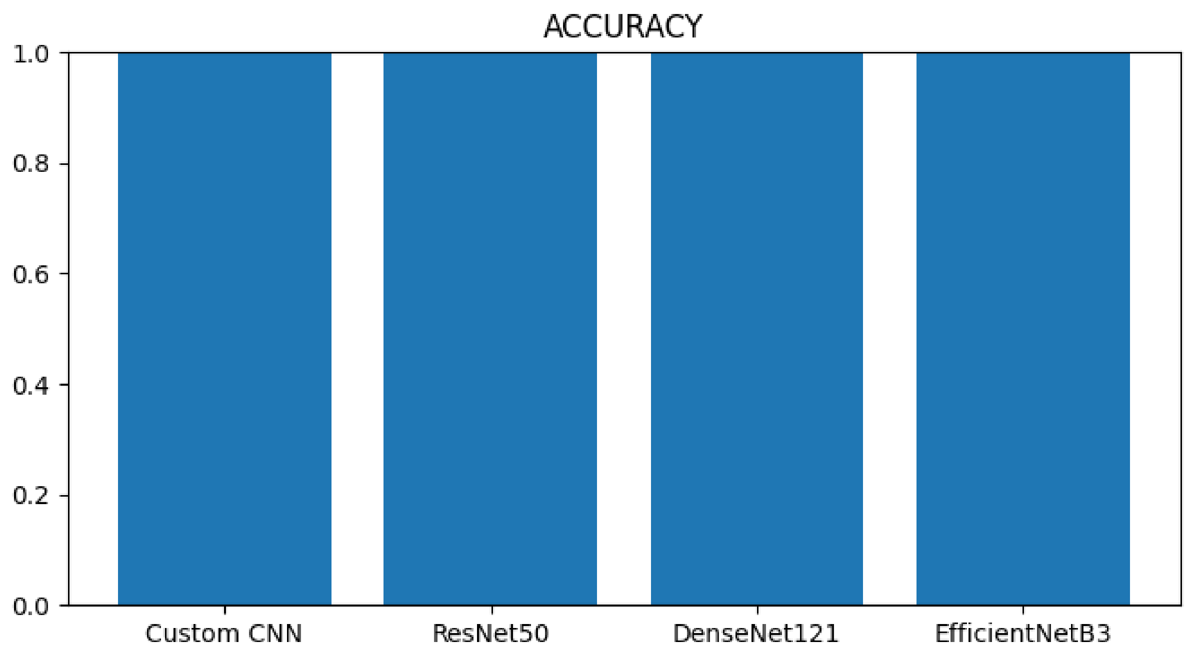


Figure 4.3: Comparison of classification accuracy across all evaluated routing architectures.

4.1.5 Training and Validation Behavior

In addition to evaluating final classification performance, the training dynamics of each architecture were analyzed to assess convergence behavior and learning stability throughout the optimization process.

Figures 4.4 illustrate the evolution of training accuracy, validation accuracy, training loss, and validation loss during the training process for all evaluated models.

Analysis of these curves provides valuable insight into convergence speed, optimization stability, and generalization behavior. The results indicate that DenseNet121 exhibited rapid convergence while maintaining stable validation performance throughout training, further supporting its selection as the final routing architecture.

4.1.6 Confusion Matrix Analysis

To further investigate classification performance, confusion matrices were generated for each evaluated architecture. Confusion matrices provide a detailed view of model predictions by illustrating the relationship between true class labels and predicted class labels.

The confusion matrix analysis is particularly important for the routing stage because misclassification at this level may cause images to be forwarded to an inappropriate downstream diagnostic model. Therefore, minimizing routing errors is essential for ensuring the reliability of the entire hierarchical framework.

Figure **??** presents the confusion matrices obtained for all evaluated architectures.

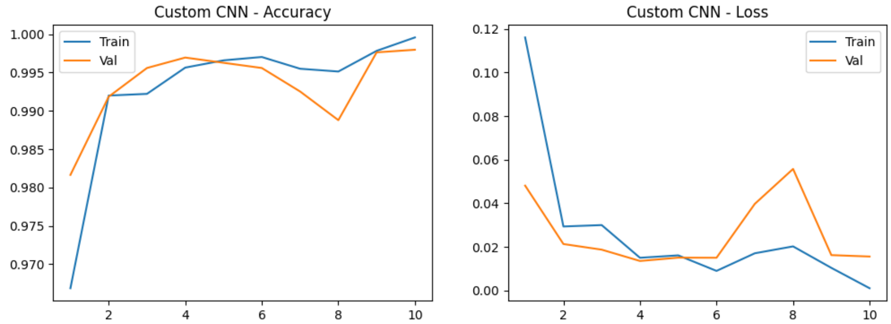
The confusion matrix corresponding to DenseNet121 demonstrated perfect classification performance across all three classes, with no observed misclassifications in the test set. This result confirms the model's ability to accurately distinguish between chest X-ray images, chest CT scans, and out-of-distribution images, making it highly suitable for deployment as the first-level routing component within the proposed framework.

4.1.7 Discussion

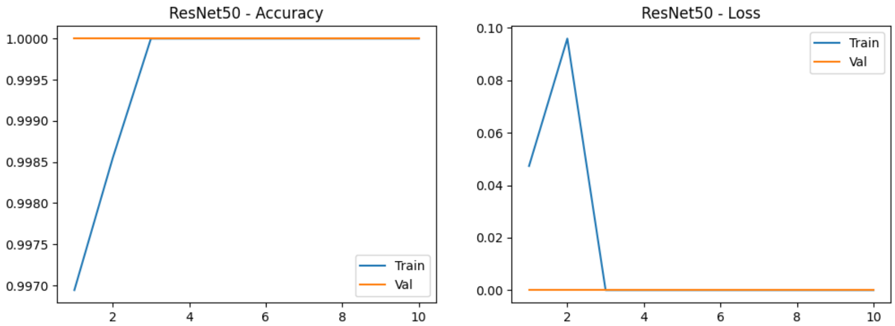
The experimental results demonstrate that the proposed routing model successfully fulfills its intended objective of accurately identifying image modalities and directing images toward the appropriate diagnostic pathway. The combination of transfer learning, data augmentation, class weighting, and hyperparameter optimization enabled the evaluated architectures to achieve strong classification performance despite the inherent challenges associated with heterogeneous medical imaging data.

Among all tested architectures, DenseNet121 consistently produced the most favorable results and achieved perfect performance on the test dataset. The model's dense connectivity structure enabled efficient feature reuse and facilitated the extraction of highly discriminative modality-specific representations. Furthermore, the confusion matrix analysis confirmed the absence of routing errors, indicating that all images were successfully assigned to their correct categories.

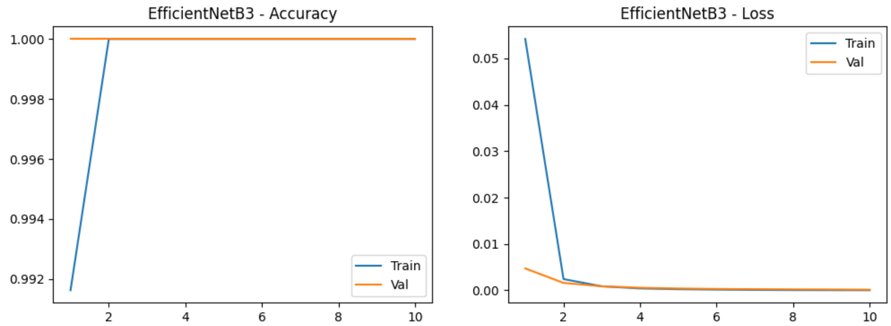
These findings validate the effectiveness of the proposed first-level routing strategy and demonstrate that the routing component provides a reliable foundation for the subsequent disease classification and lung cancer detection stages of the overall framework.



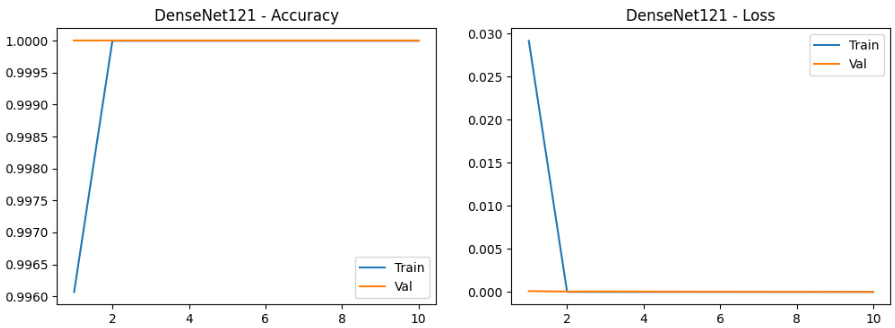
(a) Training and validation curves for the Custom CNN model.



(b) Training and validation curves for the ResNet50 model.

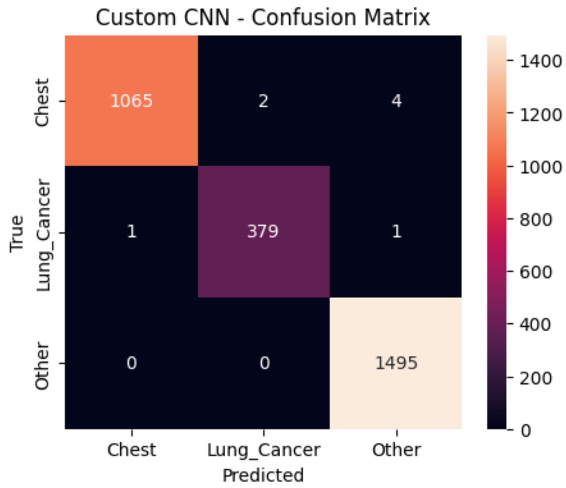


(c) Training and validation curves for the EfficientNetB3 model.

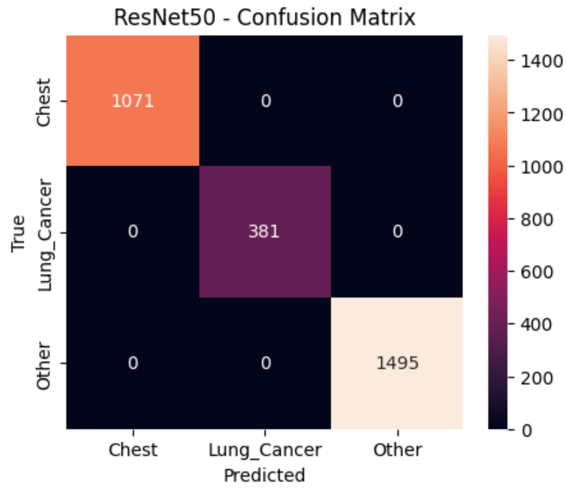


(d) Training and validation curves for the DenseNet121 model.

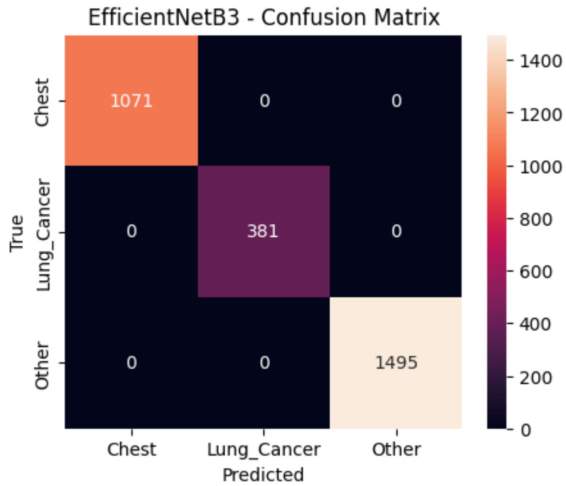
Figure 4.4: Training and validation performance curves for all evaluated routing model architectures. Each figure illustrates the evolution of accuracy and loss throughout the training process and provides insight into model convergence, optimization stability, and generalization behavior.



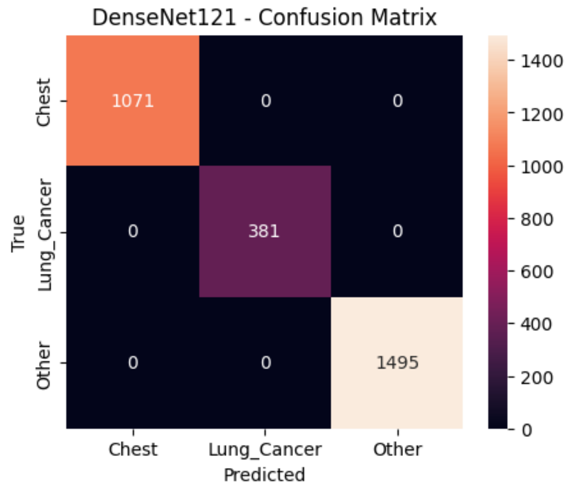
(a) Custom CNN



(b) ResNet50



(c) EfficientNetB3



(d) DenseNet121

Figure 4.5: Confusion matrices for all evaluated routing models across the three classes.

4.2 Evaluation of the Lung Cancer Detection Model

This chapter presents the experimental results of two lung cancer detection systems built on the YOLOv8n architecture: a centralized deep learning model and a federated learning (FL) model based on the FedAvg aggregation strategy. Both systems were trained on a dataset of lung CT images annotated with bounding boxes for two pathological classes — tumor and tuberculosis — and evaluated using four standard object detection metrics: Precision, Recall, mean Average Precision at IoU 0.5 (mAP@0.5), and mAP across IoU thresholds 0.5 to 0.95 (mAP@0.5:0.95).

4.2.1 Experimental Setup

The centralized model was trained locally on an Intel Core i7-8850H CPU at 2.60 GHz using Ultralytics YOLOv8n — a compact architecture comprising 3,006,038 parameters and requiring 8.1 GFLOPs per inference pass. Training ran for 50 epochs at an image size of 640×640 with a batch size of 16. The dataset was split into training (75%), validation (15%), and test (10%) subsets, yielding 253 test images with 759 annotated instances.

The federated learning system was executed on a Google Colab instance equipped with an NVIDIA Tesla T4 GPU (14 GB VRAM) using PyTorch 2.11 and Ultralytics 8.4.62. Three clients participated in 5 federated communication rounds, each performing 3 local training epochs using the AdamW optimizer ($lr = 0.001667$). Training data was partitioned randomly and equally across the 3 clients (approximately 632 images per client), while the validation set (380 images, 1,202 instances) and test set (253 images, 759 instances) were shared across all clients. The FedAvg strategy aggregated client weights proportionally to local dataset size after each round.

4.2.2 Centralized Model Results

The centralized YOLOv8n model was evaluated on the held-out test set of 253 images containing 759 annotated lesion instances. Table 4.5 reports the per-class detection metrics on the test set.

Table 4.5: Per-class detection results of the centralized YOLOv8n model on the test set (253 images, 759 instances).

Model / Class	Precision	Recall	mAP@0.5	mAP@0.5:0.95
Centralized YOLOv8n (all)	0.804	0.609	0.703	0.408
Centralized YOLOv8n (tumor)	0.804	0.609	0.703	0.408

The model achieved a Precision of 0.804 and a Recall of 0.609 on the test set, yielding an mAP@0.5 of 0.703 and an mAP@0.5:0.95 of 0.408. The high Precision indicates that the vast majority of the model's bounding box predictions correspond to genuine lesions, while the moderate Recall reflects that some ground-truth annotations were missed. This conservative detection behaviour is a known characteristic of nano-scale YOLO models trained on CPU hardware with a limited number of epochs. The mAP@0.5:0.95 score of 0.408, while lower than the mAP@0.5, is expected given the stricter localization requirements imposed by higher IoU thresholds. Overall, the centralized model demonstrates solid baseline performance for a privacy-unaware single-node training scenario, and serves as the upper-bound reference point for the federated learning comparison.

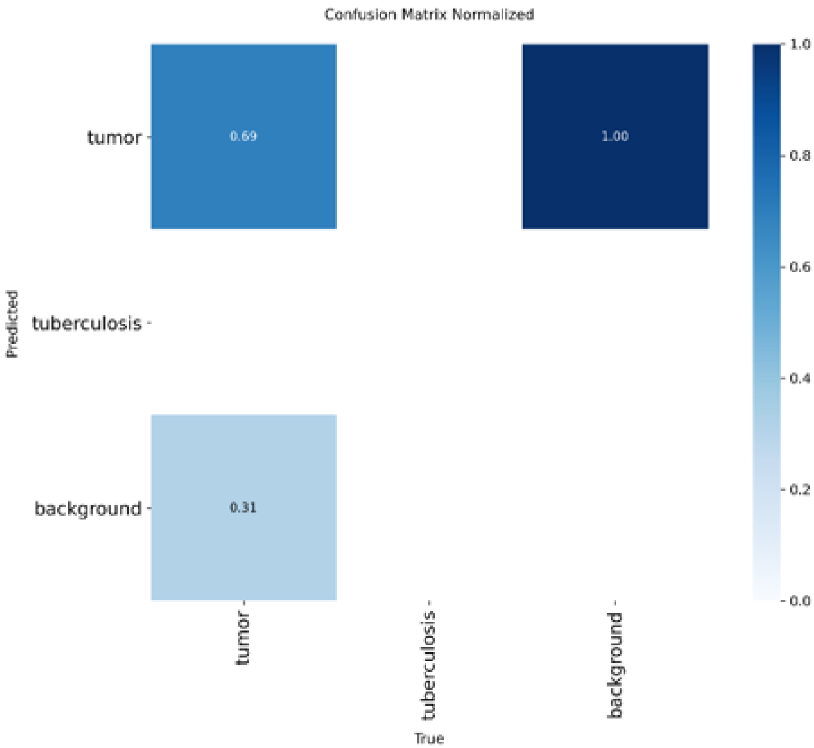


Figure 4.6: Normalized confusion matrix of the centralized YOLOv8n model.

4.2.3 Federated Learning Model Results

Following 5 federated communication rounds with 3 participating clients, the aggregated global YOLOv8n model was evaluated on both the shared validation and test sets. Table 4.6 summarizes the results across both splits.

Table 4.6: Evaluation results of the federated YOLOv8n (FedAvg) model on validation and test sets.

Split	Precision	Recall	mAP@0.5	mAP@0.5:0.95	Images / Instances
Validation	0.373	0.104	0.122	0.054	380 / 1,202
Test	0.431	0.113	0.133	0.061	253 / 759

On the test set, the federated model achieved a Precision of 0.431, a Recall of 0.113, an mAP@0.5 of 0.133, and an mAP@0.5:0.95 of 0.061. The validation and test results are broadly consistent with each other, suggesting the federated model generalizes without significant overfitting to any individual client's local distribution. However, the absolute

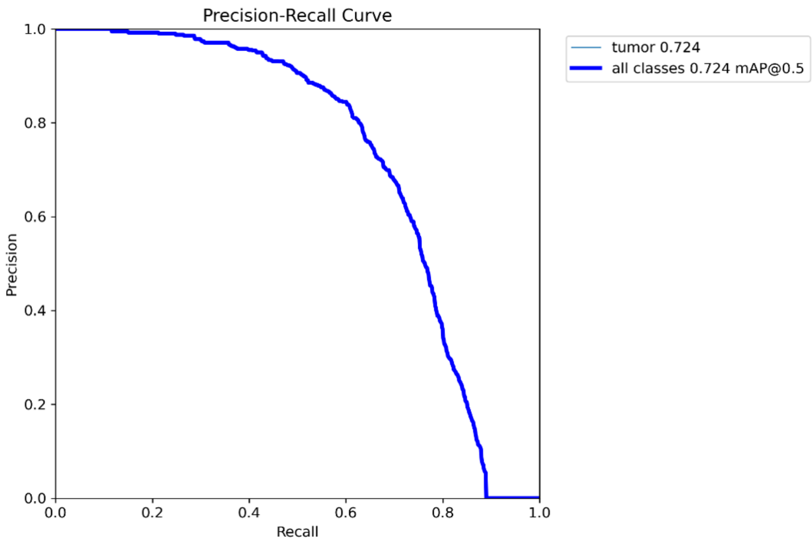


Figure 4.7: Precision-Recall curve of the centralized YOLOv8n model.

metric values are substantially lower than those of the centralized baseline, indicating that the federated training process did not converge to a comparable detection quality within the given number of rounds and local epochs.

4.2.4 Comparative Analysis

Table 4.7 presents a direct comparison of the two approaches on the test set, and Table 4.8 quantifies the performance gap in both absolute and relative terms.

Table 4.7: Side-by-side comparison of centralized vs. federated YOLOv8n on the test set.

Model / Class	Precision	Recall	mAP@0.5	mAP@0.5:0.95
Centralized YOLOv8n	0.804	0.609	0.703	0.408
Federated YOLOv8n (FedAvg)	0.431	0.113	0.133	0.061

Table 4.8: Absolute and relative performance gap between centralized and federated models (test set).

Metric	Centralized	Federated	Δ (Abs.)	Δ (%)
Precision	0.804	0.431	-0.373	-46.4%
Recall	0.609	0.113	-0.496	-81.4%
mAP@0.5	0.703	0.133	-0.570	-81.1%
mAP@0.5:0.95	0.408	0.061	-0.347	-85.0%

The federated model underperforms the centralized baseline across all four metrics, with the largest relative gap observed in Recall (-81.4%) and mAP@0.5 (-81.1%). Several factors explain this performance gap:

1. **Insufficient communication rounds:** Only 5 federated rounds were conducted, each with 3 local epochs. This amounts to 15 effective local training epochs per client, compared to 50 full epochs in the centralized setting. The global model had limited opportunity to converge through the aggregation process.

2. **Non-IID data distribution:** Although training images were randomly partitioned, the relatively small client datasets ($\approx$ 632 images each) may have introduced statistical heterogeneity that FedAvg alone cannot fully compensate for in a small number of rounds.

3. **Warm-up re-initialization overhead:** The federated client implementation performs a single-epoch warm-up before each round to resize the detection head to the correct number of classes. With only 5 rounds, this overhead represents a significant fraction of total training time and limits how much each client can specialize its local model.

4. **Hardware discrepancy:** The centralized model trained for 50 epochs on the full dataset in a single run, allowing the optimizer to traverse more of the loss landscape. Federated training on a shared Colab GPU with sequential clients introduced additional latency and restricted per-round batch count.

Despite these limitations, the federated approach offers a critical advantage: no raw patient data is transmitted beyond each client node. This makes it inherently more suitable for deployment across multiple hospitals or diagnostic centres operating under strict data protection regulations such as HIPAA or GDPR. With further tuning — more rounds, more local epochs, a larger client count, or an advanced aggregation strategy such as FedProx or SCAFFOLD — the performance gap is expected to narrow significantly.

CHAPTER 5
CONCLUSION

Chapter 5 Conclusion

The rapid advancement of artificial intelligence has transformed numerous scientific and industrial domains, with healthcare emerging as one of the most promising areas for practical application. Recent developments in deep learning have demonstrated remarkable capabilities in medical image analysis, enabling automated systems to assist healthcare professionals in disease screening, diagnosis, prognosis estimation, and treatment planning. Despite these advancements, the widespread adoption of artificial intelligence within clinical environments continues to face several challenges. Many existing solutions are developed for highly specific diagnostic tasks, often focusing on a single disease, a single imaging modality, or a single dataset. Furthermore, concerns related to model interpretability, data privacy, scalability, and real-world deployment remain significant barriers to adoption.

These challenges motivated the development of *X-Fed NeuroScan: Federated Learning with Explainable AI for Medical Image Classification*. Rather than approaching medical image analysis as a collection of isolated classification problems, this project sought to develop a unified framework capable of handling multiple imaging modalities while simultaneously incorporating explainability and privacy-preserving learning mechanisms. The resulting system combines hierarchical image routing, multi-label chest disease classification, lung cancer detection, Explainable Artificial Intelligence (XAI), Federated Learning (FL), and web-based deployment within a single integrated architecture.

This chapter concludes the thesis by reflecting upon the objectives established at the beginning of the project, evaluating the achievements of the proposed framework, discussing its practical significance, summarizing its scientific and engineering contributions, examining the challenges encountered throughout development, identifying current limitations, and outlining several promising directions for future research.

5.1 Achievement of Project Objectives

The primary objective of this project was to design and implement an intelligent medical image analysis framework capable of supporting multiple diagnostic workflows while maintaining flexibility, interpretability, and scalability. Unlike conventional systems that assume a fixed input modality and a single diagnostic task, the proposed framework was designed to accommodate heterogeneous medical images through a hierarchical processing pipeline.

The first objective involved the development of an intelligent routing mechanism capable of distinguishing between chest X-ray images, CT scans, and unsupported image categories. This objective was successfully achieved through the implementation and evaluation of several deep learning architectures, ultimately leading to the selection of DenseNet121 as the routing model. The routing stage enables modality-aware diagnosis and establishes a foundation upon which subsequent classification stages can operate efficiently.

The second objective focused on the development of a chest disease classification model capable of identifying multiple respiratory diseases simultaneously. Rather than treating disease diagnosis as a mutually exclusive classification problem, the proposed framework adopted a multi-label learning paradigm that reflects the realities of clinical practice. Through the implementation of a ResNet50-based architecture, the system was capable of recognizing COVID-19, Tuberculosis, Pneumonia, and normal cases within chest radiographs.

The third objective involved the integration of a dedicated lung cancer detection model for CT scans. This objective was accomplished through the implementation of a YOLOv8-based detector capable of identifying suspicious cancer-related findings in computed tomography images. The incorporation of a specialized detection model expanded the diagnostic capabilities of the framework and demonstrated the feasibility of combining classification and detection tasks within a unified system.

The fourth objective concerned the incorporation of explainability mechanisms. This objective was addressed through the integration of Grad-CAM visualizations that provide interpretable heatmaps highlighting image regions that contribute most significantly to model predictions. These explanations enhance transparency and support trust in automated decision-making processes.

The fifth objective involved exploring privacy-preserving collaborative learning through Federated Learning. Using the Flower framework, the project successfully demonstrated the feasibility of distributed model training without requiring centralized access to sensitive medical data. Although the experiments were conducted on a limited scale, the implementation establishes a foundation for future multi-institutional collaboration.

Finally, the project sought to demonstrate the practical deployability of the proposed system. This objective was achieved through the development of a web application that integrates all major framework components into an accessible interface suitable for demonstration and future extension.

Taken collectively, these achievements demonstrate that the project successfully fulfilled its primary objectives and established a comprehensive framework that extends beyond conventional medical image classification systems.

5.2 Discussion of the Proposed Framework

One of the defining characteristics of the proposed framework is its hierarchical architecture. Traditional medical image analysis systems often assume that all incoming images belong to a predefined modality and can therefore be processed directly by a diagnostic model. While such assumptions may simplify implementation, they do not accurately reflect real-world healthcare environments where multiple imaging modalities coexist and image sources may vary significantly.

The proposed framework addresses this challenge by introducing a dedicated routing stage that acts as an intelligent gateway between incoming data and downstream diagnostic models. This design provides several important advantages. First, it improves modularity by allowing each diagnostic component to specialize in a specific task. Second, it facilitates future expansion since additional modalities and diagnostic models can be incorporated without redesigning the entire system. Third, it improves robustness by introducing explicit handling of out-of-distribution inputs.

The architecture also demonstrates the value of combining multiple artificial intelligence paradigms within a single framework. Rather than viewing classification, detection, explainability, and federated learning as independent research topics, this project illustrates how these technologies can complement one another when integrated thoughtfully. The resulting system is therefore not merely a collection of models but rather a cohesive framework designed with practical deployment considerations in mind.

Another notable aspect of the framework is its emphasis on scalability. Healthcare institutions differ significantly in terms of available resources, data accessibility, and computational infrastructure. The modular design adopted throughout this project facilitates adaptation to different deployment scenarios and creates opportunities for future customization according to institutional needs.

5.3 Discussion of Individual System Components

5.3.1 Routing Model

The routing model represents the cornerstone of the proposed architecture. Its primary responsibility is to identify the modality of incoming images and ensure that they are directed toward the appropriate diagnostic pathway. Although modality classification may appear straightforward when compared to disease detection, its role within the overall framework is critically important.

An incorrect routing decision would propagate errors throughout the remainder of the system. For example, forwarding a CT scan to the chest disease classifier or directing an unsupported image toward a diagnostic model could compromise reliability. Consequently, substantial effort was devoted to evaluating alternative architectures and optimizing model performance.

The experimentation process highlighted the effectiveness of transfer learning for modality recognition. While a custom CNN provided a lightweight baseline solution, pretrained architectures consistently demonstrated stronger feature extraction capabilities. DenseNet121 ultimately emerged as the preferred solution due to its efficient dense connectivity structure and strong generalization characteristics.

The inclusion of an out-of-distribution category constitutes another important contribution of the routing stage. Medical imaging environments frequently contain data that fall outside the scope of a particular system. By explicitly recognizing unsupported inputs, the framework adopts a safer and more realistic approach to deployment.

5.3.2 Chest Disease Classification Model

The chest disease classification component represents one of the most clinically significant elements of the framework. Respiratory diseases continue to pose substantial public health challenges worldwide, and timely diagnosis remains essential for effective treatment and disease management.

A key design decision involved formulating the problem as a multi-label classification task. Many existing studies simplify disease diagnosis by assuming that each image belongs to a single category. While convenient from a machine learning perspective, this assumption may not accurately reflect clinical reality. Patients can present with multiple conditions simultaneously, and diagnostic systems should be capable of representing such complexity.

The ResNet50 architecture was selected due to its proven effectiveness in medical imaging applications and its ability to learn rich feature representations. Through residual connections, the model mitigates optimization difficulties associated with deep neural networks while maintaining strong feature extraction capabilities.

Beyond predictive performance, the multi-label formulation contributes to the clinical relevance of the framework. By recognizing multiple diseases independently, the model more closely aligns with real-world diagnostic requirements and provides a foundation for future expansion to additional thoracic conditions.

5.3.3 Lung Cancer Detection Model

Lung cancer remains one of the most serious health challenges worldwide and is responsible for a significant proportion of cancer-related mortality. Early detection is often critical for improving treatment outcomes and patient survival rates.

The decision to employ YOLOv8 for lung cancer detection reflects the project's emphasis on practical applicability. Unlike traditional classification models that produce only image-level predictions, object detection models provide localization information that identifies potentially suspicious regions within the image. This capability is particularly valuable in clinical settings because it enables healthcare professionals to understand not only what prediction was made but also where relevant findings were identified.

The integration of YOLOv8 demonstrates the flexibility of the proposed architecture and illustrates how detection-based approaches can complement classification-based systems. Future extensions may further leverage localization capabilities to support more advanced diagnostic workflows.

5.3.4 Explainable Artificial Intelligence Component

The integration of Explainable Artificial Intelligence represents one of the most important aspects of the proposed framework. While deep learning models have achieved impressive performance across numerous medical imaging tasks, their adoption in healthcare is often limited by concerns regarding transparency and trustworthiness.

Medical professionals are accustomed to making decisions based on observable evidence. Consequently, systems that provide predictions without explanation may encounter resistance regardless of their predictive accuracy. Explainability mechanisms address this challenge by providing insights into the reasoning process of artificial intelligence models.

Grad-CAM was selected due to its compatibility with convolutional neural networks and its ability to generate intuitive visual explanations. By highlighting image regions that influence model predictions, Grad-CAM helps bridge the gap between automated analysis and human interpretation.

The incorporation of explainability transforms the framework from a purely predictive system into a more transparent decision-support tool. This distinction is particularly important in healthcare environments where accountability and interpretability play central roles in clinical decision-making.

5.3.5 Federated Learning Component

Privacy remains one of the most significant obstacles to large-scale artificial intelligence development in healthcare. Medical institutions often possess valuable datasets that could contribute to improved model performance, yet legal, ethical, and regulatory constraints frequently restrict direct data sharing.

The federated learning component was introduced to address this challenge. Through distributed training, institutions can collaboratively contribute to model development while maintaining local control over sensitive patient information. This paradigm represents a promising alternative to conventional centralized training approaches.

The implementation of federated learning using the Flower framework demonstrated that collaborative model development can be achieved without compromising data privacy. Although the experiments conducted in this project were limited in scale and utilized homogeneous client distributions, they nevertheless provide evidence supporting the feasibility of integrating federated learning into medical imaging workflows.

The significance of this contribution extends beyond the scope of the current project. Federated learning has the potential to fundamentally reshape how medical artificial intelligence systems are developed by enabling cooperation among institutions that would otherwise be unable to share data directly.

5.4 Practical and Clinical Significance

Beyond its technical contributions, the proposed framework possesses important practical implications. Healthcare systems around the world continue to face increasing demands for efficient diagnostic support tools capable of assisting medical professionals in managing growing workloads.

The modality-aware design of the framework aligns well with realistic healthcare environments where multiple imaging modalities coexist. Furthermore, the integration of explainability mechanisms addresses one of the most commonly cited concerns regarding artificial intelligence adoption in medicine.

The federated learning component further enhances practical relevance by acknowledging the privacy requirements that characterize modern healthcare systems. By incorporating privacy-preserving collaboration from the outset, the framework anticipates challenges that frequently emerge during real-world deployment.

The development of a web-based implementation also demonstrates the feasibility of translating research prototypes into accessible software platforms. While additional validation would be required before clinical deployment, the project establishes an important bridge between research and application.

5.5 Project Contributions

The contributions of this project can be categorized into architectural, methodological, and practical contributions.

Architecturally, the project introduced a hierarchical modality-aware framework capable of integrating multiple diagnostic models within a single unified system. Methodologically, the project combined routing, multi-label classification, object detection, explainability, and federated learning into a cohesive workflow. Practically, the project demonstrated the deployability of these technologies through a functional web application.

Collectively, these contributions distinguish the proposed framework from many existing studies that focus exclusively on isolated diagnostic tasks.

5.6 Challenges Encountered During Development

The development of X-Fed NeuroScan was accompanied by numerous technical and practical challenges. One of the most significant challenges involved dataset aggregation. The datasets used throughout the project originated from multiple sources and exhibited substantial variability in image quality, resolution, labeling standards, and class distributions. Harmonizing these datasets required extensive preprocessing and validation efforts.

Class imbalance constituted another major challenge. Certain categories contained significantly fewer samples than others, creating the potential for biased learning. Data augmentation and balancing techniques were therefore incorporated to improve model robustness and reduce overfitting.

The implementation of multi-label learning introduced additional complexity. Unlike single-label classification, multi-label prediction requires the model to learn independent disease representations while simultaneously accounting for potential interactions among conditions.

The integration of explainability mechanisms presented its own challenges. Generating meaningful visual explanations requires careful alignment between model architecture and interpretation methodology. Ensuring that Grad-CAM outputs remained informative and clinically interpretable required extensive experimentation.

Federated learning introduced a different set of difficulties related to distributed training, synchronization, and communication management. Even within a controlled experimental environment, coordinating multiple clients and aggregating model updates demanded careful implementation.

Finally, integrating all system components into a unified web application required substantial engineering effort. Each component had to be designed not only for individual performance but also for compatibility with the overall framework.

5.7 Limitations

Despite its achievements, several limitations remain. Dataset diversity remains constrained relative to the enormous variability present in real-world healthcare environments. Additional validation on larger and more geographically diverse populations is necessary.

The federated learning experiments were conducted under relatively controlled conditions and therefore do not fully capture the complexities associated with large-scale multi-institutional deployments. Similarly, the explainability component relies exclusively on Grad-CAM and may benefit from complementary interpretability techniques.

The framework also focuses exclusively on imaging data and does not currently incorporate additional clinical information such as laboratory tests, patient history, pathology reports, or genomic data. These information sources may provide valuable context that could enhance diagnostic performance.

Furthermore, extensive prospective clinical validation remains necessary before deployment within healthcare settings can be considered.

5.8 Future Work

The work presented in this thesis establishes a foundation for numerous future research directions. One particularly promising avenue involves the adoption of Vision Transformers and hybrid CNN-transformer architectures. Recent advances suggest that transformer-based models may offer improved feature representation capabilities for medical imaging tasks, particularly when trained on large datasets.

Another important direction involves self-supervised and foundation-model learning. Medical image annotation is expensive and time-consuming, making self-supervised approaches particularly attractive. Future researchers may investigate how large-scale pretrained medical vision models can be adapted to the tasks explored in this project.

The lung cancer component may be extended through full three-dimensional CT volume analysis. Rather than processing individual slices, future systems could leverage volumetric information to capture richer anatomical context and improve detection performance.

Future work should also investigate advanced federated learning strategies capable of handling non-IID data distributions, heterogeneous client resources, and large-scale institutional participation. Methods such as FedProx, FedNova, Scaffold, and personalized federated learning represent promising directions.

The explainability component may be enhanced through the integration of complementary techniques including SHAP, LIME, Integrated Gradients, and attention-based visualization methods. Combining multiple explanation approaches may provide deeper insights into model behavior and improve clinical trust.

Another significant opportunity involves multimodal learning. Future versions of the framework could integrate imaging data with electronic health records, laboratory results, pathology findings, and genomic information to create more comprehensive clinical decision-support systems.

Continual learning represents another valuable research direction. Medical knowledge and imaging technologies evolve continuously, making it important for artificial intelligence systems to adapt without requiring complete retraining.

The development of Federated Explainable Artificial Intelligence systems constitutes an especially promising area of future investigation. Combining privacy-preserving collaboration with interpretable decision-making could address two of the most significant barriers to healthcare AI adoption simultaneously.

Finally, future research should focus on large-scale clinical validation and real-world deployment studies. Collaborations with hospitals, healthcare providers, and research institutions would provide valuable insights regarding usability, workflow integration, and clinical impact.

5.9 Final Remarks

This thesis has presented the design, implementation, and evaluation of X-Fed NeuroScan, a unified framework that combines hierarchical medical image classification, explainable artificial intelligence, federated learning, and web-based deployment within a single system. Through the integration of DenseNet121-based modality routing, ResNet50-based multi-label chest disease classification, YOLOv8 lung cancer detection, Grad-CAM explainability, and Flower-based federated learning, the project demonstrates how multiple advanced artificial intelligence technologies can be combined to address complex healthcare challenges.

Although significant opportunities for improvement remain, the work presented throughout this thesis demonstrates the feasibility and potential value of modality-aware, explainable, and privacy-preserving medical image analysis systems. The proposed framework contributes to the growing body of research seeking to bridge the gap between artificial intelligence innovation and practical healthcare deployment. It is hoped that the ideas, methodologies, and findings presented in this work will serve as a foundation for future research aimed at developing more capable, trustworthy, and collaborative intelligent healthcare systems.

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